

Regression Techniques

Lecture Notes

Indian Statistical Institute

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Chapter 1

Pre-requisites

1.1 Introduction to Linear Regression Models.

Data:

- Dependent variable or response (y),
- Independent variables or covariates or regressors ($x_1, x_2, \dots, x_p$)
- Usually, we have n observations on all variables.

$$\underset{\sim_n}{y} = \begin{bmatrix} y_1 \\ \vdots \\ y_n \end{bmatrix} : \text{The vector of } n \text{ observations of the response.}$$

$$\begin{aligned} X_n &= \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}^{n \times p} : \text{Design matrix or matrix of } n \text{ observations of } p \text{ covariates} \\ &= ((x_{ij})) \end{aligned}$$

$\underset{\sim_i}{x}$: i th row of X_n provides the i th observation on all variables, $i = 1, \dots, n$.

$\underset{\sim_j}{x}$: j th column of X_n provides a vector of all observations on the j th variables, $j = 1, \dots, p$.

Examples

1. *Air-quality (UCI)*. Data on 5 metal oxide chemical sensors (CO, Non-metanic hydrocarbon, Benzene, NOx, NO₂) and the temperature, humidity, relative humidity. (See the data [here](#))

Q: Does air-quality affect the temperature, or humidity, or relative humidity?

2. *Boston Housing data (Kaggle)*. See the data description [here](#). Read the data description and identify the underlying regression-related question in the data.

Regression setup. How to relate the temperature with the air pollution indicators in the AQI data?

Usually all the variables are assumed to be random. At least the response variable (Y) is assumed to be random, and the n observations $y_{\sim n} = (y_1, \dots, y_n)^\top$ are realizations of n independent samples.

In regression, one assumes that

$$E(Y) = g(x_1, \dots, x_p), \quad \text{or,} \quad E(y|X_1 = x_1, \dots, X_p = x_p) = g(x_1, \dots, x_p),$$

according as the covariates are non-random, or random.

Linear Model

In linear model, we assume

$$E(Y) = g(x_1, \dots, x_p) = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p,$$

where $\beta_0, \beta_1, \dots, \beta_p$ are unknown constants or parameters, to be estimated using the data.

Here linear means, linear in parameters. A function, which is not linear in $\{x_1, \dots, x_p\}$ can be a linear regression model. For example,

$$E(Y) = \beta_0 + \beta_1 \sin x_1 + \beta_2 e^{x_2}$$

is also a linear model.

Note: In simple linear model literature, $\text{Var}(Y) = \sigma^2$, which is free of $x_1, \dots, x_p$.

What can be said about joint distribution of $Y_1, \dots, Y_n$?

Random vectors. Let

$$Z_{\sim} = \begin{bmatrix} Z_1 \\ \vdots \\ Z_n \end{bmatrix}$$

be a random vector, i.e., $Z_1, \dots, Z_n$ are random variables and $Z_{\sim}$ is a stacked version of all of these random variables.

$$\begin{aligned}
f_{\tilde{Z}}(z) &= f_{Z_1, \dots, Z_n}(z_1, \dots, z_n) \\
&= \begin{cases} P(Z_1 = z_1, \dots, Z_n = z_n) & \text{for discrete random variables,} \\ \text{joint pdf of } \{Z_1, \dots, Z_p\} \text{ evaluated at } \{z_1, \dots, z_p\}. & \end{cases}
\end{aligned}$$

Expectation and Variance.

$$E(\tilde{Z}) = \begin{bmatrix} E(Z_1) \\ \vdots \\ E(Z_n) \end{bmatrix} = \underset{\sim Z}{\mu}$$

Let $\tilde{Z} \in \mathbb{R}^p$ and $\tilde{W} \in \mathbb{R}^q$ be two random vectors, then

$$\begin{aligned}
\text{Cov}(\tilde{Z}, \tilde{W}) &= ((\text{Cov}(Z_j, W_k)))^{p \times q}, \quad j = 1, \dots, p; \quad k = 1, \dots, q, \\
\text{Var}(\tilde{Z}) &= E \left[(\tilde{Z} - \underset{\sim Z}{\mu})(\tilde{Z} - \underset{\sim Z}{\mu})^\top \right] = ((\text{Cov}(Z_i, Z_j))) = \text{Cov}(\tilde{Z}, \tilde{Z}) = \text{Var}(\tilde{Z})
\end{aligned}$$

Note:

1. $E(\tilde{Z})$ is a non-random vectors, and $\text{Var}(\tilde{Z})$ is a non-random $p \times p$ matrix.
2. $\text{Var}(\tilde{Z})$ is a positive semi-definite matrix.

Some exercises:

Show the following:

1. $\text{Var}(\tilde{Z} - \beta) = \text{Var}(\tilde{Z})$.
2. $\text{Cov}(\underset{\sim}{a}^\top \tilde{Z}, \underset{\sim}{b}^\top \tilde{W}) = \underset{\sim}{a}^\top \text{Cov}(\tilde{Z}, \tilde{W}) \underset{\sim}{b}$.
3. $\text{Cov}(\underset{\sim}{X}_1 + \underset{\sim}{X}_2, \underset{\sim}{Z}_1 + \underset{\sim}{Z}_2) = \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_1, \underset{\sim}{Z}_2) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_1) + \text{Cov}(\underset{\sim}{X}_2, \underset{\sim}{Z}_2)$.
4. $\text{Var}(\tilde{X} \pm \tilde{Z}) = \text{Var}(\tilde{X}) + \text{Var}(\tilde{Z}) \pm \text{Cov}(\tilde{X}, \tilde{Z}) \pm \text{Cov}(\tilde{Z}, \tilde{X})$.
5. $E(\tilde{Z}^\top A \tilde{Z}) = \underset{\sim}{\mu}^\top A \underset{\sim}{\mu} + \text{tr}(A \Sigma)$, where $E(\tilde{Z}) = \underset{\sim}{\mu}$, $\text{Var}(\tilde{Z}) = \Sigma$.

Back to the linear regression model:

If $Y_i \sim F(\mu_i, \sigma^2)$ some distribution with mean μ_i and variance σ^2 , then $\mu_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p}$. Further, it is *assumed* that $Y_1, \dots, Y_n$ are independently distributed. Therefore,

$$E(\underset{\sim}{Y}_n) = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix} = X_n \underset{\sim}{\beta} \quad \text{and} \quad \text{Var}(\underset{\sim}{Y}_n) = \sigma^2 I_n,$$

where $\underset{\sim}{\beta} = (\beta_0, \dots, \beta_p)^\top$ is the vector of regression parameters, and I_n is the n -dimensional identity matrix.

Alternative representation. One can rephrase the above model in the following way:

$$Y_i = \beta_0 + \beta_1 x_{i,1} + \dots + \beta_p x_{i,p} + \varepsilon_i,$$

where $E(\varepsilon_i) = 0$ and $\text{Var}(\varepsilon_i) = \sigma^2$ for all $i = 1, \dots, n$.

1.2 Vector Differentiation

1.2.1 Gradient and Hessian

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$.

Example: i. $k = 2$, $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = x + y$. ii. For general k , let $\underline{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_k \end{bmatrix}$, $g(\underline{x}) = x_1 + \dots + x_k \in \mathbb{R}$.

- **Gradient:**

$$\nabla g = \frac{\partial g}{\partial \underline{x}} = \begin{bmatrix} \partial g / \partial x_1 \\ \vdots \\ \partial g / \partial x_k \end{bmatrix}^{k \times 1}$$

For $g(\underline{x}) = x_1 + \dots + x_k$,

$$\nabla g = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = \mathbf{1}$$

- **Second derivative / Hessian:** The matrix of second derivatives is called the Hessian matrix,

and is defined as

$$H(\underline{x}) = \left(\left(\frac{\partial^2 g}{\partial x_i \partial x_j} \right) \right) = \frac{\partial^2 g}{\partial \underline{x} \partial \underline{x}^\top} = \begin{bmatrix} \frac{\partial^2 g}{\partial x_1^2} & \cdots & \frac{\partial^2 g}{\partial x_1 \partial x_k} \\ \frac{\partial^2 g}{\partial x_2 \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_2 \partial x_k} \\ \vdots & \vdots & \vdots \\ \frac{\partial^2 g}{\partial x_k \partial x_1} & \cdots & \frac{\partial^2 g}{\partial x_k^2} \end{bmatrix}^{k \times k}.$$

• For vector-valued $g : \mathbb{R}^k \rightarrow \mathbb{R}^k$, $g(\underline{x}) = \begin{bmatrix} g_1(\underline{x}) \\ \vdots \\ g_k(\underline{x}) \end{bmatrix}$

$$\nabla g = \left[\left(\frac{\partial g_1}{\partial \underline{x}} \right) \cdots \left(\frac{\partial g_k}{\partial \underline{x}} \right) \right]^{k \times k} = \begin{bmatrix} \frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial g_k}{\partial x_1} \\ \vdots & \ddots & \vdots \\ \frac{\partial g_1}{\partial x_k} & \cdots & \frac{\partial g_k}{\partial x_k} \end{bmatrix}$$

Example: $g\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$, then

$$\nabla g = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}$$

1.2.2 Use 1: Jacobian of Transformation

Recall the transformation technique of obtaining the distribution of a one-to-one function of X , when the distribution of X is known. Now, we will extend the transformation technique to multivariate change of variables problems.

Univariate: $X \sim f_X(x)$ (pdf), $Z = h(X)$, then

$$f_Z(z) = f_X(x) \Big|_{x \rightarrow z} |J_{x \rightarrow z}| = f_X(h^{-1}(z)) \left| \frac{\partial x}{\partial z} \right|.$$

Multivariate: $\underline{X} \sim f_{\underline{X}}$ (joint pdf), $\underline{Z} = h(\underline{X})$ where $h : \mathbb{R}^p \rightarrow \mathbb{R}^p$ is one-to-one function.

The pdf of $\underline{Z}$ at $\underline{z}$ is

$$f_{\underline{Z}}(\underline{z}) = f_{\underline{X}}(\underline{x}) \Big|_{\underline{x} \rightarrow \underline{z}} |J_{\underline{x} \rightarrow \underline{z}}| = f_{\underline{X}}(h^{-1}(\underline{z})) \left| \det \left(\frac{\partial \underline{x}}{\partial \underline{z}} \right) \right|.$$

Example 1. Let $X_i \stackrel{iid}{\sim} N(0, \sigma^2)$, $i = 1, 2$. Let

$$Z = \begin{pmatrix} X_1 + X_2 \\ X_1 - X_2 \end{pmatrix}, \quad Z = AX, \quad A = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}.$$

$$f_{\underline{Z}}(\underline{x}) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2}(x_1^2 + x_2^2)\right\}$$

We have

$$\underline{X} = \begin{pmatrix} (z_1 + z_2)/2 \\ (z_1 - z_2)/2 \end{pmatrix} = B\underline{Z},$$

$$J_{x \rightarrow z} = \left| \det \left(\frac{\partial \underline{x}}{\partial \underline{z}} \right) \right| = \left| \det \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & -1/2 \end{pmatrix} \right| = \frac{1}{2}.$$

Thus,

$$f_{\underline{Z}}(\underline{z}) = \frac{1}{2\pi\sigma^2} \exp\left\{-\frac{1}{2\sigma^2} \left(\frac{(z_1 + z_2)^2}{4} + \frac{(z_1 - z_2)^2}{4} \right)\right\} \times \frac{1}{2}, \quad z_1 \in \mathbb{R}, z_2 \in \mathbb{R}, \sigma^2 > 0.$$

Identify the distribution of $\underline{Z}$.

1.2.3 Use 2: Finding maxima/minima of a multivariate function

Let $g : \mathbb{R}^k \rightarrow \mathbb{R}$ function, i.e., $g(\underline{z}) = g(z_1, \dots, z_k)$. We are interested in finding the maxima/minima of g w.r.t. $\underline{z}$.

Let g is maximized (minimized) w.r.t. $\underline{z}$ at $\underline{z}_{\sim_0}$, where $\underline{z}_{\sim_0}$ is on ‘interior point’ of the domain. The following steps may lead to the identification $\underline{z}_{\sim_0}$:

1. Differentiate g w.r.t. $\underline{z}$, and equate that to 0, $\frac{\partial g}{\partial \underline{z}} = 0$.
2. Find the points for which this equation is satisfied. Let $\underline{x}_{\sim_1}, \dots, \underline{x}_{\sim_m}$ be the points. These are called stationary points, or critical points.

[Note:] A critical point can be maxima, minima or a saddle point.

3. Next evaluate the Hessian $\frac{\partial^2 g}{\partial \underline{z} \partial \underline{z}^\top} \bigg|_{\underline{x}_{\sim_j}} = H(\underline{x}_{\sim_j})$, $j = 1, \dots, k$.

If $\underline{x}_0$ is a maxima (minima) then $H(\underline{x}_0)$ will be negative (positive) definite.

Example 2. $g : \mathbb{R}^k \rightarrow \mathbb{R}$,

$$g(\underline{x}) = \sum_{j=1}^k (x_j^4 - 4x_j^2).$$

i)

$$\nabla g = \begin{bmatrix} 4x_1^3 - 8x_1 \\ \vdots \\ 4x_k^3 - 8x_k \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}.$$

Solutions to the first order equations: $x_j \in \{0, \sqrt{2}, -\sqrt{2}\}$. So total 3^k critical points:

$$\underline{x}_1^* = \underline{0}, \quad \underline{x}_2^* = \sqrt{2}\underline{1}, \quad \underline{x}_3^* = \begin{pmatrix} \sqrt{2} \\ \vdots \\ -\sqrt{2} \end{pmatrix}, \quad \dots, \quad x_m^* = -\sqrt{2}\underline{1}, \quad x_{m+1}^* = \begin{pmatrix} 0 \\ \sqrt{2} \\ \vdots \\ \sqrt{2} \end{pmatrix}, \quad \dots$$

ii)

$$H(\underline{x}) = \text{Diag}(12x_1^2 - 8, \dots, 12x_k^2 - 8)$$

Therefore, $H(\mathbf{0}) = -8I$: negative definite, which implies that $\mathbf{0}$ is a local maxima.

For all combinations with $x_j = \pm\sqrt{2} \forall j$, $H = 16I$: positive definite, and therefore, all these points are local minimas.

Note: Points with mix of 0 and $\pm\sqrt{2}$ give indefinite Hessian.

Some exercises:

1. $g(\underline{x}) = \|\underline{y} - A\underline{x}\|^2$ where A is a p.d. invertible matrix.
2. $g(\underline{x}) = \|\underline{x}\|^2 + \exp\{\underline{a}^T \underline{x}\}$

1.3 Multivariate Normal Distribution

Multivariate normal distribution plays a key role in multivariate analysis. To model a multivariate data, multivariate normal is most commonly used. The motivations behind this predominance can be stated below.

- [1] As already stated, providing a multivariate generalization to any univariate distribution may not be easy. The generalization may not be unique. For normal distribution easy extension is available.
- [2] The p -mvt. normal distribution is completely specified by the first 2 moments. The interpretable parameters, and availability of efficient estimators of the parameters, make multivariate normal more popular.
- [3] The fact that for jointly normally distributed variables, uncorrelatedness $\Leftrightarrow$ independence, makes verification of independence far less complicated than for normal distribution.

Defn (Multivariate Normal)

A p -variate RV $\underline{X}$ is said to follow a multivariate normal distribution with parameters $\underline{\mu}$ and $\Sigma \geq 0$ iff

$$\underline{a}^T \underline{X} \sim \mathcal{N}(\underline{a}^T \underline{\mu}, \underline{a}^T \Sigma \underline{a}) \text{ for all } \underline{a} \in \mathbb{R}^p$$

Here $\underline{\mu} \in \mathbb{R}^p$, and Σ is a $p \times p$ symmetric PSD matrix.

We write $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$ if $\underline{x}$ follows a multivariate normal with parameters $\underline{\mu}$ and Σ .

Note: $X_i \sim N(\mu_i, \sigma_{ii})$ for all $i = 1, \dots, p$ does not imply $\underline{X}$ is normal. Let $\underline{e}_i$ be the vector with all zero elements except the i -th one, which is one. If $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$ where $\underline{\mu} = (\mu_1, \dots, \mu_p)^T$ and $\Sigma = ((\sigma_{ij}))$, then

$$\underline{e}_i^T \underline{X} = X_i \sim N(\mu_i, \sigma_{ii}) \quad \forall i.$$

However, the following example shows a case where the marginals are normal but the joint is not.

Let (X, Y) is a bivariate random variable with joint pdf

$$f(x, y) = \frac{1}{2\pi} \exp\left\{-\frac{x^2 + y^2}{2}\right\} \left\{1 + xy \exp\left\{-\frac{x^2 + y^2}{2}\right\}\right\}$$

Properties :**[1] Characteristic function:**

Recall that the characteristic function (ch.fn.) of univariate normal with parameters η and σ^2 is $\exp(i\eta t - \frac{1}{2}t^2\sigma^2)$, at the point t . Thus,

$$\phi_{\underline{X}}(\underline{t}) = E[\exp\{i\underline{t}^T \underline{X}\}] = E[\exp\{iY\} \mid Y \sim \mathcal{N}(\underline{t}'\underline{\mu}, \underline{t}'\Sigma\underline{t})] = \exp\left(i\underline{t}'\underline{\mu} - \frac{1}{2}\underline{t}'\Sigma\underline{t}\right).$$

[2] Linear transformation:

Let $\underline{Y} = A\underline{X} + \underline{b}$, where A is a $q \times p$ matrix, and $\underline{b} \in \mathbb{R}^q$. Then $\underline{y} \sim \mathcal{N}_q(A\underline{\mu} + \underline{b}, A\Sigma A')$

Proof: The ch fn. of $\underline{Y}$ is

$$\begin{aligned} \phi_{\underline{Y}}(\underline{t}) &= E[\exp\{i\underline{t}'(A\underline{X} + \underline{b})\}] \\ &= \exp\{i\underline{t}'\underline{b}\} E[\exp\{i\underline{t}'A\underline{X}\}] \\ &= \exp\{i\underline{t}'\underline{b}\} E[\exp\{i\underline{s}'\underline{X}\}] \quad \text{where } \underline{s} = A^T \underline{t} \\ &= \exp\{i\underline{t}'\underline{b}\} \exp\left\{i\underline{s}'\underline{\mu} - \frac{1}{2}\underline{s}'\Sigma\underline{s}\right\} \\ &= \exp\left\{i\underline{t}'(A\underline{\mu} + \underline{b}) - \frac{1}{2}\underline{t}'(A\Sigma A')\underline{t}\right\}. \end{aligned}$$

By uniqueness property of characteristic function $\underline{Y} \sim \mathcal{N}_q(A\underline{\mu} + \underline{b}, A\Sigma A^\top)$. ■

Example: $\underline{X} \sim N(\underline{\mu}, \Sigma)$; $\underline{Y} = A\underline{X} + \underline{b}$.

$$\underline{Z} = \begin{bmatrix} \underline{Y} \\ \underline{X} \end{bmatrix} \quad \underline{Z} = C\underline{X} + \underline{d} \quad C = \begin{bmatrix} A \\ I_p \end{bmatrix}_{(q+p) \times p} \quad \underline{d} = \begin{bmatrix} \underline{b} \\ 0 \end{bmatrix}$$

$$C\Sigma C^T = \begin{bmatrix} A\Sigma \\ \Sigma \end{bmatrix} \begin{bmatrix} A^T & I_p \end{bmatrix} \quad \underline{Z} \sim N_{q+p} \left(\begin{bmatrix} A\underline{\mu} + \underline{b} \\ \underline{\mu} \end{bmatrix}, \begin{bmatrix} A\Sigma A^\top & A\Sigma \\ \Sigma A^\top & \Sigma \end{bmatrix} \right)$$

[3] Expectation and Variance. $E(\underline{X}) = \underline{\mu}$ and $\text{Var}(\underline{X}) = \Sigma$.

Proof: By defn. $\underline{e}_j^\top \underline{X} = X_j \sim \mathcal{N}(\mu_j, \sigma_{jj})$ and

$$(\underline{e}_j + \underline{e}_k)^\top \underline{X} = (X_j + X_k) \sim \mathcal{N}(\mu_j + \mu_k, \sigma_{jj} + \sigma_{kk} + 2\sigma_{jk})$$

From the above two facts, it is clear that—

$$E(X_j) = \mu_j \quad \text{and} \quad \text{Var}(X_j) = \sigma_{jj} \quad \forall j = 1, \dots, p$$

Further,

$$\text{Var}(X_j + X_k) = \text{Var}(X_j) + \text{Var}(X_k) + 2\text{Cov}(X_j, X_k) = \sigma_{jj} + \sigma_{kk} + 2\sigma_{jk}$$

Thus, $\text{Cov}(X_j, X_k) = \sigma_{jk}$. ■

[4] Standardization. Let $\underline{X} \sim \mathcal{N}(\underline{\mu}, \Sigma)$, and $\Sigma = BB^T$ for some matrix B . $\Leftrightarrow \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$, where $\underline{Z} \sim \mathcal{N}(\underline{0}, I)$.

Proof: $[\Sigma > 0]$ $\Sigma = U\Lambda U^T$ with $UU^T = U^T U = I_{p \times p}$, and $\Lambda = \text{Diag}(\lambda_1, \dots, \lambda_p)$ $\lambda_j > 0 \forall j$ as $\Sigma > 0$.

Define $\Lambda^{1/2} = \text{Diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p})$

The choices of B can be $B = U\Lambda^{1/2}$ (non-symmetric) or $B = U\Lambda^{1/2}U^\top$ (symmetric).

In both the cases, B^{-1} exists. For symmetric B , $B^{-1} = U\Lambda^{-1/2}U^T$ where $\Lambda^{-1/2} = \text{Diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_p})$, while for the non-symmetric B , $B^{-1} = \Lambda^{-1/2}U^T$.

($\Leftarrow$ part) $\underline{Z} \sim N_p(0, I)$

$$B\underline{Z} + \underline{\mu} \sim N_p(\underline{\mu}, BB^T = \Sigma) \quad \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$$

($\Rightarrow$ part) $\underline{X} \sim N_p(\underline{\mu}, \Sigma)$

$$\underline{Z} = B^{-1}(\underline{X} - \underline{\mu}) \sim N_p(0, I) \implies \underline{X} \stackrel{D}{=} B\underline{Z} + \underline{\mu}$$

[5] **Independence.** Let $\underline{X} = \begin{bmatrix} X_1 \\ \vdots \\ X_p \end{bmatrix} \sim \mathcal{N}_p(\underline{\mu}, \Sigma)$, $p_1 + p_2 = p$, then $\underline{X}_1 \perp \underline{X}_2 \Leftrightarrow \Sigma_{12} = \underline{0}$, where

$$\Sigma = \begin{bmatrix} \Sigma_{11}^{p_1 \times p_1} & \Sigma_{12}^{p_1 \times p_2} \\ \Sigma_{21}^{p_2 \times p_1} & \Sigma_{22}^{p_2 \times p_2} \end{bmatrix}, \quad \text{and} \quad \Sigma_{21} = \Sigma'_{12}.$$

Proof: (4 parts) We can easily verify this by property [3] of ch-fn.,

$$\text{as } \phi_{\underline{X}}(\underline{t}) = \exp \left\{ i \underline{t}' \underline{\mu} - \frac{1}{2} \underline{t}' \Sigma \underline{t} \right\} = \exp \left\{ i \underline{t}'_1 \underline{\mu} - \frac{1}{2} \underline{t}'_1 \Sigma_{11} \underline{t}_1 \right\} \times \exp \left\{ i \underline{t}'_2 \underline{\mu} - \frac{1}{2} \underline{t}'_2 \Sigma_{22} \underline{t}_2 \right\}.$$

As $\underline{X}_i \sim \mathcal{N}(\underline{\mu}_i, \Sigma_{ii})$ marginally, $i = 1, 2$, this implies

$$\phi_{\underline{X}}(\underline{t}) = \phi_{\underline{X}_1}(\underline{t}_1) \times \phi_{\underline{X}_2}(\underline{t}_2) \Rightarrow \underline{X}_1 \perp \underline{X}_2.$$

($\Rightarrow$ part) If $\underline{X}_1 \perp \underline{X}_2$, then $\text{Cov}(X_{ij}, X_{2k}) = 0$ for each $j = 1, \dots, p_1$, $k = 1, \dots, p_2$. Thus, $\Sigma_{12} = 0$.

[6] **Density.**

Case I: ($\Sigma > 0$)

By property [4], in this case $\underline{Z} = \Sigma^{-1/2}(\underline{X} - \underline{\mu}) \sim \mathcal{N}(\underline{0}, I_p)$.

By property [5], the p -components of $\underline{Z}$ are independent and identically distributed (IID) following $\mathcal{N}(0, I)$ distribution. Thus,

$$f_{\underline{Z}}(\underline{z}) = (2\pi)^{-p/2} \exp \left\{ -\frac{1}{2} \underline{z}^T \underline{z} \right\}.$$

As the transformation $\underline{Z} \rightarrow \underline{X} = \underline{\mu} + \Sigma^{1/2} \underline{Z}$ is one-one, the density of $\underline{X}$ can be obtained as:

$$\begin{aligned} f_{\underline{X}}(\underline{x}) &= f_{\underline{Z}}(\underline{z}) \Big|_{\underline{Z}=\Sigma^{-1/2}(\underline{X}-\underline{\mu})} \times J_{\underline{Z} \rightarrow \underline{X}} \\ &= |\Sigma|^{-1/2} (2\pi)^{-p/2} \exp \left\{ -\frac{1}{2} (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) \right\}. \end{aligned}$$

as

$$J_{\underline{Z} \rightarrow \underline{X}} = \left| \det \left(\frac{\partial \underline{Z}}{\partial \underline{X}} \right) \right| = \|B^{-1}\| = \frac{1}{\|B\|} = \frac{1}{|B|}, \quad \text{and} \quad |\Sigma| = |BB^T| = |B|^2 \quad \Rightarrow \quad \frac{1}{|B|} = \frac{1}{|\Sigma|^{1/2}}.$$

Further

$$\sum_{i=1}^p z_i^2 = \underline{z}^T \underline{z} = (\underline{x} - \underline{\mu})^T (B^{-1})^T B^{-1} (\underline{x} - \underline{\mu}) = (\underline{x} - \underline{\mu})^T (BB^T)^{-1} (\underline{x} - \underline{\mu}) = (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}).$$

Equi-probability contour plots.

The contour plots or contours are the collection of points of equal probability density. Thus, the set

of points with density equal to q is:

$$S_q = \left\{ \underline{x} : f_{\underline{x}}(\underline{x}) = q \right\} = \left\{ \underline{x} : (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = c_q \right\},$$

where

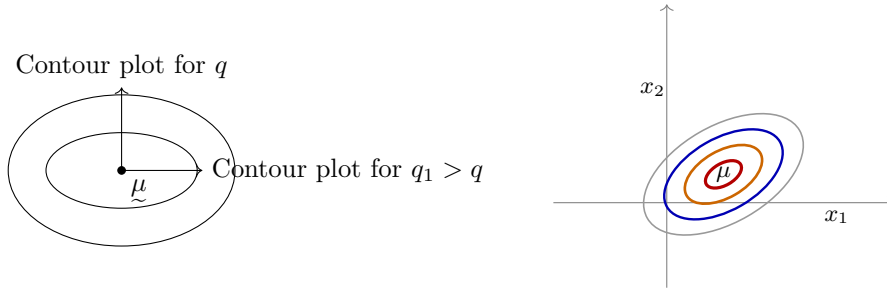
$$c_q = -2 \log \left\{ |\Sigma|^{1/2} (2\pi)^{p/2} q \right\}.$$

Clearly, S_q contains points on an ellipsoid, characterized by parameters $(\underline{\mu}, \Sigma, c_q)$, centered at $\underline{\mu}$. Also, note that the set of high density region $\rho(q) = \left\{ \underline{x} : f_{\underline{x}}(\underline{x}) \geq q \right\}$ is the interior of the ellipse.

For any $q_1 > q$, $S_{q_1} = \left\{ \underline{x} : (\underline{x} - \underline{\mu})^T \Sigma^{-1} (\underline{x} - \underline{\mu}) = c_{q_1} \right\}$, where $c_{q_1} < c_q$, and therefore the points form a concentric ellipsoid, with smaller lengths of principal axes.

When $\underline{\mu} = \underline{0}$, and $\Sigma = I$. In this case, $S_q = \left\{ \underline{x} : \underline{x}^T \underline{x} = c_q \right\}$, and thus the equi-probability contours are spherical and centered at $\underline{0}$.

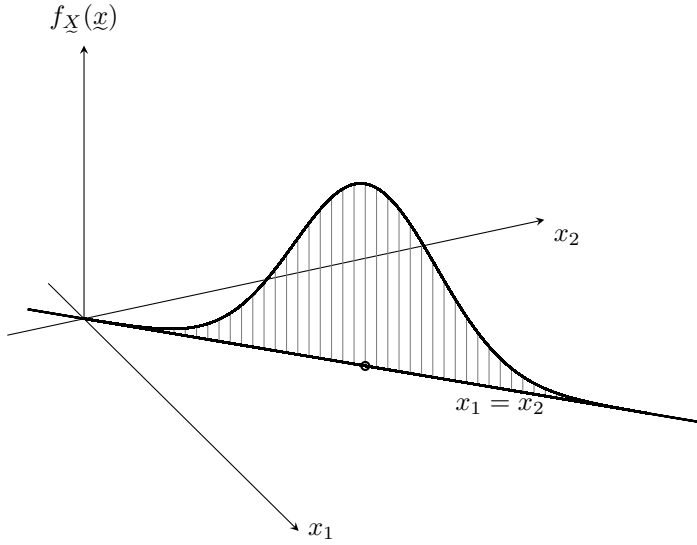
When $\underline{\mu} \neq \underline{0}$, and $\Sigma \neq I$. In this case, S_q is elliptical.



Case II: When Σ is singular, then density exists only in hyperplane which is orthogonal to the null-space of Σ , i.e., in the row-space of Σ .

An example of the case with $\det(\Sigma) = 0$: Let $\underline{X} \sim \mathcal{N} \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \right)$. Here, X_1 and X_2 are perfectly correlated. We will show that here $P(X_1 = X_2) = 1$.

Let $\underline{a} = (1, -1)^T$, then $\underline{a}^T \underline{X} = X_1 - X_2 \sim N(0, 0)$, which implies that $E(X_1 - X_2) = 0$ and $\text{var}(X_1 - X_2) = 0$, which happen only if $P(X_1 = X_2) = 1$. Then density of $\underline{X}$ exists on the hyperplane $x_1 = x_2$, as $(1, -1)\Sigma = (0 \ 0)$.



In general, if Σ is singular, then the density will exist on the line $\Sigma y = 0$ ($y \neq 0$), as

$$y^T X \sim N(y^T \mu, y^T \Sigma y), \quad \text{which implies} \quad P(y^T X = 0) = 1$$

.

[7] **Independence of two linear forms.** Let $\underline{X} \sim N(\mu, \Sigma)$; $\Sigma \geq 0$. For what choices of A and B , $\underline{Y} = A\underline{X} \perp B\underline{X} = \underline{Z}$?

First, we see that $(\underline{Y}, \underline{Z})^\top$ is jointly normally distributed, as

$$\begin{bmatrix} \underline{Y} \\ \underline{Z} \end{bmatrix} = \begin{bmatrix} A \\ B \end{bmatrix} \underline{X} = C\underline{X} \sim N(C\underline{\mu}, C\Sigma C^\top).$$

Therefore, $\underline{Y} \perp \underline{Z}$, if and only if $\text{Cov}(\underline{Y}, \underline{Z}) = 0$.

When is $\text{Cov}(\underline{Y}, \underline{Z}) = 0$? Answer: $A\Sigma B' = 0$. Hence $A\Sigma B' = 0 \iff AX \perp BX$.

[7] (**Marginal and Conditional distributions**)

Marginal: From property (2), it is straightforward to obtain the marginal distributions. Let

$$\underline{X} = \begin{pmatrix} \underline{X}_1 \\ \underline{X}_2 \end{pmatrix},$$

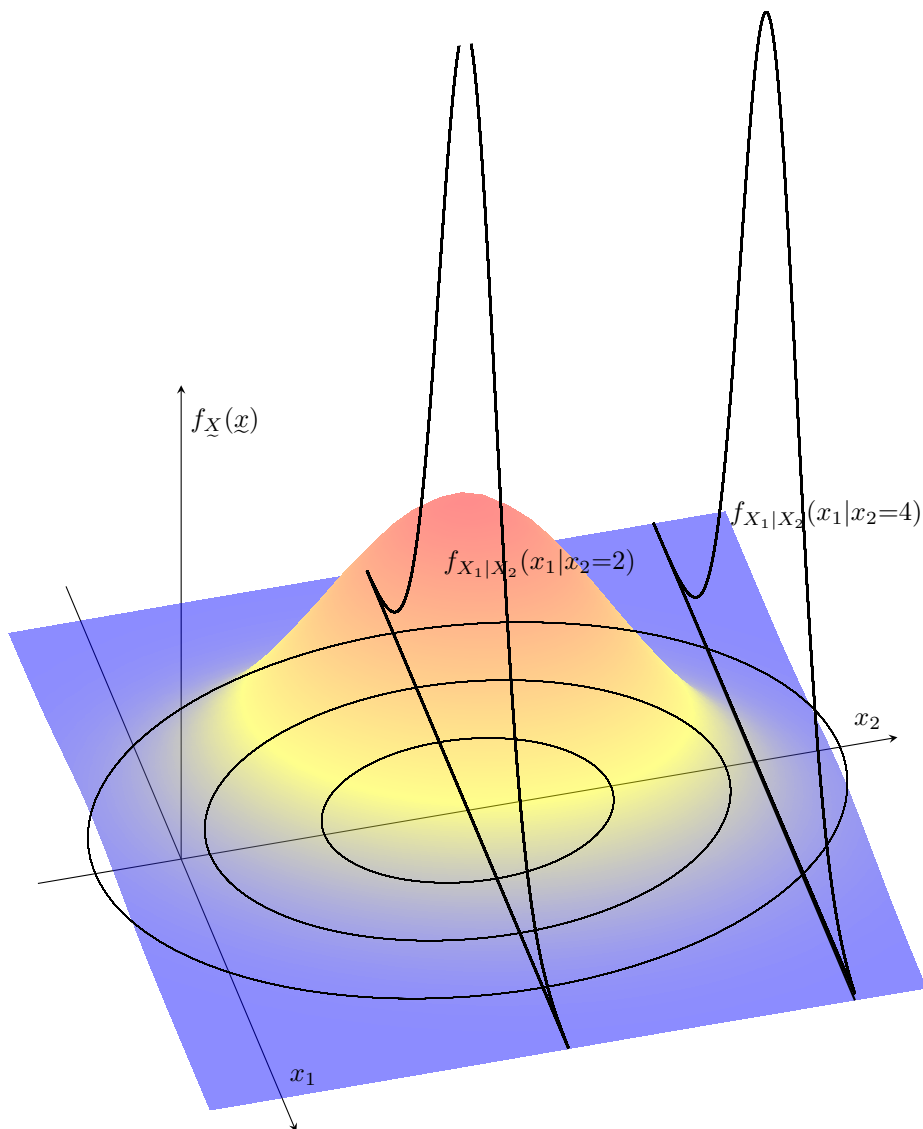
then the marginal of $\underline{X}_2 = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \underline{X}$ is normally distributed with parameters

$$\underline{\mu}_2 \text{ and } \Sigma_{22} = \begin{pmatrix} 0 & I_{p_2} \end{pmatrix} \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{pmatrix} 0 \\ I_{p_2} \end{pmatrix}.$$

Conditional distribution: (when $\Sigma_{22} > 0$)

Question. What is the conditional distribution of $\underline{X}_1$ given $\underline{X}_2 = \underline{x}_2$?

It is the distribution of $\underline{X}_1$ informed by the value of $\underline{X}_2 = \underline{x}_2$.



To obtain this, we follow the steps below:

- i. We will find B such that $\underline{Z} = \underline{X}_1 - B\underline{X}_2 \perp \underline{X}_2$.

As $\underline{Z} \perp \underline{X}_2$, the conditional distribution of $\underline{Z}$ given $\underline{X}_2 = \underline{x}_2$ is the same as the marginal distribution.

- ii. Using this fact and the marginal distribution of $\underline{Z}$, we will find the conditional distribution of $\underline{Z} + B\underline{X}_2 (= \underline{X}_1)$ given $\underline{X}_2 = \underline{x}_2$.

Step i. Define $\underline{Z} := [I_{p_1} \ -B] \begin{bmatrix} \underline{X}_1 \\ \underline{X}_2 \end{bmatrix}$, $\underline{X}_2 = [0 \ I_{p_2}] \begin{bmatrix} \underline{X}_1 \\ \underline{X}_2 \end{bmatrix}$

$$[I_{p_1} \ -B] \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix} \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0 \iff [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} 0 \\ I_{p_2} \end{bmatrix} = 0 \iff \Sigma_{12} - B\Sigma_{22} = 0$$

$$\iff B = \Sigma_{12}\Sigma_{22}^{-1}$$

Therefore, for $\underline{Z} = \underline{X}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{X}_2$, we have $\underline{Z} \perp \underline{X}_2$, and $\underline{Z} \sim N(\underline{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}\underline{\mu}_2, \Sigma_{11.2})$, where

$$\begin{aligned} \Sigma_{11.2} &= [\Sigma_{11} - B\Sigma_{21} \quad \Sigma_{12} - B\Sigma_{22}] \begin{bmatrix} I_{p_1} \\ -B' \end{bmatrix} = \Sigma_{11} - B\Sigma_{21} - \Sigma_{12}B' + B\Sigma_{22}B' \\ &= \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} + \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{22}\Sigma_{22}^{-1}\Sigma_{21} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21}. \end{aligned}$$

Step ii. Given $\underline{X}_2 = \underline{x}_2$, $\underline{Z}$ has same distribution as its marginal. Hence, $\underline{X}_2 = \underline{x}_2$,

$$\underline{X}_1 = \underline{Z} + \Sigma_{12}\Sigma_{22}^{-1}\underline{x}_2 \sim N(\underline{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\underline{x}_2 - \underline{\mu}_2), \Sigma_{11.2})$$

1.4 Distribution of Quadratic Forms

Definition 1 (Idempotent Matrix). A is idempotent if $A^2 = A$.

A is symmetric idempotent $\iff$ the eigenvalues of A are 1 with multiplicity $\rho(A)$ [rank of A] and 0 with multiplicity $p - \rho(A)$.

Property: $\rho(A) = \text{tr}(A)$.

Result 1 (Cochran's Theorem). Let $\underline{Y} \sim N(0, I_p)$ and let A be a symmetric matrix.

If A is idempotent then

$$\underline{Y}^T A \underline{Y} \sim \chi_{(q)}^2 \quad \text{where } q = \rho(A).$$

Proof Consider the spectral decomposition of A as follows:

$$A = U \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} U^T = \begin{bmatrix} U_1 & U_2 \end{bmatrix} \begin{pmatrix} I_q & 0 \\ 0 & 0 \end{pmatrix} \begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} = U_1 U_1^T \quad \text{where } UU^T = U^T U = I$$

$$\begin{bmatrix} U_1^T \\ U_2^T \end{bmatrix} \begin{bmatrix} U_1 & U_2 \end{bmatrix} = \begin{bmatrix} U_1^T U_1 & U_1^T U_2 \\ U_2^T U_1 & U_2^T U_2 \end{bmatrix} = \begin{bmatrix} I_q & 0 \\ 0 & I_{p-q} \end{bmatrix} \Rightarrow \begin{cases} U_1^T U_1 = I, \\ U_1^T U_2 = 0, \quad U_2^T U_2 = I. \end{cases}$$

Now

$$\underline{Y}^T A \underline{Y} = (U_1^T \underline{Y})^T (U_1^T \underline{Y}) = \|\underline{Z}\|^2 \quad \text{where } \underline{Z} = U_1^T \underline{Y} \sim N_q(0, I)$$

Therefore,

$$= \sum_{i=1}^q Z_i^2 \sim \chi_{(q)}^2 \quad \Longleftrightarrow \quad Z_i \stackrel{\text{iid}}{\sim} N(0, 1).$$

Example. Let $\underline{Y} \sim N_n(0, \sigma^2 I)$, $S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$.

To show that $\frac{(n-1)S_y^2}{\sigma^2} \sim \chi_{(n-1)}^2$. Observe that

$$(n-1)S_y^2 = \sum_{i=1}^n Y_i^2 - n\bar{Y}^2, \quad n\bar{Y} = \sum_{i=1}^n Y_i = \underline{Y}^T \underline{1}.$$

Therefore

$$(n-1)S_y^2 = \underline{Y}^T \underline{Y} - \frac{n}{n^2} \underline{Y}^T \underline{1} \underline{1}^T \underline{Y} = \underline{Y}^T \left(I - \frac{1}{n} \underline{1} \underline{1}^T \right) \underline{Y}.$$

Let $H_n = \left(I - \frac{1}{n} \underline{1} \underline{1}^T \right)$, then $H_n^2 = H_n$ (verify), and $\text{tr}(H_n) = \rho(H_n) = q = n-1$. Therefore, by the previous theorem

$$\underline{Z} = \frac{1}{\sigma} \underline{Y} \sim N(\underline{0}, I), \quad \text{and} \quad \frac{(n-1)S_y^2}{\sigma^2} = \underline{Z}^T H_n \underline{Z} = \frac{\underline{Y}^T H_n \underline{Y}}{\sigma^2} \sim \chi_{(n-1)}^2.$$

Result 2. $\underline{Y} \sim N_p(\underline{0}, \Sigma)$, A is symmetric and $A\Sigma$ has eigenvalues 0 or 1.

Then $\underline{Y}^T A \underline{Y} \sim \chi_r^2$ where $r = \rho(A\Sigma)$.

Proof. $\Sigma = BB^T$ and $\underline{Y} \stackrel{D}{=} B\underline{Z}$ where $\underline{Z} \sim N(\underline{0}, I)$.

$$\underline{Y}^T A \underline{Y} \stackrel{D}{=} \underline{Z}^T B^T A B \underline{Z}.$$

Note that the non-zero eigenvalues of $B^T A B$ are the same as that of $AB B^T = A\Sigma$.

So, $B^T A B$ has the eigenvalue 1 with multiplicity $r = \rho(A\Sigma)$, and $(p-r)$ zero-eigenvalues.

This implies that the spectral decomposition of $B^T A B$ will be of the form:

$$B^T A B = U U^T \quad \text{with} \quad U^T U = I_r.$$

Thus,

$$\underline{Y}^T A \underline{Y} = \underline{Z}^T B^T A B \underline{Z} = \underline{Z}^T U U^T \underline{Z} = \|U^T \underline{Z}\|^2.$$

$$\underline{Z} \sim N(\underline{0}, I) \Rightarrow U^T \underline{Z} \sim N(\underline{0}, I_r) \Rightarrow \|U^T \underline{Z}\|^2 \sim \chi_r^2.$$

□

Result 3 (Homework). $Y \sim N_p(\mu, \Sigma)$, $\Sigma > 0$. Then $(Y - \mu)^T \Sigma^{-1} (Y - \mu) \sim \chi_p^2$.

Result 4 (Craig's Theorem). $Y \sim N_p(0, I)$ and A, B are two symmetric and idempotent matrices.

Then $Y^T A Y \perp\!\!\!\perp Y^T B Y$ iff $AB = 0$.

Proof. $Y^T A Y = \|AY\|^2$ and $Y^T B Y = \|BY\|^2$.

Enough to show that $AY \perp\!\!\!\perp BY \Leftrightarrow AB = 0$. [apply Property 6 of normal distribution] □

Example (continued). Recall that $X \sim N_n(\mu 1, \sigma^2 I)$, $\sigma^2 > 0$, and

$$S_y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (n-1)S_y^2 = Y^\top H_n Y. \quad \text{where } H_n = \left(I - \frac{1}{n} 11^\top \right) \text{ is idempotent.}$$

Let $Z = \sigma^{-1}(Y - \mu 1) \sim N(0, I_n)$, then $Z^\top H_n Z \sim \chi_{n-1}^2$, that is, $\sigma^{-2}(Y - \mu)^T H_n (Y - \mu) \sim \chi_{n-1}^2$.

Now,

$$(Y - \mu)^T H_n (Y - \mu 1) = Y^\top H_n Y - 2\mu 1^\top H_n Y + \mu^2 1^\top H_n 1 = Y^\top H_n Y,$$

as $H_n 1 = 0$ then we get the same conclusion

$$\frac{(n-1)S_Y^2}{\sigma^2} \sim \chi_{n-1}^2.$$

Chapter 2

Linear Models (Revisit)

Linear Regression

Recall the linear regression model with response y , and regressors/covariates $\{x_1, \dots, x_p\}$ and n observations as follows:

$$\underset{\sim}{Y} = \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where

$$\underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad \underset{\sim}{X} = \begin{bmatrix} 1 & x_{11} & \dots & x_{1p} \\ \vdots & \vdots & & \vdots \\ 1 & x_{n1} & \dots & x_{np} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}.$$

We assume $\varepsilon_1, \dots, \varepsilon_n$ are independent and identically distributed (i.i.d.), $E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}$ and $\text{var}(\underset{\sim}{\varepsilon}) = \sigma^2 I$.

Parameters in the linear regression model: $(\underset{\sim}{\beta}, \sigma^2)$, where

- $\underset{\sim}{\beta}$: vector of regression coefficients. In particular, β_j measures the marginal increment in y when the covariate x_j is increased by one unit, keeping other covariates fixed at a certain level, $j = 1, \dots, p$. Further, the intercept β_0 is the mean effect of y , when all the covariates are set to 0.
- σ^2 : error variance. This parameter measures the average deviation of y from the explainable part (i.e., the part of y which can be systematically explained by $\{x_1, \dots, x_p\}$).

Difference between the interpretations of the multiple regression model, compared to individual simple regression models. Consider the follow to system of models:

$$Y_i = \beta_0 + \beta_1 x_{1i} + \cdots + \beta_p x_{pi} + \varepsilon_i, \quad i = 1, \dots, n, \quad (2.1)$$

and

$$\begin{cases} Y_i = \gamma_0 + \gamma_1 x_{1i} + \varepsilon_{1i} \\ \vdots \\ Y_i = \gamma_0 + \gamma_p x_{pi} + \varepsilon_{pi} \end{cases} \quad (2.2)$$

In (2.1), the parameters $\{\beta_1, \dots, \beta_p\}$ measures the effect of $\{x_1, \dots, x_p\}$ on y . In (2.2), the parameters $\{\gamma_1, \dots, \gamma_p\}$ measures the effect of $\{x_1, \dots, x_p\}$ on y .

Q. Does the two set of estimates have same interpretation? Also, will the least squares estimates of β_j be same as that of γ_j ?

A. No. Here β_j signifies the effect of x_j keeping other covariates fixed at a certain level, whereas, γ_j signifies the effect of x_j irrespective of the other covariates. For example, suppose $p = 2$, for some commodity $x_1 \equiv \text{demand}$, $x_2 \equiv \text{supply}$ and $y \equiv \text{price}$. Here $\gamma_1 = 2$ implies that if demand increases by one unit then price increases by 2 unit, irrespective of the supply. However, $\beta_1 = 4$ indicates, if demand increases by one unit, but the supply remains fixed at a certain level, then price increase by 4 unit. Observe that, in this example, it is in fact intuitive that $\beta_1 > \gamma_1$.

Simpson's Paradox: This is a statistical phenomenon where the sign of β_j and γ_j reverses. This occurs because the second variable is related to both the first variable and the response, masking the true relationship. For example, suppose

$$y = \begin{cases} 1 & \text{an exam is taken} \\ & \text{in the class,} \\ 0 & \text{otherwise,} \end{cases}, \quad x_1 = \text{class attendance}, \quad x_2 = \begin{cases} 1 & \text{the class activity is included} \\ & \text{in final grading,} \\ 0 & \text{otherwise.} \end{cases},$$

and we have data on all classes in all days during the semester. Usually exams are graded, so marginally, γ_1 will be positive indicating that more students appears in a date of exam. However, if we keep x_2 fixed, then β_1 may be negative, since students tend to avoid exams, if possible the topic is not relevant for final grading.

2.1 Least Squares Method

Here we find an estimate of β by minimizing

$$\varepsilon_{\sim}^T \varepsilon_{\sim} = (Y_{\sim} - X_{\sim} \beta)^T (Y_{\sim} - X_{\sim} \beta) = \sum_{i=1}^n (Y_i - \beta_0 - \cdots - \beta_p x_{pi})^2 = S(\beta),$$

w.r.t. $\underset{\sim}{\beta}$.

To minimize $S(\underset{\sim}{\beta}) : \mathbb{R}^{p+1} \rightarrow \mathbb{R}$ we differentiate $S(\underset{\sim}{\beta})$ w.r.t. $\underset{\sim}{\beta}$.

$$\frac{\partial S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}} = \frac{\partial}{\partial \underset{\sim}{\beta}} \left(\|\underset{\sim}{Y}\|^2 + \underset{\sim}{\beta}^T X^T X \underset{\sim}{\beta} - 2 \underset{\sim}{Y}^T X \underset{\sim}{\beta} \right) = 0 + 2X^T X \underset{\sim}{\beta} - 2X^T \underset{\sim}{Y}.$$

First order condition:

$$\frac{\partial S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta}} = 0 \Rightarrow X^T X \underset{\sim}{\beta} = X^T \underset{\sim}{Y}.$$

$$\underset{\sim}{\hat{\beta}} = (X^T X)^{-1} X^T \underset{\sim}{Y} \quad \text{when } (X^T X)^{-1} \text{ exists.}$$

Second order condition:

$$\frac{\partial^2 S(\underset{\sim}{\beta})}{\partial \underset{\sim}{\beta} \partial \underset{\sim}{\beta}^T} = X^T X.$$

When $(X^T X)^{-1}$ exists, then $X^T X$ is positive definite, $\Rightarrow \underset{\sim}{\hat{\beta}}$ is the unique minima.

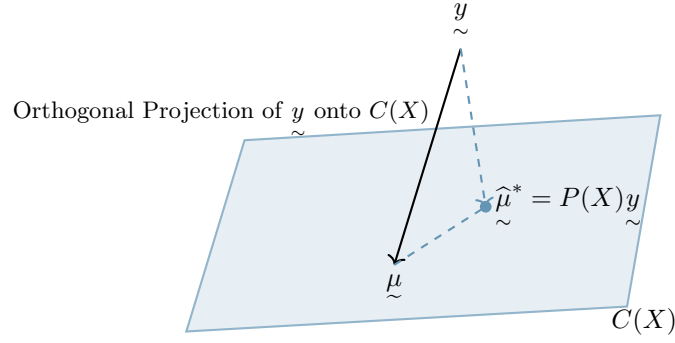
Useful derivatives: Derive the following:

1. $\frac{\partial}{\partial \underset{\sim}{x}} (\underset{\sim}{x}^T A \underset{\sim}{x}) = \frac{\partial}{\partial x_j} \left(\sum_{i,k} a_{ik} x_i x_k \right)_{j=1, \dots, p} = (A + A^T)x = 2Ax \quad (\text{when } A \text{ is symmetric}).$
2. $\frac{\partial \underset{\sim}{a}^T \underset{\sim}{x}}{\partial \underset{\sim}{x}} = \underset{\sim}{x}.$
3. $\frac{\partial A \underset{\sim}{x}}{\partial \underset{\sim}{x}} = A$ when A is symmetric.

Geometric interpretation

The problem is to find $\underset{\sim}{\hat{\beta}}$ s.t. $\|y - X \underset{\sim}{\hat{\beta}}\|^2$ is minimum.

i.e., to find the component $\underset{\sim}{\hat{\mu}}^*$ in the column space of X , $C(X)$, which is closest to $\underset{\sim}{y}$, where $C(X) = \{\underset{\sim}{z} : \underset{\sim}{z} = a_0 \underset{\sim(0)}{1} + a_1 \underset{\sim(1)}{x} + \dots + a_p \underset{\sim(p)}{x} = X \underset{\sim}{a}\}.$



Orthogonal projection of y onto $C(X)$; $C(X)$ is the plane.

We will show that $\hat{\mu}_{\sim}^* = \hat{\mu}_{\sim\text{OLS}}$ is the closest point of y in $C(X)$.

Properties of Orthogonal projection matrix

Let $P(X)$ be the orthogonal projection matrix onto the column space of X . Then the following properties are satisfied:

- $P^2(X) = P(X) \Leftrightarrow P(X)[I - P(X)] = 0$ and also symmetric.
- If $\mu \in C(X)$ then $P(X)\mu = \mu$.
- $\rho(P(X)) = \rho(X) = \text{tr}(P(X))$.
- If $(X'X)^{-1}$ exists then $P(X) = X(X'X)^{-1}X'$.
If $(X'X)$ is singular then $P(X) = X(X'X)^-X'$, where A^- is any g-inverse of A .

Least Square solution (Geometrically)

Let

$$\hat{\mu}_{\sim}^* = \hat{\mu}_{\sim\text{OLS}} = P(X)y \in C(X).$$

We will show that $\hat{\mu}_{\sim}^*$ is the closest point of y in $C(X)$.

Take any $\mu \in C(X)$. Observe that

$$\|y - \mu\|^2 = \|y - \hat{\mu}_{\sim}^* + \hat{\mu}_{\sim}^* - \mu\|^2 = \|y - \hat{\mu}_{\sim}^*\|^2 + \|\hat{\mu}_{\sim}^* - \mu\|^2.$$

The product term vanishes due to the following:

$$\begin{aligned}
 (\underset{\sim}{y} - \underset{\sim}{\hat{\mu}^*})^T (\underset{\sim}{\hat{\mu}^*} - \underset{\sim}{\mu}) &= (\underset{\sim}{y} - P(X)\underset{\sim}{y})^T (P(X)\underset{\sim}{y} - \underset{\sim}{\mu}) \\
 &= \underset{\sim}{y}^T (I - P(X)) (P(X)\underset{\sim}{y} - \underset{\sim}{\mu}) \\
 &= \underset{\sim}{y}^T \underbrace{(I - P(X))P(X)}_0 \underset{\sim}{y} - \underset{\sim}{y}^T \underbrace{(I - P(X))\underset{\sim}{\mu}}_{\underset{\sim}{\mu} - P(X)\underset{\sim}{\mu}} \\
 &= 0 \quad \text{on } \underset{\sim}{\mu} \in C(X).
 \end{aligned}$$

Thus, the distance is minimized if $\underset{\sim}{\hat{\mu}^*} = \underset{\sim}{\hat{\mu}}_{\text{OLS}} = \underset{\sim}{\mu}$. So, the optimum point in $C(X)$ is $\underset{\sim}{\hat{\mu}^*}$.

Q. For which choice of $\underset{\sim}{\beta}$, $P(X)\underset{\sim}{y} = X\underset{\sim}{\beta}$? That is, what is the solution to the following equation

$$\Rightarrow X(X'X)^-X'\underset{\sim}{y} = X\underset{\sim}{\beta} \quad (\star)$$

Case 1: $(X'X)^{-1}$ exists

$(X'X)^{-1}$ exists $\Leftrightarrow X$ has full column rank, $\rho(X) = p + 1$

$\Leftrightarrow$ There exists a matrix Z such that $ZX = I_{p+1}$ (i.e., left inverse of X exists)

So, pre-multiply $(\star)$ by Z ,

$$\underbrace{ZX}_{I_{p+1}}(X'X)^{-1}X'\underset{\sim}{y} = \underbrace{ZX}_{I_{p+1}}\underset{\sim}{\beta} = \underset{\sim}{\hat{\beta}}_{\text{OLS}}.$$

Case II: $X'X$ is singular

Recall the problem of finding solution of the system of equations $A\underset{\sim}{z} = \underset{\sim}{b}$ where $\rho(A) < q$, q being the number of columns of A . In this case, a solution exists if $\underset{\sim}{b} \in C(A)$, and as $\rho(A) < q$, there are infinite solutions.

Here $A = X$ and $\underset{\sim}{z} = \underset{\sim}{\beta}$ and $\underset{\sim}{b} = P(X)\underset{\sim}{y} \in C(X)$. So, a solution exists. Further, as $\rho(X) < p + 1$, the solution is not unique.

One possible solution is

$$\underset{\sim}{\hat{\beta}} = (X'X)^-X'\underset{\sim}{y}.$$

Definition 2 (Residual). The minimized error after inserting the least square solution is called the **residual**, and is denoted by $\underset{\sim}{e} = \underset{\sim}{y} - X\underset{\sim}{\hat{\beta}}_{\text{OLS}} = [I - P(X)]\underset{\sim}{y}$.

The minimized error sum of squares, equivalently, the minimized $\underset{\sim}{\varepsilon}^T \underset{\sim}{\varepsilon}$ is called the **residual sum of squares** (RSS)

$$\text{RSS} = \|\underset{\sim}{y} - X\underset{\sim}{\hat{\beta}}_{\text{OLS}}\|^2 = \underset{\sim}{y}^T (I - P(X))\underset{\sim}{y} \quad (\text{RSS residual Sum of Squares}).$$

Note that, $\underset{\sim}{e}$ and $\underset{\sim}{\varepsilon}$ are NOT the same. While $\underset{\sim}{\varepsilon}$ is an unobservable quantity indicating the part of $\underset{\sim}{y}$ which can not be explained by a linear function of $\{x_1, \dots, x_p\}$, given the data, $\underset{\sim}{e}$ is an observable

quantity indicating the (least-square) minimized estimation of ε .

Remark: When $X'X$ is singular, then the solution to least squares equation (i.e., β in $(*)$) is not unique.

However, $P(X)$ is uniquely defined in this case as well. Therefore, the terms depending on $P(X)$, such as $\hat{\mu}_{\text{OLS}} = P(X)\tilde{y}$, $\varepsilon = (I - P(X))\tilde{y}$ and RSS are also unique even if $X'X$ is singular.

Estimation of σ^2

Result

$\frac{\text{RSS}}{n - r}$, where $\rho(X) = r$, is an unbiased estimator of σ^2 .

Proof. Observe that

$$\begin{aligned} E\left(\frac{\text{RSS}}{n - r}\right) &= E\left(\frac{\tilde{y}^T (I - P(X))\tilde{y}}{n - r}\right) \\ &= \frac{1}{n - r} \left\{ (X\tilde{\beta})^T (I - P(X))(X\tilde{\beta}) + \text{tr}(\sigma^2 I (I - P(X))) \right\} \end{aligned}$$

[As $X\tilde{\beta} \in e(X)$, $(I - P(X))X\tilde{\beta} = 0$ indicating that the first term in the above equation is 0.]
Also, recall from PS 1 that $E(\tilde{x}^T A \tilde{x}) = \mu^T A \mu + \text{tr}(A \Sigma)$ when $E(\tilde{x}) = \mu$, $\text{var}(\tilde{x}) = \Sigma$. Thus,

$$E\left(\frac{\text{RSS}}{n - r}\right) = \frac{1}{n - r} \left\{ 0 + \sigma^2 (\text{tr}(I_n) - \text{tr}(P(X))) \right\} = \sigma^2,$$

since $\text{tr}(P(X)) = \rho(P(X)) = \rho(X) = r$. □

We write $\hat{\sigma}^2 = \frac{\text{RSS}}{n - r}$.

Properties of $\hat{\beta}_{\text{OLS}}$

Result 5. Let X have full column rank, and $\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X'y$. Then

- i) $E_{\beta}(\hat{\beta}_{\text{OLS}}) = \beta$.
- ii) $\text{Var}_{\beta}(\hat{\beta}_{\text{OLS}}) = \sigma^2(X'X)^{-1}$.
- iii) Let $\tilde{\mu}^T = g(\mu)$ is the parameter of interest, where $\mu = X\tilde{\beta}$, and $\hat{\mu}_{\text{OLS}} = X\hat{\beta}_{\text{OLS}}$. Then $\tilde{\mu}_{\text{OLS}}^T$ is the (unique) BLUE (best linear unbiased estimator) of $g(\mu)$
[This result remains unchanged when $X'X$ is singular.]

Proof. i) Trivial.

ii) Observe that

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \text{Var}((X'X)^{-1}X'y) = (X'X)^{-1}X' \text{var}(y) X(X'X)^{-1} = \sigma^2(X'X)^{-1}.$$

iii) Observe that, $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ is a linear unbiased estimator, as

$$\mathcal{C}^T \hat{\mu}_{\text{OLS}} = \mathcal{C}^T X(X'X)^{-1}X'y \quad (\text{linear in } y),$$

and

$$E(\mathcal{C}^T \hat{\mu}_{\text{OLS}}) = \mathcal{C}^T \underbrace{X(X'X)^{-1}X'}_{P(X)} \underbrace{X\beta}_{\mu \in C(X)} = \mathcal{C}^T \mu \quad (\text{unbiased}).$$

It remains to show that $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ has the lowest variance among all other linear unbiased estimators.

Let $\mathcal{Q}^T y$ be another LUE of $\mathcal{C}^T \mu$. As $\mathcal{Q}^T y$ is unbiased, $E(\mathcal{Q}^T y) = \mathcal{Q}^T \mu = \mathcal{C}^T \mu$.

$$\begin{aligned} \Rightarrow (\mathcal{Q} - \mathcal{C})^T \mu &= (\mathcal{Q} - \mathcal{C})^T X\beta = 0 \\ \Rightarrow (\mathcal{Q} - \mathcal{C})^T X &= \mathbf{0}^T \quad (\text{as } \beta \in \mathbb{R}^{p+1} \text{ is any arbitrary vector}) \\ \Rightarrow (\mathcal{Q} - \mathcal{C})^T P(X) &= \mathbf{0}^T \\ \Rightarrow P(X)(\mathcal{Q} - \mathcal{C}) &= \mathbf{0} \quad (*). \end{aligned}$$

Now, consider the difference in variance of $\mathcal{C}^T \hat{\mu}_{\text{OLS}}$ compared to $\mathcal{Q}^T y$ as

$$\begin{aligned} \text{Var}(\mathcal{Q}^T y) - \text{Var}(\mathcal{C}^T \hat{\mu}_{\text{OLS}}) &= (\mathcal{Q}^T \mathcal{Q})\sigma^2 - \mathcal{C}^T \text{Var}(P(X)y)\mathcal{C} \\ &= \sigma^2 [\mathcal{Q}^T \mathcal{Q} - \mathcal{C}^T P(X)\mathcal{C}] \quad ((P(X)\mathcal{C})^T (P(X)\mathcal{C}) = \mathcal{C}^T P(X)\mathcal{C}) \\ &= \sigma^2 [\mathcal{Q}^T \mathcal{Q} - \mathcal{Q}^T P(X)\mathcal{Q}] \quad (P(X)\mathcal{C} = (P(X)\mathcal{Q}) \text{ by } (*)) \\ &= \sigma^2 \mathcal{Q}^T (I - P(X))\mathcal{Q} = \sigma^2 \|(I - P(X))\mathcal{Q}\|^2 \\ &\geq 0. \end{aligned}$$

Equality holds, $\mathcal{Q} = P(X)\mathcal{Q} \stackrel{(*)}{=} P(X)\mathcal{C}$. Then $\mathcal{Q}^T y = \mathcal{C}^T P(X)y = \mathcal{C}^T \hat{\mu}_{\text{OLS}}$. □

2.2 Distribution Theory

Let us now assume

$$\varepsilon \sim N_n(\mathbf{0}, \sigma^2 I) \Leftrightarrow \varepsilon_i \stackrel{\text{iid}}{\sim} N(0, \sigma^2) \Leftrightarrow Y = X\beta + \varepsilon \sim N_n(X\beta, \sigma^2 I).$$

Result

Let $\underline{Y} \sim N(X\underline{\beta}, \sigma^2 I)$, and $\rho(X) = p + 1$. Then

- i. $\hat{\underline{\beta}}_{\text{OLS}} \sim N_{p+1}(\underline{\beta}, \sigma^2(X'X)^{-1}) \quad ((X'X)^{-1} \text{ exists}).$
- ii. $\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2.$

Proof. i. Trivial.

ii. $\text{RSS} = \sigma^{-2} \underline{\varepsilon}^T \underline{\varepsilon} = \sigma^{-2} \underline{y}^T (I - P(X)) \underline{y}$. Note that $\underline{Z} = \sigma^{-1} \underline{Y} - X\underline{\beta} \sim N(\underline{0}, I)$.

Therefore,

$$\begin{aligned} \text{RSS} &= \frac{1}{\sigma^2} (\underline{Y} - X\underline{\beta} + X\underline{\beta})^T (I - P(X)) (\underline{Y} - X\underline{\beta} + X\underline{\beta}) \\ &= \frac{\underline{\varepsilon}^T (I - P(X)) \underline{\varepsilon}}{\sigma^2} + 0 \quad [(I - P(X))X\underline{\beta} = 0, \text{ as } X\underline{\beta} = \mu \in C(X)] \\ &= \underline{Z}^T (I - P(X)) \underline{Z}. \end{aligned}$$

Recall that $\underline{Z} \sim N(\underline{0}, I)$, and A is symmetric, idempotent with $\rho(A) = r, \implies \underline{Z}^T A \underline{Z} \sim \chi_r^2$. As $(I - P(X))$ is symmetric, idempotent with $\rho(I - P(X)) = n - \text{tr}(P(X)) = n - (p + 1)$, the result follows. □

HW: What happens if $\rho(X) = r < p + 1$? Show that, in this case $\frac{\text{RSS}}{\sigma^2} \sim \chi_{n-r}^2$.

2.3 Maximum Likelihood Estimation

Log-likelihood,

$$\ell(\underline{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} (\underline{Y} - X\underline{\beta})^T (\underline{Y} - X\underline{\beta}).$$

Maximizing the likelihood w.r.t. $\underline{\beta}$, is same as minimizing the negative of last part w.r.t. $\underline{\beta}$, which is $\|\underline{Y} - X\underline{\beta}\|^2, \sigma^2 > 0$.

Thus,

$$\ell(\underline{\beta}, \sigma^2) \leq \ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2).$$

Now, we can simply minimize $\ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2)$ w.r.t. σ^2 .

$$\ell(\hat{\underline{\beta}}_{\text{OLS}}, \sigma^2) = h(\sigma^2) = C - \frac{n}{2} \log \sigma^2 - \frac{\text{RSS}}{2\sigma^2}.$$

The first order condition provides

$$\frac{\partial}{\partial \sigma^2} \ell(\hat{\beta}_{\text{OLS}}, \sigma^2) = 0 \quad \Rightarrow \quad \sigma^2 = \frac{\text{RSS}}{n}.$$

Second order conditions provides

$$\begin{aligned} \left. \frac{\partial^2}{\partial (\sigma^2)^2} \ell(\hat{\beta}_{\text{OLS}}, \sigma^2) \right|_{\sigma^2 = n^{-1} \text{RSS}} &= \left. \frac{\partial}{\partial \sigma^2} \left(-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \text{RSS} \right) \right|_{\sigma^2 = n^{-1} \text{RSS}} = \left. \left(\frac{n}{2\sigma^4} - \frac{\text{RSS}}{\sigma^6} \right) \right|_{\sigma^2 = n^{-1} \text{RSS}} \\ &= \left(\frac{n^3}{2 \text{RSS}^2} - \frac{n^3}{\text{RSS}^2} \right) = -\frac{n^3}{2 \text{RSS}^2} < 0. \end{aligned}$$

Therefore, the maximum likelihood estimates of (β, σ^2) are:

$$\hat{\beta}_{\text{ML}} = \hat{\beta}_{\text{OLS}}, \quad \text{and} \quad \hat{\sigma}_{\text{ML}}^2 = \frac{\text{RSS}}{n}.$$

While the MLE of β is unbiased, MLE of σ^2 is a biased estimate.

2.4 Effects of centering and/or scaling the covariates on the OLS/MLE.

A. Centering

We can reparametrize the linear model as follows:

$$\begin{aligned} Y_i &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i \\ &= (\beta_0 + \beta_1 \bar{x}_1 + \cdots + \beta_p \bar{x}_p) + \beta_1 (x_{i1} - \bar{x}_1) + \cdots + \beta_p (x_{ip} - \bar{x}_p) + \varepsilon_i \\ &= \alpha_0 + \beta_1 x_{i1}^c + \cdots + \beta_p x_{ip}^c + \varepsilon_i \end{aligned}$$

Does the OLS/MLS estimators of $\{\beta_1, \dots, \beta_p\}$ remain same if we center the covariates?

Recall that

$$X = \begin{bmatrix} \mathbf{1} & \mathbf{x}_{(1)} & \cdots & \mathbf{x}_{(p)} \end{bmatrix}, \quad X_c = \begin{bmatrix} \mathbf{1} & \mathbf{x}_{(1)}^c & \cdots & \mathbf{x}_{(p)}^c \end{bmatrix} \quad \text{where} \quad x_{(j)}^c = x_{(j)} - \bar{x}_j \mathbf{1}, \quad x_{(j)} = \begin{bmatrix} x_{1j} \\ \vdots \\ x_{nj} \end{bmatrix}.$$

Further,

$$\hat{\beta}_{\text{OLS}}^c = (X_c^\top X_c)^{-1} X_c^\top Y, \quad \hat{\beta}_{\text{OLS}} = (X^\top X)^{-1} X^\top Y.$$

Observe that,

$$X_c = X \begin{bmatrix} 1 & -\bar{x}_1 & \dots & -\bar{x}_p \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{bmatrix} = X \begin{bmatrix} 1 & -(\bar{x}_1 & \dots & \bar{x}_p) \\ 0 & I_p \end{bmatrix} = XA.$$

As $X_c = XA$ the column-spaces of X and X_c are same. To see this, observe that

$$\begin{aligned} \text{if } \underline{y} \in C(X_c), \text{ then } \quad \underline{y} &= X_c \underline{a} = XA \underline{a} = X \underline{b} \Rightarrow C(X_c) \subseteq C(X). \\ \text{Similarly, if } \underline{z} \in C(X), \text{ then } \quad X \underline{a} &= X_c A^{-1} \underline{a} = X_c \underline{d} \Rightarrow C(X) \subseteq C(X_c). \end{aligned}$$

Therefore, the orthogonal projection matrix is the same: $P(X_c) = P(X)$. This also implies that, the estimated $\underline{Y}$, $\hat{\underline{Y}}_{OLS} = P(X)\underline{Y}$, residual $\hat{\underline{e}} = (1 - P(X))\underline{Y}$ and the RSS $\|\hat{\underline{e}}\|^2$ remains same before and after centering.

When X is of full rank, this also implies that

$$P(X_c) = X_c(X_c^\top X_c)^{-1}X_c^\top = X(X^\top X)^{-1}X^\top = P(X).$$

Let Z be the left-inverse of X , i.e., $ZX = I_{p+1}$, then

$$P(X) = P(X_c) \Rightarrow A(X_c^\top X_c)^{-1}X_c^\top \underline{Y} = (X^\top X)^{-1}X^\top \underline{Y}$$

$$\text{Now, pre-multiplying } \underline{e}_k = \underline{e}_k = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} \leftarrow k\text{th in both sides of the above expression, for } k = 2, \dots, p+1,$$

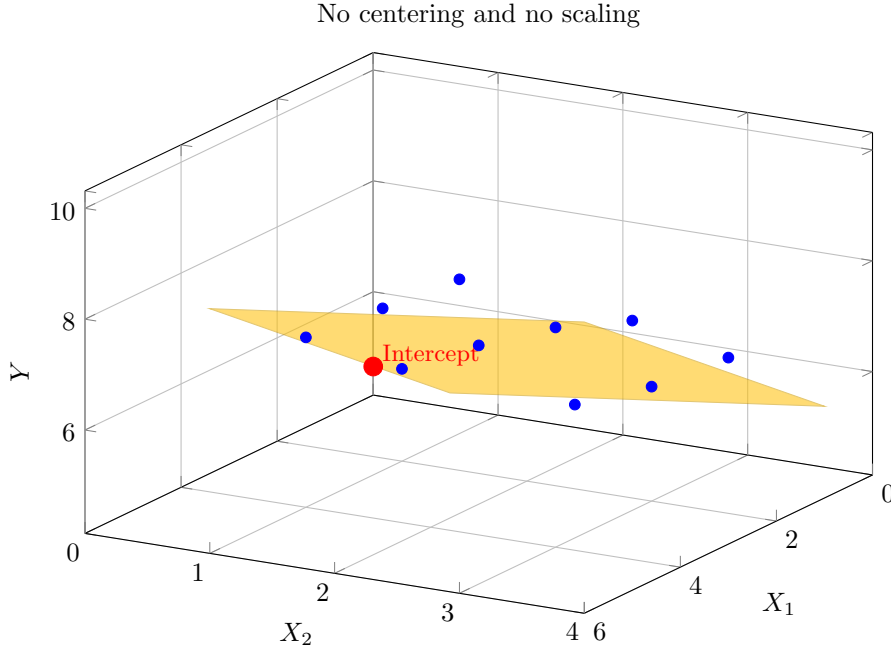
we get:

$$\underline{e}_k^\top A \hat{\underline{\beta}}_{OLS}^c = \hat{\beta}_{k,OLS} \Rightarrow \hat{\beta}_{k,OLS}^c = \hat{\beta}_{k,OLS}, \quad \text{for } k = 2, \dots, p.$$

Also,

$$\begin{aligned} \underline{e}_1^\top A \hat{\underline{\beta}}_{OLS}^c &= \hat{\beta}_{0,OLS} \Rightarrow (1 \quad -\bar{x}_1 \quad \dots \quad -\bar{x}_p) \begin{pmatrix} \hat{\alpha}_0 = \hat{\beta}_{0,OLS}^c \\ \hat{\beta}_{1,OLS}^c \\ \vdots \\ \hat{\beta}_{p,OLS}^c \end{pmatrix} = \hat{\beta}_{0,OLS}. \\ \Rightarrow \hat{\beta}_{0,OLS} &= \hat{\alpha}_0 + \bar{x}_1 \hat{\beta}_{1,OLS}^c + \dots + \bar{x}_p \hat{\beta}_{p,OLS}^c = \hat{\alpha}_0 + \bar{x}_1 \hat{\beta}_{1,OLS} + \dots + \bar{x}_p \hat{\beta}_{p,OLS}, \end{aligned}$$

which is consistent with the reparametrization.



Next, observe that after centering the orthogonal projection matrix of X_c decomposes as a sum of two orthogonal projection matrices.

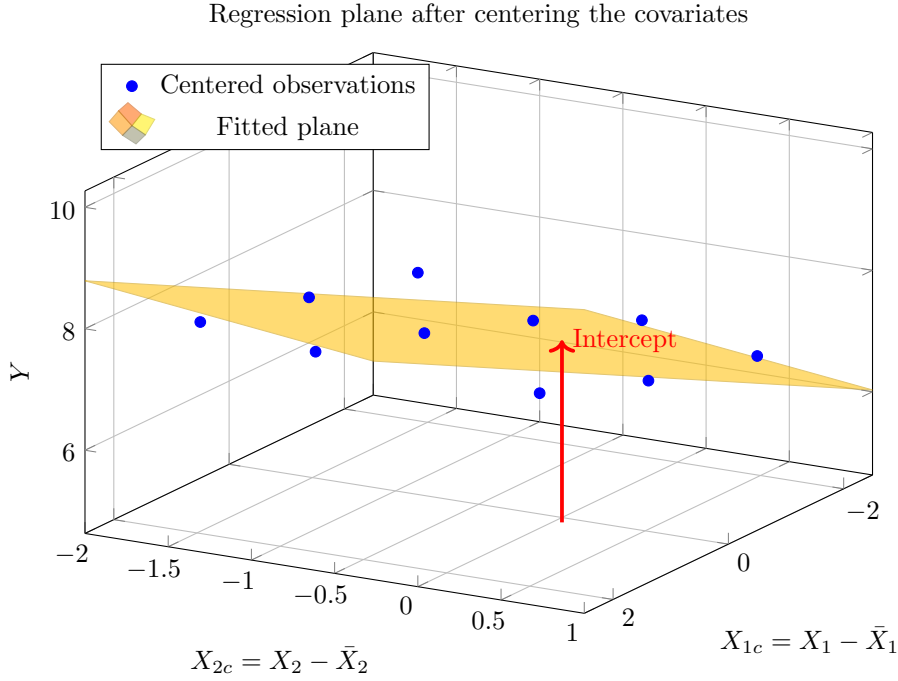
Recall that $X_c = [\underline{1} \ x_{(1)}^c \ \dots \ x_{(p)}^c] = [\underline{1} \ Z_c]$, and $\underline{1}^\top Z_c = \underline{0}^\top$ as $\underline{1}^\top x_{(j)}^c = \underline{1}^\top (x_{(j)} - \bar{x}_j \underline{1}) = \sum_{i=1}^n x_{ij} - n\bar{x}_j = 0$. Therefore,

$$\begin{aligned}
 X_c^\top X_c &= \begin{bmatrix} \underline{1}^\top \\ Z_c^\top \end{bmatrix} \begin{bmatrix} \underline{1} & Z_c \end{bmatrix} = \begin{bmatrix} n & \underline{0}^\top \\ \underline{0} & Z_c^\top Z_c \end{bmatrix}, & \Rightarrow (X_c^\top X_c)^{-1} &= \begin{pmatrix} \frac{1}{n} & \underline{0}^\top \\ \underline{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix}, \\
 \Rightarrow P(X_c) &= X_c (X_c^\top X_c)^{-1} X_c^\top = \begin{bmatrix} \underline{1} & Z_c \end{bmatrix} \begin{pmatrix} 1/n & \underline{0}^\top \\ \underline{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix} \begin{bmatrix} \underline{1}^\top \\ Z_c^\top \end{bmatrix} \\
 &\stackrel{\text{verify}}{=} \frac{1}{n} \underline{1} \underline{1}^\top + P(Z_c) = P(\underline{1}) + P(Z_c).
 \end{aligned}$$

Therefore, the RSS can be expressed as

$$\begin{aligned}
 \text{RSS} &= \underline{Y}^\top (I - P(X_c)) \underline{Y} = \underline{Y}^\top \left(I - \frac{1}{n} \underline{1} \underline{1}^\top - P(Z_c) \right) \underline{Y} = \underline{Y}^\top (I - P(Z_c)) \underline{Y} - n \bar{Y}^2 \\
 &= (\underline{Y} - \bar{Y} \underline{1})^\top (I - P(Z_c)) (\underline{Y} - \bar{Y} \underline{1}),
 \end{aligned}$$

as $\underline{1}^\top P(Z_c) = \underline{0}^\top$. This also explains that if the response $\underline{Y}$ is centered, then the intercept of the fitted model would be *zero*.



B. Scaling

The design matrix corresponding to the scaled covariates can be obtained by post-multiplying original design matrix by a diagonal matrix of scaling factors:

$$X_s = \begin{bmatrix} 1 & x_{(1)}/s_1 & \dots & x_{(p)}/s_p \end{bmatrix} = X \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1/s_1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1/s_p \end{bmatrix} = XD,$$

where

$$s_j = \text{SD}(x_{1j}, \dots, x_{nj}), \quad D = \text{Diag}\left(1, \frac{1}{s_1}, \dots, \frac{1}{s_p}\right).$$

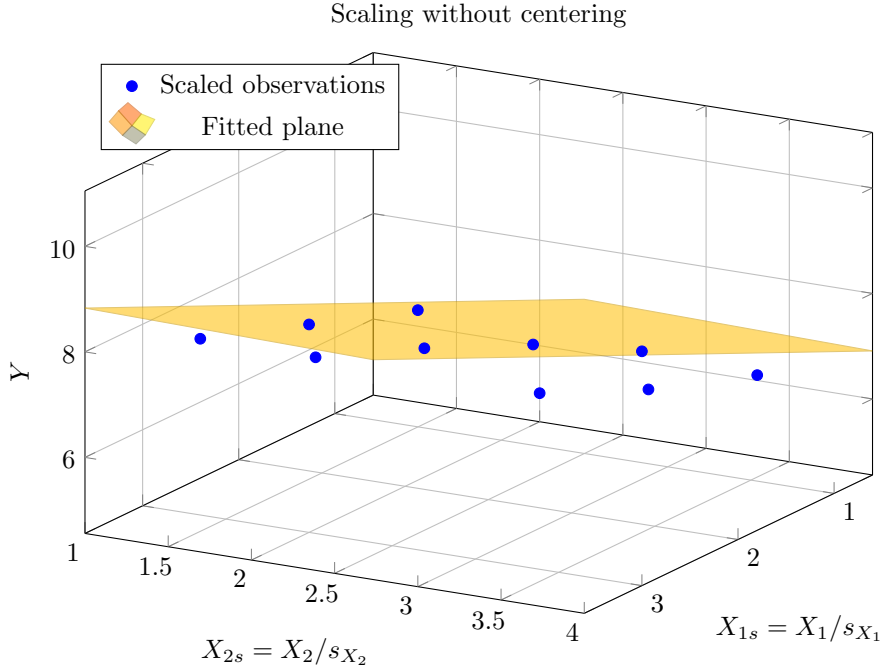
This as before implies that

$$C(X_s) = C(X) \iff P(X_s) = P(X).$$

Therefore, the fitted values, $\hat{Y}$, residuals and RSS remains the same. When $X^\top X$ is invertible, then it can be easily verified that

$$D^{-1} \hat{\beta}_{\text{OLS}} = \hat{\beta}_{\text{OLS}}^s \Rightarrow \hat{\beta}_{j,\text{OLS}}^s = s_j \hat{\beta}_{j,\text{OLS}} \text{ for } j = 1, \dots, p.$$

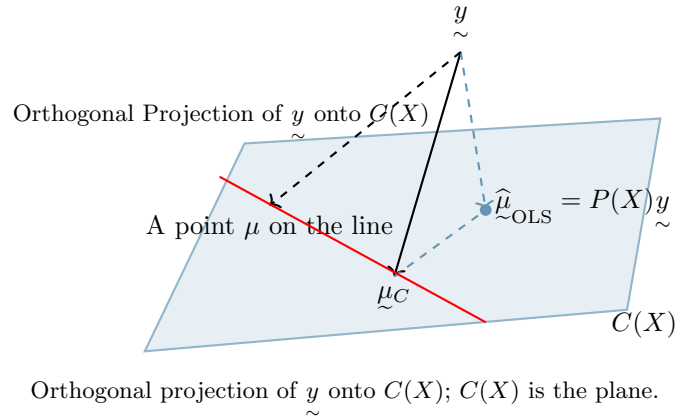
Therefore, the regression coefficients change after scaling.



Observe in the above figure that, the slope of the regression plane has changed after scaling.

2.5 Finding MLE/OLS estimate of β under linear constraint

Consider the problem of estimating (β, σ^2) under a constant $A\beta = c$, where A is a $r \times (p+1)$ matrix of rank r .



Recall that, any point μ in the column space of $C(X)$ can be expressed as $X\beta$ for some β and its distance from $\underline{y}$ can be expressed as the sum of distance of $\underline{y}$ and $\hat{\mu}_{OLS} = P(X)\underline{y} = X\hat{\beta}_{OLS}$ and that of $\hat{\mu}_{OLS}$ and μ , i.e.,

$$\|\underline{Y} - X\underline{\beta}\|^2 = \|\underline{Y} - X\hat{\underline{\beta}}_{\text{OLS}}\|^2 + \|X\hat{\underline{\beta}}_{\text{OLS}} - X\underline{\beta}\|^2. \quad (*)$$

Here the problem is to find the $\mu = X\underline{\beta}$ on the constrained line (i.e., the line satisfying $A\underline{\beta} = \underline{c}$) that is closest to $\underline{Y}$. Therefore, it is enough to find $\underline{\beta}$ satisfying $A\underline{\beta} = \underline{c}$, such that $X\underline{\beta}$ is closest to $X\hat{\underline{\beta}}_{\text{OLS}}$. Both the MLE and the OLS estimators of $\underline{\beta}$ minimizes $\|\underline{Y} - X\underline{\beta}\|^2$. In particular, we consider MLE here, and the OLS solution can be obtained similarly.

MLE Estimates under Constraints

Consider the log likelihood of $(\underline{\beta}, \sigma^2)$ and constraint $A\underline{\beta} = \underline{c}$ where A is an $r \times (p+1)$ matrix of rank r .

$$\ell_n(\underline{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{1}{2\sigma^2} \|\underline{Y} - X\underline{\beta}\|^2.$$

For constraint optimization we use the method of Lagrange's multiplier, where the critical points are obtained by differentiating the Lagrangian function $g(\underline{\beta}, \sigma^2, \underline{\lambda})$ stated below with respect to $\underline{\beta}$, σ^2 and Lagrange's constant $\underline{\lambda} \in \mathbb{R}^r$:

$$g(\underline{\beta}, \sigma^2, \underline{\lambda}) = \ell_n(\underline{\beta}, \sigma^2) - \underline{\lambda}^\top (A\underline{\beta} - \underline{c}),$$

Differentiating we get:

$$\frac{\partial g}{\partial \underline{\beta}} = \frac{2}{\sigma^2} X^\top \underline{Y} - \frac{2}{\sigma^2} X^\top X \underline{\beta} - A^\top \underline{\lambda} = 0, \Rightarrow \underline{\beta} = \hat{\underline{\beta}}_{\text{OLS}} - \frac{\sigma^2}{2} (X^\top X)^{-1} A^\top \underline{\lambda}, \quad (2.3)$$

$$\frac{\partial g}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \|\underline{Y} - X\underline{\beta}\|^2 = 0, \Rightarrow \sigma^2 = \frac{\|\underline{Y} - X\underline{\beta}\|^2}{n} \quad (2.4)$$

$$\frac{\partial g}{\partial \underline{\lambda}} = A\underline{\beta} - \underline{c} = 0, \Rightarrow A\underline{\beta} = \underline{c}. \quad (2.5)$$

From (2.3) and (2.5) we get:

$$A\underline{\beta} = \underline{c} = A\hat{\underline{\beta}}_{\text{OLS}} + \frac{\sigma^2}{2} A(X^\top X)^{-1} A^\top \underline{\lambda}.$$

As A has full row rank, $A(X^\top X)^{-1} A^\top$ is invertible, which implies

$$\frac{\sigma^2}{2} \underline{\lambda} = -[A(X^\top X)^{-1} A^\top]^{-1} (A\hat{\underline{\beta}}_{\text{OLS}} - \underline{c}) \quad (2.6)$$

Inserting (2.6) into (2.3) we get the solution to (2.3) and (2.5) $\hat{\underline{\beta}}_{\text{H}}$ as:

$$\hat{\beta}_{\sim H} = \hat{\beta}_{\sim \text{OLS}} + (X^\top X)^{-1} A^\top [A(X^\top X)^{-1} A^\top]^{-1} (A\hat{\beta}_{\sim \text{OLS}} - \mathfrak{c}) \quad (2.7)$$

Inserting the solution of β in (2.6) we get:

$$\hat{\sigma}_H^2 = \frac{\|Y - X\hat{\beta}_{\sim H}\|^2}{n}. \quad (2.8)$$

next, we need to show that $(\hat{\beta}_{\sim H}, \hat{\sigma}_H^2)$ maximizes the likelihood subject to $A\beta = \mathfrak{c}$.

Consider any β satisfying $A\beta = \mathfrak{c}$. Then we can show

$$\begin{aligned} \|X\beta - X\hat{\beta}_{\sim \text{OLS}}\|^2 &= \|X\beta - X\hat{\beta}_{\sim H} + X\hat{\beta}_{\sim H} - X\hat{\beta}_{\sim \text{OLS}}\|^2 \\ &= \|X\beta - X\hat{\beta}_{\sim H}\|^2 + \|X\hat{\beta}_{\sim H} - X\hat{\beta}_{\sim \text{OLS}}\|^2. \end{aligned}$$

The product term

$$(\hat{\beta}_{\sim \text{OLS}} - \hat{\beta}_{\sim H})^\top X^\top X (\hat{\beta}_{\sim H} - \beta) = (A\hat{\beta}_{\sim \text{OLS}} - \mathfrak{c})^\top [A(X^\top X)^{-1} A^\top]^{-1} A(\hat{\beta}_{\sim H} - \beta) = 0,$$

vanishes as $A(\hat{\beta}_{\sim H} - \beta) = 0$ for any β satisfying the constraint.

Therefore, from (*), we get, for any β satisfying $A\beta = \mathfrak{c}$

$$\begin{aligned} \|Y - X\beta\|^2 &= \|Y - X\hat{\beta}_{\sim \text{OLS}}\|^2 + \|X\beta - X\hat{\beta}_{\sim H}\|^2 + \|X\hat{\beta}_{\sim H} - X\hat{\beta}_{\sim \text{OLS}}\|^2 \\ &\geq \|Y - X\hat{\beta}_{\sim \text{OLS}}\|^2 + \|X\hat{\beta}_{\sim H} - X\hat{\beta}_{\sim \text{OLS}}\|^2, \end{aligned} \quad (2.9)$$

and the equality holds only if $\beta = \hat{\beta}_{\sim \text{OLS}}$. Thus, $\hat{\beta}_{\sim \text{OLS}}$ minimizes the distance between Y and any point $X\beta$ satisfying $A\beta = \mathfrak{c}$.

Therefore, under the constraint $A\beta = \mathfrak{c}$

$$\ell(\beta, \sigma^2) \leq \ell(\hat{\beta}_{\sim H}, \sigma^2).$$

Exercise.

1. Next, show that $\ell(\widehat{\beta}_H, \sigma^2) \leq \ell(\widehat{\beta}_H, \sigma_H^2)$, where σ_H^2 is as defined in (2.8).
2. Show that, instead of maximizing the likelihood function, if one would minimize $\|X\widehat{\beta}_{\text{OLS}} - X\widetilde{\beta}\|^2$ w.r.t. $\widetilde{\beta}$ and subject to the constraint $A\widetilde{\beta} = \underline{c}$, then same solution of $\widetilde{\beta}, \widehat{\beta}_H$ would result in.
3. Consider the normal linear regression equation

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i,$$

where $\varepsilon_i \stackrel{i.i.d.}{\sim} N(0, \sigma^2)$. Then find the MLE of (β, σ^2) under the constraint $\beta_1 = 2\beta_2 + 1$.

2.6 Testing of Hypothesis

Consider the normal linear regression model

$$\mathbf{Y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N(\mathbf{0}, \sigma^2 \mathbf{I}).$$

Want to test for special structure of β , which can be expressed as

$$H_0 : A\beta = \mathbf{c} \quad \text{vs} \quad H_1 : \text{No constraint.}$$

Likelihood Ratio Testing (LRT)

$L^*(\cdot)$ and $\ell^*(\cdot)$ denote the maximized log-likelihood. IN particular,

$L^*(\hat{\beta}_{\sim H_0}, \hat{\sigma}_{H_0}^2)$ and $\ell^*(\hat{\beta}_{\sim H_0}, \hat{\sigma}_{H_0}^2)$ denote maximized likelihood and log-likelihood under H_0 ,
and $L^*(\hat{\beta}, \hat{\sigma}^2)$ and $\ell^*(\hat{\beta}, \hat{\sigma}^2)$ denote maximized likelihood and log-likelihood without any constraint.

The LRT test statistic is given by

$$\Lambda(\mathbf{Y}) = \frac{L^*(\hat{\beta}_{\sim H_0}, \hat{\sigma}_{H_0}^2)}{L^*(\hat{\beta}, \hat{\sigma}^2)}.$$

If $\Lambda(\mathbf{Y})$ is small, or equivalently, $\ell^*(\hat{\beta}_{\sim H_0}, \hat{\sigma}_{H_0}^2) - \ell^*(\hat{\beta}, \hat{\sigma}^2)$ is small, then we reject H_0 .

Under H_0 : The MLEs of β and σ^2 under H_0 are obtained in the previous section as

$$\hat{\beta}_{\sim H_0} = \hat{\beta}_{\text{OLS}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top [\mathbf{A}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}^\top]^{-1} (\mathbf{A} \hat{\beta}_{\text{OLS}} - \mathbf{c}), \quad \text{and} \quad \hat{\sigma}_{H_0}^2 = \frac{\|\mathbf{Y} - \mathbf{X} \hat{\beta}_{\sim H_0}\|^2}{n}.$$

Therefore, the maximized log-likelihood is:

$$\ell^*(\hat{\beta}_{\sim H_0}, \hat{\sigma}_{H_0}^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_{H_0}^2 - \frac{1}{2\hat{\sigma}_{H_0}^2} \|\mathbf{Y} - \mathbf{X} \hat{\beta}_{\sim H_0}\|^2 = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_{H_0}^2 - \frac{n}{2}.$$

Unconstrained: The unconstrained MLEs of β and σ^2 was obtained before in Section 2.3 as

$$\hat{\beta} = \hat{\beta}_{\text{MLE}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}, \quad \text{and} \quad \hat{\sigma}^2 = \hat{\sigma}_{\text{MLE}}^2 = \frac{\|\mathbf{Y} - \mathbf{X} \hat{\beta}_{\text{MLE}}\|^2}{n}.$$

Therefore, the maximized log-likelihood is:

$$\ell^*(\hat{\beta}, \hat{\sigma}^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \hat{\sigma}_{\text{MLE}}^2 - \frac{n}{2}.$$

The LRT test statistic is thus obtained as

$$\Lambda(\mathcal{Y}) = \left(\frac{\hat{\sigma}_H^2}{\hat{\sigma}^2} \right)^{-n/2}.$$

To understand how small a value the test statistic should take in order to reject H_0 at $100\alpha\%$ level, we need to know the distribution of the test statistic under H_0 .

Below we will find the distribution of a one-one function of $\Lambda(\mathcal{Y})$:

$$\begin{aligned} \Lambda(\mathcal{Y})^{-2/n} &= \frac{\hat{\sigma}_H^2}{\hat{\sigma}^2} = \frac{\|Y - X\hat{\beta}_H\|^2}{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2} = \frac{\|\mathcal{Y} - X\hat{\beta}_{OLS} + X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2} \\ &= \frac{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2 + \|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2} = 1 + \frac{\|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2}{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2} \\ &= 1 + \frac{RSS_{H_0} - RSS}{RSS}. \end{aligned}$$

Next, we will find the distribution of a scaled version of the ratio $(RSS_{H_0} - RSS)/RSS$.

Denominator:

$$\frac{RSS}{\sigma^2} = \frac{\|\mathcal{Y} - X\hat{\beta}_{OLS}\|^2}{\sigma^2} = \frac{\mathcal{Y}^\top (I - P(X)) \mathcal{Y}}{\sigma^2} \sim \chi_{n-p-1}^2.$$

Numerator:

$$\begin{aligned} RSS_{H_0} - RSS &= \|X(\hat{\beta}_H - \hat{\beta}_{OLS})\|^2 \\ &= (A\hat{\beta}_{OLS} - \mathcal{C})^\top [A(X^\top X)^{-1}A^\top]^{-1} A(X^\top X)^{-1}A^\top [A(X^\top X)^{-1}A^\top]^{-1} (A\hat{\beta}_{OLS} - \mathcal{C}) \\ &= (A\hat{\beta}_{OLS} - \mathcal{C})^\top [A(X^\top X)^{-1}A^\top]^{-1} (A\hat{\beta}_{OLS} - \mathcal{C}). \end{aligned} \quad (2.10)$$

Observe that,

$$\begin{aligned} A\hat{\beta}_{OLS} &= A(X^\top X)^{-1}X^\top \mathcal{Y} \stackrel{H_0}{\sim} N(A\beta = \mathcal{C}, \sigma^2 A(X^\top X)^{-1}A^\top), \quad \text{under } H_0 \\ (A\hat{\beta}_{OLS} - \mathcal{C}) &\stackrel{H_0}{\sim} N(\mathbf{0}, \sigma^2 A(X^\top X)^{-1}A^\top), \quad \text{under } H_0. \end{aligned}$$

Define

$$\mathcal{Z} = \frac{1}{\sigma} [A(X^\top X)^{-1}A^\top]^{-1/2} (A\hat{\beta}_{OLS} - \mathcal{C}) \stackrel{H_0}{\sim} N_r(0, I), \quad \Rightarrow \quad \mathcal{Z}^\top \mathcal{Z} \stackrel{H_0}{\sim} \chi_r^2.$$

Observe that, the last expression in (2.10) is in the form $\sigma^2 \mathcal{Z}^\top \mathcal{Z}$. Therefore,

$$\frac{RSS_{H_0} - RSS}{\sigma^2} \stackrel{H_0}{\sim} \chi_r^2.$$

At this point, note the following:

For some p.d. matrix Σ having spectral decomposition

$$\begin{aligned} \Sigma &= UDU^\top, \quad D = \text{Diag}(d_1, \dots, d_r), \quad \text{with } d_j > 0 \forall j, \quad \text{and } U^\top U = I, \\ \text{we define } \Sigma^{1/2} &= UD^{1/2}U^\top, \quad \text{with } D^{1/2} = \text{Diag}(\sqrt{d_1}, \dots, \sqrt{d_r}), \\ \text{and } \Sigma^{-1/2} &= UD^{-1/2}U^\top, \quad \text{where } D^{-1/2} = \text{Diag}(1/\sqrt{d_1}, \dots, 1/\sqrt{d_r}), \\ \text{so that } \Sigma^{-1/2}\Sigma^{-1/2} &= \Sigma^{-1}. \end{aligned}$$

Moving back to the original discussion, recall that $m\chi_n^2/n\chi_n^2 \sim F_{n,m}$, provided both the chi squares are independent. Therefore, it remains to show that RSS and RSS_{H_0} are independently distributed. Observe, from (2.10), that $RSS_{H_0} = \|X\hat{\beta}_{OLS} - X\hat{\beta}_H\|^2$ is a function of $\hat{\beta}_{OLS}$ only, and $\hat{\beta}_{OLS} \perp\!\!\!\perp RSS$ (see PS 2). So, RSS and RSS_{H_0} are independently distributed.

Combining these facts, we get

$$\frac{(n-p-1)(RSS_{H_0} - RSS)}{rRSS} \stackrel{H_0}{\sim} F_{r, n-p-1} \text{ distribution.} \quad (2.11)$$

2.7 Some Important Testing Problems

(A) Y is a white noise

$$H_0 : \beta_0 = \beta_1 = \dots = \beta_p = 0 \quad A = I_{p+1}, \quad \mathcal{L} = \emptyset$$

$$RSS_{H_0} - RSS = \hat{\beta}_{OLS}^\top (X^\top X) \hat{\beta}_{OLS} = \mathcal{Y}^\top X (X^\top X)^{-1} X^\top \mathcal{Y} = \mathcal{Y}^\top P(X) \mathcal{Y} \sim \sigma^2 \chi_{p+1}^2$$

$$RSS = \mathcal{Y}^\top (I - P(X)) \mathcal{Y} \sim \sigma^2 \chi_{n-p-1}^2$$

$$\frac{(n-p-1)(RSS_{H_0} - RSS)}{(p+1)RSS} \stackrel{H_0}{\sim} F_{p+1, n-p-1}$$

Observe that, under H_0 , $\sigma^{-1}\mathcal{Y} \sim N(\emptyset, I_n)$, and $RSS_{H_0} - RSS = \mathcal{Y}^\top P(X) \mathcal{Y} \sim \sigma^2 \chi_{p+1}^2$, independent of $RSS = \mathcal{Y}^\top (I - P(X)) \mathcal{Y} \sim \sigma^2 \chi_{n-p-1}^2$, as $P(X)(I - P(X)) = 0$.

(B) None of the covariates affect Y significantly

$$H_0 : \beta_1 = \dots = \beta_p = 0 \quad A^{p \times p+1} = \begin{bmatrix} 0 & : & I_p \end{bmatrix}, \quad \mathcal{L} = \emptyset.$$

Under H_0 :

$$\underline{Y} \stackrel{H_0}{\sim} N(\beta_0 \underline{1}, \sigma^2 I)$$

Note that, in this case it is difficult to obtain $RSS_{H_0} - RSS$ directly from formula. One may approach this problem in two ways:

Recall that, the OLS/MLE estimates of the regression estimates remain unaltered if the covariates are centered. However, in that case the design matrix with centered covariates, X_c satisfies

$$X_c^\top X_c = \begin{bmatrix} n & \underline{0}^\top \\ \underline{0} & Z_c^\top Z_c \end{bmatrix} \quad \text{where} \quad X_c = [\underline{1} : Z_c] \quad \text{and} \quad Z_c = [\underline{x}_{(1)} - \bar{x}_1 \underline{1} \quad \cdots \quad \underline{x}_{(p)} - \bar{x}_p \underline{1}]$$

Transforming the original testing problem to that for the centered regression coefficients we get:

$$(A_{\underline{\beta}_{OLS}^c}^c - \underline{c}) = \begin{pmatrix} \underline{0} & I_p \end{pmatrix} \begin{pmatrix} 1/n & \underline{0}^\top \\ \underline{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix} \begin{pmatrix} \underline{1}^\top \\ Z_c^\top \end{pmatrix} \underline{Y} = (Z_c^\top Z_c)^{-1} Z_c^\top \underline{Y}$$

Therefore,

$$\begin{aligned} & (A_{\underline{\beta}_{OLS}^c}^c - \underline{c})^\top [A(X_c^\top X_c)^{-1} A^\top]^{-1} (A_{\underline{\beta}_{OLS}^c}^c - \underline{c}) \\ &= \underline{Y}^\top Z_c (Z_c^\top Z_c)^{-1} \left[\begin{pmatrix} \underline{0} & I_p \end{pmatrix} \begin{pmatrix} 1/n & \underline{0}^\top \\ \underline{0} & (Z_c^\top Z_c)^{-1} \end{pmatrix} \begin{pmatrix} \underline{0}^\top \\ I_p \end{pmatrix} \right]^{-1} (Z_c^\top Z_c)^{-1} Z_c^\top \underline{Y} \\ &= \underline{Y}^\top P(Z_c) \underline{Y} \end{aligned}$$

Under H_0 , $(\underline{Y} - \beta_0 \underline{1})^\top P(Z_c) (\underline{Y} - \beta_0 \underline{1}) = \underline{Y}^\top P(Z_c) \underline{Y} \sim \sigma^2 \chi_p^2$ [WHY?]. Thus,

$$\frac{(n - p - 1)(RSS_{H_0} - RSS)}{pRSS} \sim F_{p, n-p-1}.$$

Alternatively, one may directly find the RSS_{H_0} . Since under H_0 , Y is explained by a single predictor $\underline{1}$, this can be considered as a special case of the normal linear regression model, and the general theory follows with $n \times 1$ design matrix $\underline{1}$.

Under the normal linear regression model under H_0 with design matrix $\underline{1}$, we have

$$RSS_{H_0} = \underline{Y}^\top (I - P(\underline{1})) \underline{Y} = \underline{Y}^\top (I - n^{-1} \underline{1} \underline{1}^\top) \underline{Y}, \quad (2.12)$$

and

$$RSS_{H_0} - RSS = \underline{Y}^\top (P(X) - n^{-1} \underline{1} \underline{1}^\top) \underline{Y} = \underline{Y}^\top P(Z_c) \underline{Y},$$

as

$$P(X) = P(X_c) = P(\underline{1}) + P(Z_c) \quad [\text{see Section 2.4}].$$

The testing follows as before.

2.7.1 Multiple correlation coefficient

Multiple correlation coefficient (MCC): Y on $\{x_1, \dots, x_p\}$, R , is defined as follows:

$$R = \text{simple correlation between } \mathcal{Y} \text{ and } \hat{\mathcal{Y}} = P(X)\mathcal{Y} = \frac{\sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{\hat{Y}})}{\sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2 \sum_{i=1}^n (\hat{Y}_i - \bar{\hat{Y}})^2}} \quad (2.13)$$

- R^2 is called the *Coefficient of Determination*. R^2 measures the strength of association of Y and $\{x_1, \dots, x_p\}$, given the data. If $\mathcal{Y}$ is explained well by X_n , then $\hat{\mathcal{Y}}$ will be closely aligned with $\mathcal{Y}$, and the correlation between $\hat{\mathcal{Y}}$ and $\mathcal{Y}$, i.e., R must be close to *one*.

Thus, it is natural that R^2 is related to the test statistics for the testing problem in (B) above. The following result establishes the relation between the LRT test statistic under testing (B) and R^2 .

Result 6. Let $\bar{\hat{Y}}$ be the mean of $\hat{\mathcal{Y}}$, R as defined in (2.13), RSS_{H_0} under testing (B) (see (2.12)), and $\Lambda(\mathcal{Y})$ be the LRT test statistic for the testing (B). Then

$$(a) \bar{Y} = \bar{\hat{Y}}, \quad (b) RSS_{H_0} = RSS + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2, \quad \text{and} \quad (c) R^2 = 1 - \frac{RSS}{RSS_{H_0}} = 1 - (\Lambda(\mathcal{Y}))^{2/n}.$$

Note that, R^2 is inversely proportional to the LRT test statistic $\Lambda(\mathcal{Y})$. This is also intuitive, since $\Lambda(\mathcal{Y})$ would be small if $\mathcal{Y}$ is not explained significantly under H_0 compared to H_1 , i.e., when X_n explains $\mathcal{Y}$ well, i.e., when R^2 is high.

Proof of Result 6. (a) Observe that

$$\bar{Y} = \frac{1}{n} \mathbf{1}^\top Y, \quad \hat{\bar{Y}} = \frac{1}{n} \mathbf{1}^\top \hat{Y} = \frac{1}{n} \mathbf{1}^\top P(X)Y = \frac{1}{n} \mathbf{1}^\top Y = \bar{Y},$$

since

$$P(X) = P(X_c) = \frac{1}{n} \mathbf{1} \mathbf{1}^\top + P(Z_c) \quad \text{and} \quad \mathbf{1}^\top P(X) = \mathbf{1}^\top, \quad \mathbf{1}^\top Z_c = 0.$$

(b) Consider the right hand side (RHS) of the expression in (b) as

$$\begin{aligned} \text{RHS} &= \text{RSS} + \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2 = Y^\top (I - P(X))Y + Y^\top P(X) (I - n^{-1} \mathbf{1} \mathbf{1}^\top) P(X)Y \\ &= Y^\top \left(I - P(X) + P(X) - n^{-1} P(X) \mathbf{1} \mathbf{1}^\top P(X) \right) Y \\ &= Y^\top \left(I - n^{-1} \mathbf{1} \mathbf{1}^\top \right) Y = \sum_{i=1}^n (Y_i - \bar{Y})^2 = \text{RSS}_{H_0} = \text{LHS}. \end{aligned}$$

(c) Finally, observe that

$$\begin{aligned} \sum_{i=1}^n (Y_i - \bar{Y})(\hat{Y}_i - \bar{Y}) &= \langle (I - n^{-1} \mathbf{1} \mathbf{1}^\top)Y, (I - n^{-1} \mathbf{1} \mathbf{1}^\top)P(X)Y \rangle \\ &= Y^\top (I - n^{-1} \mathbf{1} \mathbf{1}^\top) (P(X) - n^{-1} \mathbf{1} \mathbf{1}^\top) Y \\ &= Y^\top (P(X) - n^{-1} \mathbf{1} \mathbf{1}^\top) Y = \|(P(X) - P(\mathbf{1}))Y\|^2 = \sum_{i=1}^n (\hat{Y}_i - \bar{Y})^2. \end{aligned}$$

Thus,

$$R^2 = \frac{[\sum_i (\hat{Y}_i - \bar{Y})^2]^2}{[\sum_i (\hat{Y}_i - \bar{Y})^2] [\sum (Y_i - \bar{Y})^2]} = \frac{\sum_i (\hat{Y}_i - \bar{Y})^2}{\text{RSS}_{H_0}} = \frac{\text{RSS}_{H_0} - \text{RSS}}{\text{RSS}_{H_0}} = 1 - \frac{\text{RSS}}{\text{RSS}_{H_0}}.$$

□

(C) Some of the variables are not important

Suppose we want to test if the regressors $x_{q+1}, \dots, x_p$ have significant impact of Y . We split the design matrix X as

$$X_n = [X_{(1)} : X_{(2)}] = \underbrace{[\mathbf{1}, X_{(1)}, \dots, X_{(q)}]}_{X_{(1)}} \underbrace{[X_{(q+1)}, \dots, X_{(p)}]}_{X_{(2)}}$$

and test

$$H_0 : \beta_{q+1} = \dots = \beta_p = 0 \Leftrightarrow \beta_{q+1} = 0, \beta_{q+2} = 0, \dots, \beta_p = 0 \quad \text{against} \quad H_1 : \text{No restriction}$$

Here $A^{p-q \times p+1} = \begin{bmatrix} 0^{p-q \times q+1} & I_{p-q} \end{bmatrix}$, and $\mathcal{C} = \mathbf{0}$.

Here as well, one may directly calculate $RSS_{H_0} - RSS$ from the formula. However, it is much easier to calculate RSS_{H_0} directly, without deriving the formula.

Observe that, under H_0 , $\underline{Y}$ is effectively explained by only $X_{(1)}$ (i.e., by $\{1, x_1, \dots, x_q\}$). Therefore, the RSS under H_0 would be

$$RSS_{H_0} = \underline{Y}^T (I - P(X_{(1)})) \underline{Y}, \quad \text{while, as before} \quad RSS = \underline{Y}^T (I - P(X)) \underline{Y}.$$

Therefore,

$$F = \frac{RSS_{H_0} - RSS}{RSS} = \frac{\underline{Y}^T (P(X) - P(X_{(1)})) \underline{Y}}{\underline{Y}^T (I - P(X)) \underline{Y}}, \quad \text{and} \quad \frac{n-p-1}{p-q} F \stackrel{H_0}{\sim} F_{p-q, n-p-1}.$$

Note, here that $P(X) - P(X_{(1)})$ is also an orthogonal projection matrix (i.e., $P(X) - P(X_{(1)})$ is symmetric and idempotent), due to the fact $P(X)P(X_{(1)}) = P(X_{(1)})P(X) = P(X_{(1)})$ as $C(X_{(1)}) \subseteq C(X)$.

To see the last result observe that, if $C(X_{(1)}) \subseteq C(X)$, then $P(X)X_{(1)} = X_{(1)}$. Therefore

$$P(X_{(1)})P(X) = X_{(1)}(X_{(1)}^T X_{(1)})^{-1} X_{(1)}^T P(X) = X_{(1)}(X_{(1)}^T X_{(1)})^{-1} X_{(1)}^T = P(X_{(1)}),$$

and

$$P(X)P(X_{(1)}) = P(X)X_{(1)}(X_{(1)}^T X_{(1)})^{-1} X_{(1)}^T = X_{(1)}(X_{(1)}^T X_{(1)})^{-1} X_{(1)}^T = P(X_{(1)}).$$

(C*) A special case.

Consider the special case where we are interested in testing if the j -th covariate is significant or not, where $j \in \{1, \dots, p\}$. Without loss of generality (WLoG), assume $j = p$, then the relevant testing problem is $H_0 = \beta_p = 0$.

From the previous testing, we find that the F statistics for this problem reduces to

$$T^2 := \frac{RSS_{H_0} - RSS}{RSS} = \frac{\underline{Y}^T (P(X) - P(X_{(1)})) \underline{Y}}{\underline{Y}^T (I - P(X)) \underline{Y}}, \quad \text{and} \quad (n-p-1)T^2 \stackrel{H_0}{\sim} F_{1, n-p-1}.$$

where $X_{(1)} = \begin{bmatrix} 1 & \underline{x}_{(1)} & \cdots & \underline{x}_{(p-1)} \end{bmatrix}$.

2.7.2 Partial correlation coefficient

Partial correlation coefficient (PCC) measures the correlation between the response Y and one predictor, say x_p , after removing the linear effects of the other predictors, i.e., $\{1, x_1, \dots, x_{p-1}\}$, from both variables. It is denoted by $r_{Yp \cdot 1 \dots p-1}$.

Steps for calculating the PCC.

- Regress Y on $\{x_1, \dots, x_{p-1}\}$; obtain the residuals $\underline{e}_{Y \cdot 1 \dots p-1}$.
- Regress x_p on $\{x_1, \dots, x_{p-1}\}$; obtain the residuals $\underline{e}_{p \cdot 1 \dots p-1}$.
- The simple correlation between $\underline{e}_{Y \cdot 1 \dots p-1}$ and $\underline{e}_{p \cdot 1 \dots p-1}$, is called the partial correlation coefficient of Y and x_p given $\{x_1, \dots, x_{p-1}\}$, i.e., $r_{Yp \cdot 1 \dots p-1} = \text{Corr}(\underline{e}_{Y \cdot 1 \dots p-1}, \underline{e}_{p \cdot 1 \dots p-1})$

As MCC is related to the testing problem (B), it is natural that PCC is related to the testing problem (C*). The result below explores the relation in details.

Result 7. Let $\underline{e}_{Y \cdot 1 \dots p-1}$, $\underline{e}_{p \cdot 1 \dots p-1}$ and $r_{Yp \cdot 1 \dots p-1}$ be the residual of $\underline{Y}$ and $\underline{x}_{(p)}$ when regressed on $\underline{x}_{(1)}, \dots, \underline{x}_{(p-1)}$, and the partial correlation coefficient between $\underline{Y}$ and $\underline{x}_{(p)}$, given $\underline{x}_{(1)}, \dots, \underline{x}_{(p-1)}$. Further, let RSS_{H_0} be the residual sum of squares under the hypothesis $H_0 : \beta_p = 0$, and $X_{(1)} = \begin{bmatrix} 1 & \underline{x}_{(1)} & \cdots & \underline{x}_{(p-1)} \end{bmatrix}$. Then the following results hold:

- The means of $\underline{e}_{Y \cdot 1 \dots p-1}$ and $\underline{e}_{p \cdot 1 \dots p-1}$ are zeros.
- $P(X) - P(X_{(1)}) = \underline{z}\underline{z}^\top / \|\underline{z}\|^2$ where $\underline{z} = (I - P(X_{(1)}))\underline{x}_{(p)}$, and $P(X)$, $P(X_{(1)})$ being the orthogonal projection matrices onto the column spaces of X and $X_{(1)}$, respectively.
- $r_{Yp \cdot 1 \dots p-1}^2 = 1 - \frac{RSS}{RSS_{H_0}} = 1 - \{\Lambda(\underline{Y})\}^{2/n}$, where $\Lambda(\underline{Y})$ is the LRT test statistic for testing $H_0 : \beta_p = 0$.

Observe that, the LRT test statistics for testing $H_0 = \beta_p = 0$ has exactly same relation with PCC, as the relation of LRT test statistics for testing $H_0 = \beta_1 = \dots = \beta_p = 0$ with MCC. Similar explanation of the inverse relation of $\Lambda(\underline{Y})$ and PCC holds here as well.

Proof. (a) Observe that, $\underline{e}_{Y \cdot 1 \dots p-1} = [I - P(X_{(1)})]\underline{Y}$ and $\underline{e}_{p \cdot 1 \dots p-1} = [I - P(X_{(1)})]\underline{x}_{(p)}$. As in the proof of Result 6, one can show that $\underline{1}^\top P(X_{(1)}) = \underline{1}^\top$, and therefore $\underline{1}^\top [I - P(X_{(1)})] = \underline{0}^\top$, and the mean of $\underline{e}_{Y \cdot 1 \dots p-1}$, $\underline{1}^\top \underline{e}_{Y \cdot 1 \dots p-1} / n = 0$. Similarly, mean of $\underline{e}_{p \cdot 1 \dots p-1}$ is also zero.

(b) This part is a consequence of the following more general result:

Let $X^{n \times p} = [V^{n \times q} : W^{n \times p-q}]$, then $P(X) = P(V) + P(Z)$ where $Z = [I - P(V)]W$. The proof of this result is deferred to a later chapter. Here $V = X_{(1)}$ and $W = \underline{x}_{(p)}$, and therefore, $Z = \underline{z}$, and $P(Z) = \underline{z}\underline{z}^\top / \|\underline{z}\|^2$, which proves the result.

(c) From part (a), the covariance between $\underline{e}_{Y \cdot 1 \dots p-1}$ and $\underline{e}_{p \cdot 1 \dots p-1}$ is

$$ncov(\underline{e}_{Y \cdot 1 \dots p-1}, \underline{e}_{p \cdot 1 \dots p-1}) = \langle [I - P(X_{(1)})]\underline{Y}, [I - P(X_{(1)})]\underline{x}_{(p)} \rangle = \underline{Y}^\top [I - P(X_{(1)})]\underline{x}_{(p)}.$$

Further,

$$nvar(\underline{e}_{Y \cdot 1 \dots p-1}) = \underline{Y}^\top [I - P(X_{(1)})]\underline{Y} = RSS_{H_0}, \quad \text{and} \quad nvar(\underline{e}_{p \cdot 1 \dots p-1}) = \underline{x}_{(p)}^\top [I - P(X_{(1)})]\underline{x}_{(p)}.$$

Next, observe that

$$\begin{aligned} RSS_{H_0} - RSS &= \underline{Y}^\top [P(X) - P(X_{(1)})]\underline{Y} \stackrel{[\text{from (b)}]}{=} \underline{Y}^\top \frac{\underline{z}\underline{z}^\top}{\|\underline{z}\|^2} \underline{Y} \\ &= \frac{\underline{Y}^\top [I - P(X_{(1)})]\underline{x}_{(p)}\underline{x}_{(p)}^\top [I - P(X_{(1)})]\underline{Y}}{\underline{x}_{(p)}^\top [I - P(X_{(1)})]\underline{x}_{(p)}} \\ &= \frac{n^2 cov^2(\underline{e}_{Y \cdot 1 \dots p-1}, \underline{e}_{p \cdot 1 \dots p-1})}{nvar(\underline{e}_{p \cdot 1 \dots p-1})}. \end{aligned}$$

Hence,

$$\frac{RSS_{H_0} - RSS}{RSS_{H_0}} = 1 - \{\Lambda(\underline{Y})\}^{2/n} = \frac{cov^2(\underline{e}_{Y \cdot 1 \dots p-1}, \underline{e}_{p \cdot 1 \dots p-1})}{var(\underline{e}_{Y \cdot 1 \dots p-1}) var(\underline{e}_{p \cdot 1 \dots p-1})} = r_{Y p \cdot 1 \dots p-1}^2,$$

which completes the proof. □

(D) Effects of all the covariates are same

Suppose we want to test if effects of the covariates does not differ significant. If this hypothesis is accepted, then the design matrix reduces drastically from $n \times p + 1$ to $n \times 2$, where the only covariate is $x_1 + x_2 + \dots + x_p$. More formally, here we want to test:

$H_0 : \beta_1 = \beta_2 = \dots = \beta_p$, equivalently, $\beta_1 - \beta_2 = \dots = \beta_{p-1} - \beta_p = 0$, against $H_1 : \text{No restriction}$

If we want to write the null hypothesis in the form $A\beta = \underline{c}$, then

$$A^{p-1 \times p+1} = \begin{bmatrix} 0 & 1 & -1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & -1 & \cdots & 0 \\ \vdots & & & \ddots & \ddots & \\ 0 & 0 & \cdots & 0 & 1 & -1 \end{bmatrix}, \quad \text{and} \quad \underline{c} = \underline{0}.$$

Observe that, under H_0 , $\underline{Y} \sim N(\beta_0 \underline{1} + \beta \underline{z}, \sigma^2 I)$ where $\underline{z} = \underline{x}_{(1)} + \cdots + \underline{x}_{(p)}$ and β is the common regression coefficient. So, the design matrix is $X_{H_0} = \begin{bmatrix} \underline{1} & \underline{z} \end{bmatrix}$.

As before, $RSS_{H_0} = \underline{Y}^T (I - P(X_{H_0})) \underline{Y}$ and $RSS = \underline{Y}^T (I - P(X)) \underline{Y}$, which implies

$$F = \frac{RSS_{H_0} - RSS}{RSS} = \frac{\underline{Y}^T (P(X) - P(X_{H_0})) \underline{Y}}{\underline{Y}^T (I - P(X)) \underline{Y}}, \quad \text{and} \quad \frac{n-p-1}{p-1} F \sim F_{p-1, n-p-1}.$$

Show that, $P(X) - P(X_{H_0})$ is an orthogonal projection matrix, as $\mathcal{C}(X_{H_0}) \subseteq \mathcal{C}(X_n)$, and verify the distribution of F under H_0 using Craig's theorem.

(E) A non-canonical testing problem: Sum of regression coefficients is k

Suppose we want to test if the sum of the regression coefficients is a known constant k , i.e., we want to test

$$H_0 : \beta_1 + \cdots + \beta_p = k \quad \text{against} \quad H_1 : \text{No constraint.}$$

If we want to write the null hypothesis in the form $A\beta = \underline{c}$, then

$$A \text{ is a vector, say } \underline{a} = \begin{pmatrix} 0 & 1 & \cdots & 1 \end{pmatrix}^T, \quad c = k.$$

Note: When $\underline{c} = \underline{0}$, then corresponding testing is called a canonical testing. Testing (A)-(D) are canonical testing problem, while (E) is a non-canonical testing problem.

Since H_0 concerns $\{\beta_1, \dots, \beta_p\}$, and not the intercept, we may transform the testing problem to that in the normal linear regression with centered covariates here as well.

By the formula of F statistic

$$RSS_{H_0} - RSS = \frac{(\underline{q}^T \hat{\underline{\beta}}_{\text{OLS}} - k)^2}{\underline{q}^T (X_c^\top X_c)^{-1} \underline{q}} = \frac{(\hat{\beta}_{1,\text{OLS}} + \cdots + \hat{\beta}_{p,\text{OLS}} - k)^2}{\underline{q}^T (X_c^\top X_c)^{-1} \underline{q}}$$

where $(X_c^\top X_c)^{-1} = \begin{bmatrix} n^{-1} & \underline{0}^T \\ \underline{0} & (Z_c^\top Z_c)^{-1} \end{bmatrix}$, and $\underline{q}^T (X_c^\top X_c)^{-1} \underline{q} = \underline{1}^T (Z_c^\top Z_c)^{-1} \underline{1}$.

Finally,

$$\frac{(n-p-1)(RSS_{H_0} - RSS)}{RSS} = \sim F_{1, n-p-1} \equiv t_{n-p-1}^2.$$

2.8 Interval Estimation

Univariate: Recall that, for an univariate parameter of interest β , by interval estimator we mean a pair of statistics $[L(Y), U(Y)]$, which satisfies

$$\inf_{\beta} P_{\beta}(L(Y) \leq \beta \leq U(Y)) = (1 - \alpha), \quad \text{which implies,} \quad P_{\beta}(L(Y) \leq \beta \leq U(Y)) \geq (1 - \alpha) \quad \forall \beta.$$

Multivariate: When the parameter of interest, say $\underline{\beta}$, is in $\mathbb{R}^{p+1}$, the idea of confidence intervals can be extended to confidence sets. A set valued random variable, $S(Y)$, is said to form a confidence set if

$$\inf_{\underline{\beta}} P_{\underline{\beta}}(\underline{\beta} \in S(Y)) = (1 - \alpha), \quad \text{which implies,} \quad P_{\underline{\beta}}(\underline{\beta} \in S(Y)) \geq (1 - \alpha) \quad \forall \underline{\beta}.$$

2.8.1 Confidence Ellipsoid

In the context of normal linear regression model, *confidence ellipsoids* provide confidence sets for any linear combinations of $\underline{\beta}$.

Consider the problem of finding confidence sets for r linear combinations of $\underline{\beta}$, say, $\underline{a}_j^{\top} \underline{\beta}$ for $j = 1, \dots, r$. We assume no redundancy in the parameters, in other words, the vectors $\underline{a}_1, \dots, \underline{a}_r$ are assumed to be

linearly independent. Define $A = \begin{bmatrix} \underline{a}_1^{\top} \\ \vdots \\ \underline{a}_r^{\top} \end{bmatrix}$ as the $r \times (p+1)$ matrix of constraints of rank r . The problem is to find a $100(1 - \alpha)\%$ confidence set for the r -variate parameter $A\underline{\beta}$.

Recall that

$$\widehat{\beta}_{\text{OLS}} \sim N_r(A\beta, \sigma^2 A(X'X)^{-1}A^T).$$

Observe that, $A(X'X)^{-1}A^T$ is invertible, and define $[A(X'X)^{-1}A^T]^{-1/2}$ as before, and then

$$\frac{1}{\sigma} [A(X'X)^{-1}A^T]^{-1/2} (\widehat{\beta}_{\text{OLS}} - A\beta) \sim N_r(0, I).$$

Then

$$\frac{1}{\sigma^2} (\widehat{\beta}_{\text{OLS}} - A\beta)^T [A(X'X)^{-1}A^T]^{-1} (\widehat{\beta}_{\text{OLS}} - A\beta) \sim \chi_r^2.$$

Further, as before

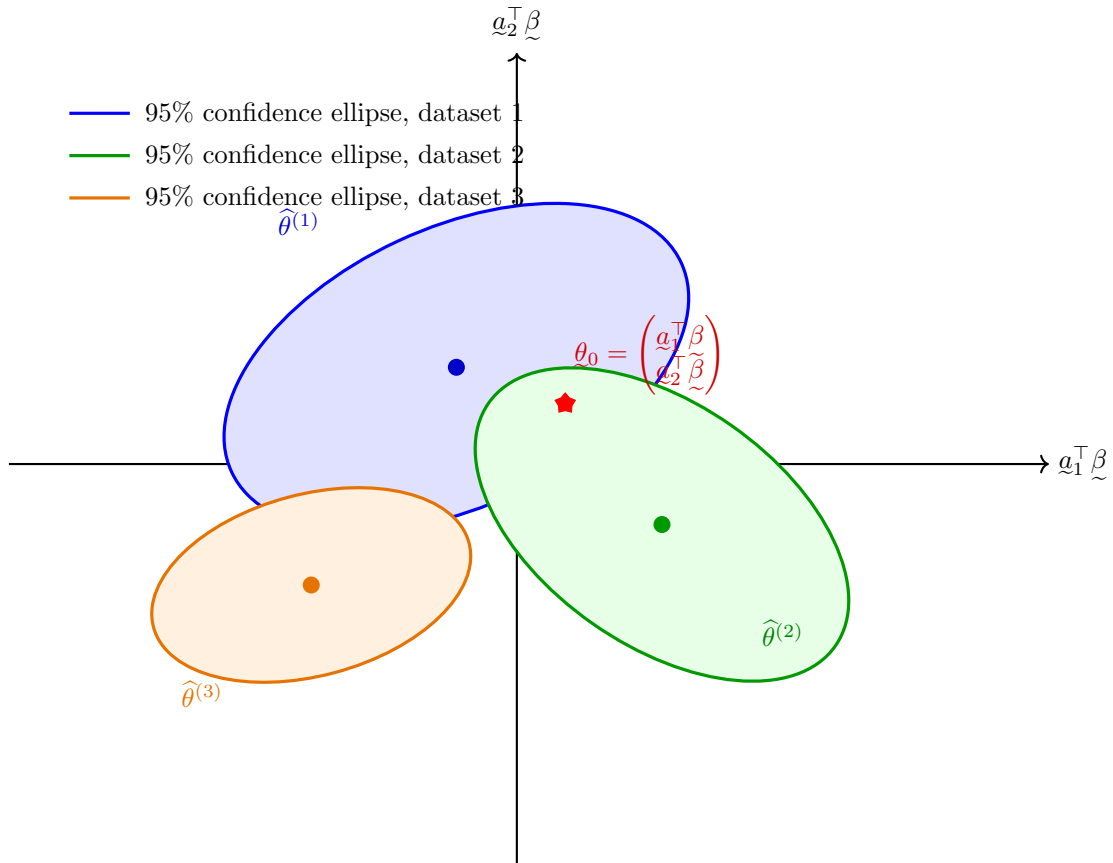
$$\frac{RSS}{\sigma^2} \sim \chi_{n-p-1}^2 \quad \text{independently, i.e., } \widehat{\beta}_{\text{OLS}} \perp\!\!\!\perp RSS.$$

Therefore,

$$\frac{(\widehat{\beta}_{\text{OLS}} - A\beta)^T [A(X'X)^{-1}A^T]^{-1} (\widehat{\beta}_{\text{OLS}} - A\beta)}{rRSS/(n-p-1)} \sim F_{r, n-p-1}.$$

Let $F_{\alpha; r, n-p-1}$ be the upper- α point of $F_{r, n-p-1}$ distribution, then

$$P_{\beta} \left(\frac{n-p-1}{r \cdot RSS} (\widehat{\beta}_{\text{OLS}} - \beta)^T A^T [A(X'X)^{-1}A^T]^{-1} A (\widehat{\beta}_{\text{OLS}} - \beta) \leq F_{\alpha; r, n-p-1} \right) = (1 - \alpha).$$



2.8.2 Simultaneous Confidence Interval

When the dimension r of the linear parametric function $A\beta$ is high, then the confidence ellipsoids are not much informative. In such cases, one is rather interested in simultaneous confidence intervals, i.e., a list of r intervals, $[L_j(\underline{Y}), U_j(\underline{Y})]$ for $j = 1, \dots, r$, such that,

$$P_{\underline{\beta}} \left(L_j(\underline{Y}) < \underline{a}_j^T \underline{\beta} \leq U_j(\underline{Y}) \quad \forall \quad j = 1, \dots, r \right) \geq (1 - \alpha) \quad \forall \underline{\beta}.$$

Examples. Let $\underline{\beta} \in \mathbb{R}^{p+1}$ and we are interested in finding simultaneous confidence interval of $(\beta_1 - \beta_2, \beta_2 - \beta_3) \in \mathbb{R}^2$, or $(\beta_1, \beta_2, \beta_3) \in \mathbb{R}^3$.

In the first example, $r = 2$ and $\underline{a}_1 = (0, 1, -1, 0, \dots, 0)^T$, $\underline{a}_2 = (0, 0, 1, -1, 0, \dots, 0)^T$, while in the second example, $r = 3$ and $\underline{a}_j = \underline{e}_{j+1}$ for $j = 1, 2, 3$.

Simultaneous Coverage Intervals

Consider a set of r linear combinations:

$$\underline{a}_i^T \underline{\beta}, \quad i = 1, \dots, r.$$

Simultaneous Coverage Intervals are a set of r pairs of statistics

$$\{L_i(\underline{Y}), U_i(\underline{Y})\}, \quad \text{such that} \quad L_i(\underline{Y}) \leq U_i(\underline{Y}) \quad \forall i, \text{ w.p. } 1, \quad \text{and}$$

$$\inf_{\underline{\beta}} P_{\underline{\beta}} \left(\underbrace{L_1(\underline{Y}) \leq \underline{a}_1^T \underline{\beta} \leq U_1(\underline{Y})}_{E_1}, \dots, \underbrace{L_r(\underline{Y}) \leq \underline{a}_r^T \underline{\beta} \leq U_r(\underline{Y})}_{E_r} \right) = (1 - \alpha)$$

$$\Rightarrow P_{\underline{\beta}} (E_1 \cap E_2 \cap \dots \cap E_r) \geq (1 - \alpha) \quad \forall \underline{\beta}.$$

In general, the confidence ellipsoids are not much helpful in finding the simultaneous confidence intervals, in particular in higher dimensions.

What if we use the individual confidence intervals? Using

$$\underline{a}^T \hat{\underline{\beta}}_{\text{OLS}} \sim N \left(\underline{a}^T \underline{\beta}, \sigma^2 \underline{a}^T (X'X)^{-1} \underline{a} \right), \quad \text{and} \quad \frac{RSS}{\sigma^2} \sim \chi_{n-p-1}^2, \quad \text{independently,}$$

we get the following individual confidence intervals:

Individual CI Coverage

$$P_{\underline{\beta}} \left(\underbrace{\underline{a}^T \underline{\beta} \in \underline{a}^T \hat{\underline{\beta}}_{\text{OLS}} \pm \sqrt{\frac{RSS}{n-p-1}} \sqrt{\underline{a}^T (X'X)^{-1} \underline{a}} \cdot t_{\alpha/2; n-p-1}}_{E_a} \right) = (1 - \alpha) \quad \forall \underline{\beta} \quad (2.14)$$

For each $j = 1, \dots, r$, we have an individual confidence interval for $\underline{a}_j^\top \underline{\beta}$, and correspondingly an event E_j such that $P_{\underline{\beta}}(E_j) = (1 - \alpha)$ for all $\underline{\beta}$.

However, if we consider the probability that all the events $E_1, \dots, E_r$ occurs simultaneously, then the joint probability

$$\begin{aligned} P_{\underline{\beta}}(E_1 \cap E_2 \cap \dots \cap E_r) &= P_{\underline{\beta}}\left(\bigcap_{j=1}^r E_j\right) = 1 - P_{\underline{\beta}}\left(\left(\bigcap_{j=1}^r E_j\right)^c\right) = 1 - P_{\underline{\beta}}\left(\bigcup_{j=1}^r E_j^c\right) \\ &\geq 1 - \sum_{j=1}^r P_{\underline{\beta}}(E_j^c) = 1 - r\alpha. \end{aligned}$$

Example

Ex: $r = 10$; $\alpha = 0.05$.

$$P_{\underline{\beta}}(E_1 \cap \dots \cap E_r) \geq 0.5$$

Even if the events $E_1, \dots, E_r$ are independent, then also we have

$$P_{\underline{\beta}}(E_1 \cap \dots \cap E_r) = \prod_{j=1}^r P_{\underline{\beta}}(E_j) = (1 - \alpha)^r.$$

Example Independent Case

Ex: $r = 10$; $\alpha = 0.05$, $P_{\underline{\beta}}(E_1 \cap \dots \cap E_r) \approx 0.6$.

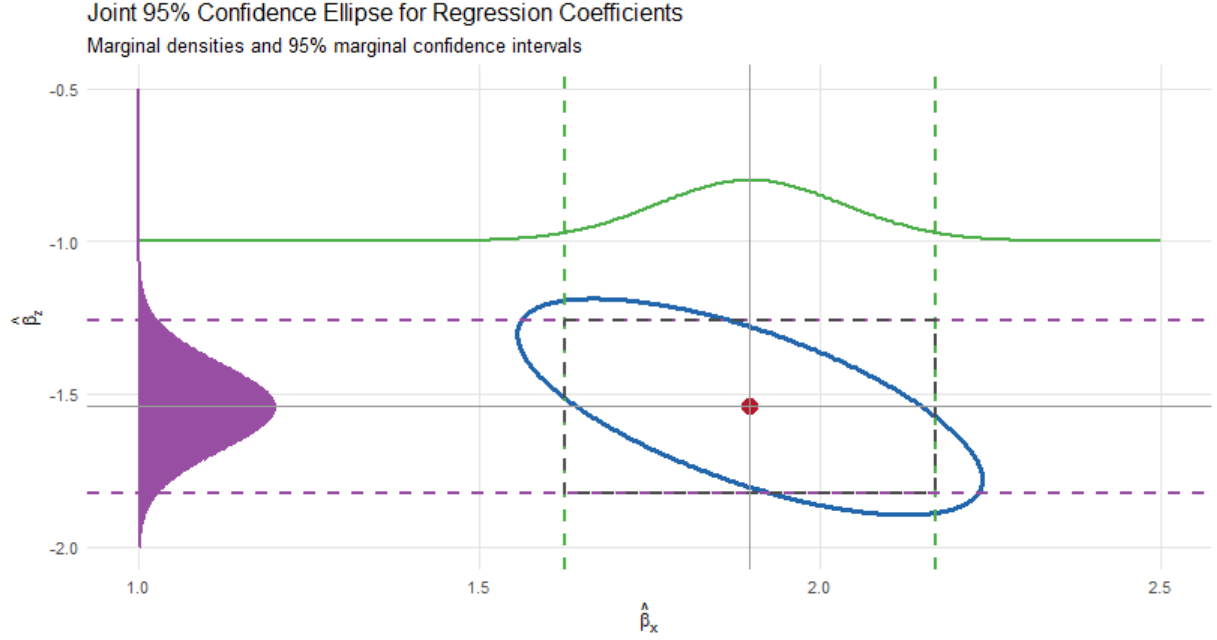
We have to make sure that,

$$P_{\underline{\beta}}(E_1 \cap \dots \cap E_r) \geq (1 - \alpha) \quad \forall \underline{\beta}.$$

We will learn two methods of doing that.

Bonferroni's correction

Bonferroni suggested considering the individual confidence coefficients to be α/r instead of α , and considering the intersection of all individual confidence coefficients to obtain a simultaneous confidence interval ensuring confidence coefficient at least $1 - \alpha$.



Bonferroni's method

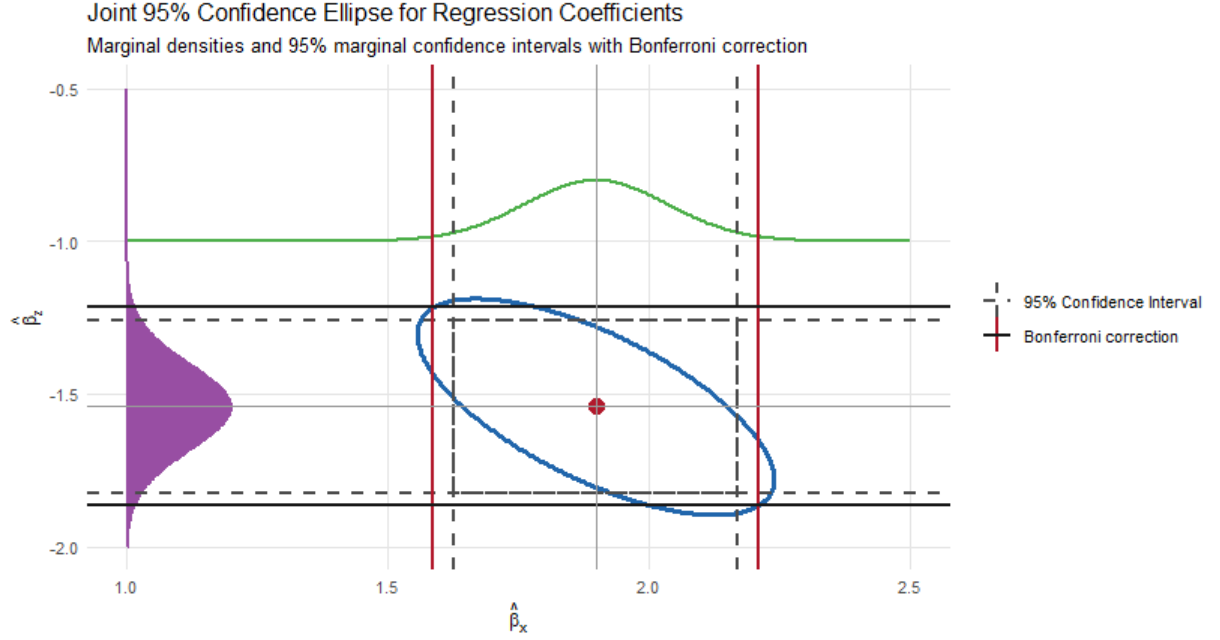
1. Take $E_j = \underline{a}_j^T \underline{\beta} \in \underline{a}_j^T \hat{\underline{\beta}}_{OLS} \pm \sqrt{RSS \underline{a}_j^T (X'X)^{-1} \underline{a}_j / (n-p-1)} \cdot t_{\alpha/2r; n-p-1}$, so that $P_{\underline{\beta}}(E_j) = (1 - \alpha/r)$ for all $\underline{\beta}$ and $j = 1, \dots, r$.
2. Consider $E_1 \cap \dots \cap E_r$ as the simultaneous confidence interval (SCI), such that

$$\Rightarrow P_{\underline{\beta}}(E_1 \cap \dots \cap E_r) \geq 1 - r \cdot \frac{\alpha}{r} = (1 - \alpha) \quad \forall \underline{\beta}.$$

Note: When r is large, then α/r is very small, so the resultant SCI is over-inclusive, hence is not useful. For example, if $r = 10$ and $\alpha = 0.05$, then the individual confidence coefficients are 99.5%, which is quite large.

Scheffé's S-method

Without loss of generality, we consider the set of r linear functions of $\underline{\beta}$, say, $\underline{a}_1^T \underline{\beta}, \dots, \underline{a}_r^T \underline{\beta}$, such that $\{\underline{a}_1, \dots, \underline{a}_r\}$ are linearly independent.



Scheffé's S-method

Define

$$A_{r \times p+1} = \begin{bmatrix} \underline{a}'_1 \\ \vdots \\ \underline{a}'_r \end{bmatrix}. \text{ Clearly, } A \text{ has rank } r.$$

Recall

$$A\hat{\underline{\beta}}_{\text{OLS}} \sim N_r \left(A\beta, \sigma^2 A(X'X)^{-1}A' \right) \text{ and } \text{RSS} \sim \sigma^2 \chi^2_{n-p-1} \text{ independently.}$$

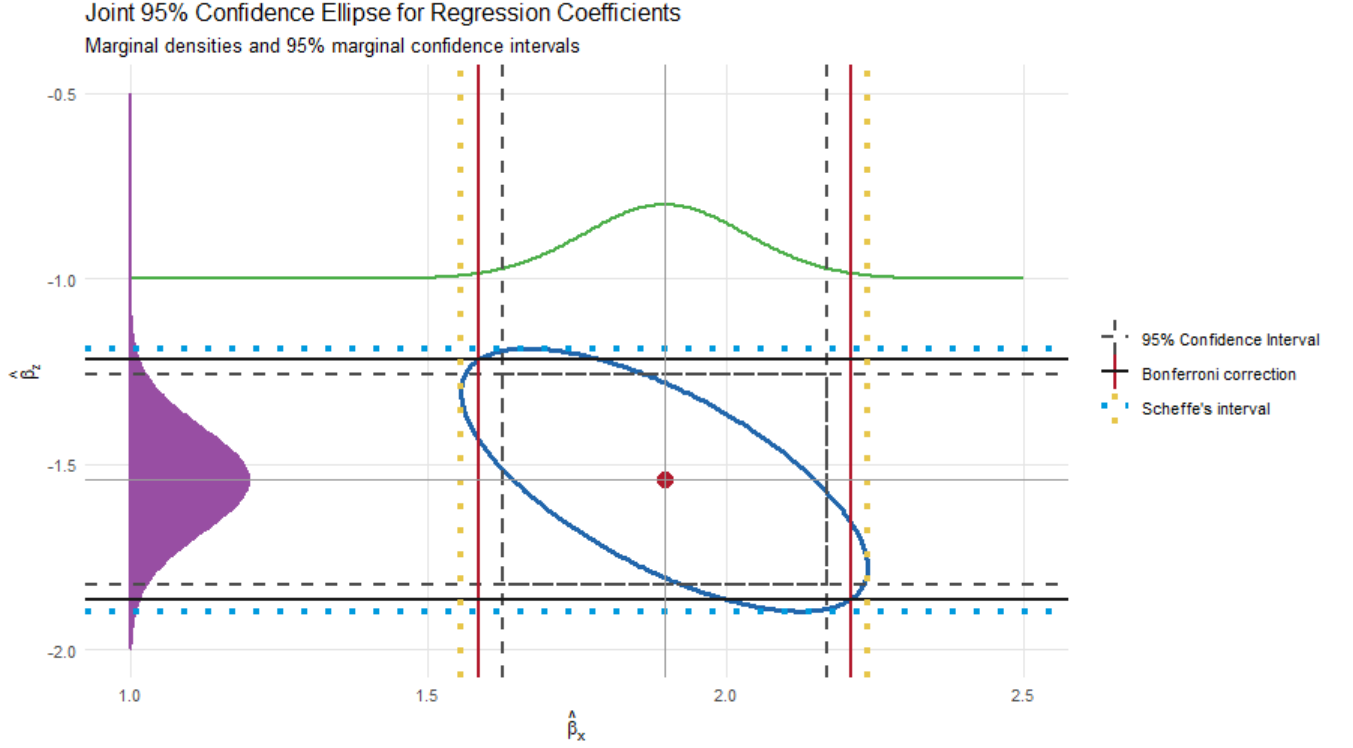
Using this we get, as before

$$P_{\underline{\beta}} \left(\underbrace{[A(\hat{\underline{\beta}}_{\text{OLS}} - \underline{\beta})]^T}_{\underline{b}^T} \underbrace{[A(X'X)^{-1}A']^{-1}}_{L^{-1}} \underbrace{A(\hat{\underline{\beta}}_{\text{OLS}} - \underline{\beta})}_{\underline{b}} \leq \underbrace{r \frac{\text{RSS}}{n-p-1} \cdot F_{\alpha; r, n-p-1}}_{\mathcal{E}_{\alpha}} \right) = (1 - \alpha) \quad \forall \underline{\beta}$$

$$\iff P_{\underline{\beta}} (\underline{b}^T L^{-1} \underline{b} \leq \mathcal{E}_{\alpha}) = (1 - \alpha). \quad (2.15)$$

Result 8. Let L be a positive definite matrix, and $\underline{b}$ be any vector in $\mathbb{R}^k$, then

$$\underline{b}^T L^{-1} \underline{b} = \sup_{\underline{w} \neq 0} \frac{(\underline{w}^T \underline{b})^2}{\underline{w}^T L \underline{w}}$$



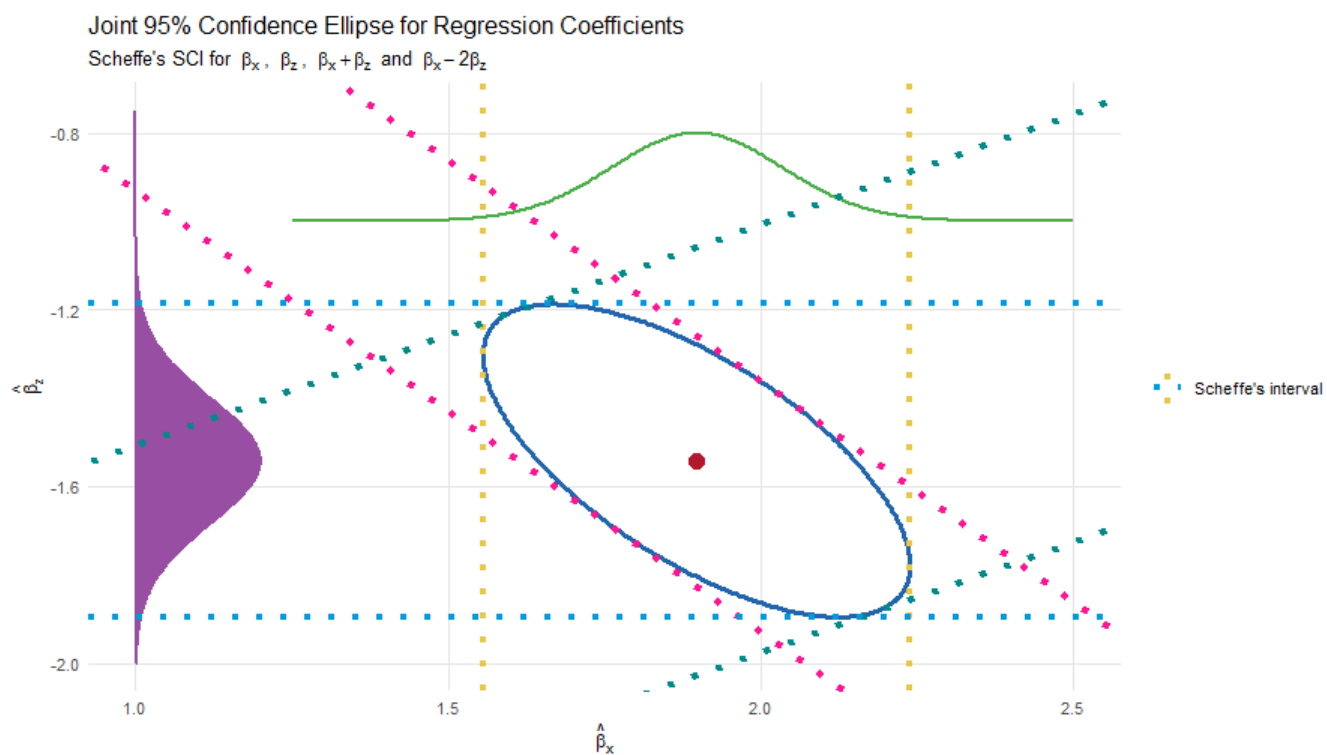
Scheffé's S-method (cont..)

From (2.15) and Result 8 we get:

$$\begin{aligned}
 P_{\underline{\beta}} \left(\sup_{w \neq 0} \frac{(w^T \underline{\hat{\beta}})^2}{w^T L w} \leq \mathcal{E}_\alpha \right) &= (1 - \alpha) \quad \forall \underline{\beta} \\
 \iff P_{\underline{\beta}} \left(w^T A(\underline{\hat{\beta}}_{\text{OLS}} - \underline{\beta}) \leq \pm \sqrt{w^T L w \cdot \mathcal{E}_\alpha} \quad \forall w \neq 0 \right) &= (1 - \alpha) \quad \forall \underline{\beta} \\
 \iff P_{\underline{\beta}} \left(w^T A \underline{\beta} \in w^T A \underline{\hat{\beta}}_{\text{OLS}} \pm \sqrt{w^T L w \cdot \mathcal{E}_\alpha} \quad \forall w \neq 0 \right) &= (1 - \alpha) \quad \forall \underline{\beta} \\
 \iff P_{\underline{\beta}} \left(\underline{a}_j^T \underline{\beta} \in \underline{a}_j^T \underline{\hat{\beta}}_{\text{OLS}} \pm \sqrt{\underline{e}_j^T L \underline{e}_j \cdot \mathcal{E}_\alpha} \quad \forall w \neq 0, \text{ and } \forall j = 1, \dots, r \right) &\geq (1 - \alpha), \quad \forall \underline{\beta},
 \end{aligned}$$

in particular, taking w as $\underline{e}_1, \dots, \underline{e}_r$. Note that, $\underline{e}_j^T L \underline{e}_j = \ell_{j,j} = \underline{a}_j^T (X'X)^{-1} \underline{a}_j$. Therefore, we get Scheffé's simultaneous SCI as

$$P_{\underline{\beta}} \left(\underbrace{\underline{a}_1^T \underline{\hat{\beta}}_{\text{OLS}} - \underline{a}_1^T \underline{\beta} \leq \pm \sqrt{\mathcal{E}_\alpha \cdot \ell_{11}}}_{\text{taking } w=\underline{e}_1}, \dots, \overbrace{\underline{a}_r^T \underline{\hat{\beta}}_{\text{OLS}} - \underline{a}_r^T \underline{\beta} \leq \pm \sqrt{\ell_{rr} \cdot \mathcal{E}_\alpha}}^{\text{taking } w=\underline{e}_r} \right) \geq (1 - \alpha) \quad \forall \underline{\beta}.$$



Proof of Result 8

Define $\underline{w}^* = L^{1/2}\underline{w} \Leftrightarrow \underline{w} = L^{-1/2}\underline{w}^*$ where $L = \underbrace{A(X'X)^{-1}A'}_{\text{p.d. matrix}}$

Consider the RHS =

$$\begin{aligned} \sup_{\underline{w} \neq 0} \frac{\underline{w}^T \underline{b} \underline{b}^T \underline{w}}{\underline{w}^T L \underline{w}} &= \sup_{\underline{w}^* \neq 0} \frac{\underline{w}^{*T} L^{-1/2} \underline{b} \underline{b}^T L^{-1/2} \underline{w}^*}{\underline{w}^{*T} \underline{w}^*} \\ &= \lambda_{\max} \left(\underline{L}^{-1/2} \underline{b} \underline{b}^T \underline{L}^{-1/2} \right) = \underline{b}^T L^{-1} \underline{b}. \end{aligned}$$

2.9 Confidence Interval on Regression Function

In the previous section have learnt ways to find simultaneous confidence interval (SCI) on different linear functions of β . Next, we use this theory to predict the regression surface or regression curve for a future observation.

One such linear function is the regression curve itself for a future observation $\underline{x}^*$, i.e., $\mu^* = \underline{x}^{*T} \beta$.

Let $(y^*, \underline{x}^*)$ be a future observation, where y^* is not observed. We assume that

$$Y^* = \underline{x}^{*T} \beta + \varepsilon_* = \mu^* + \varepsilon_*, \quad \text{i.e.,} \quad E(y^*) = \mu^*$$

is the quantity of interest.

Point estimator of μ^* is: $\underline{x}^{*T} \hat{\beta}_{\text{OLS}}$.

Confidence interval

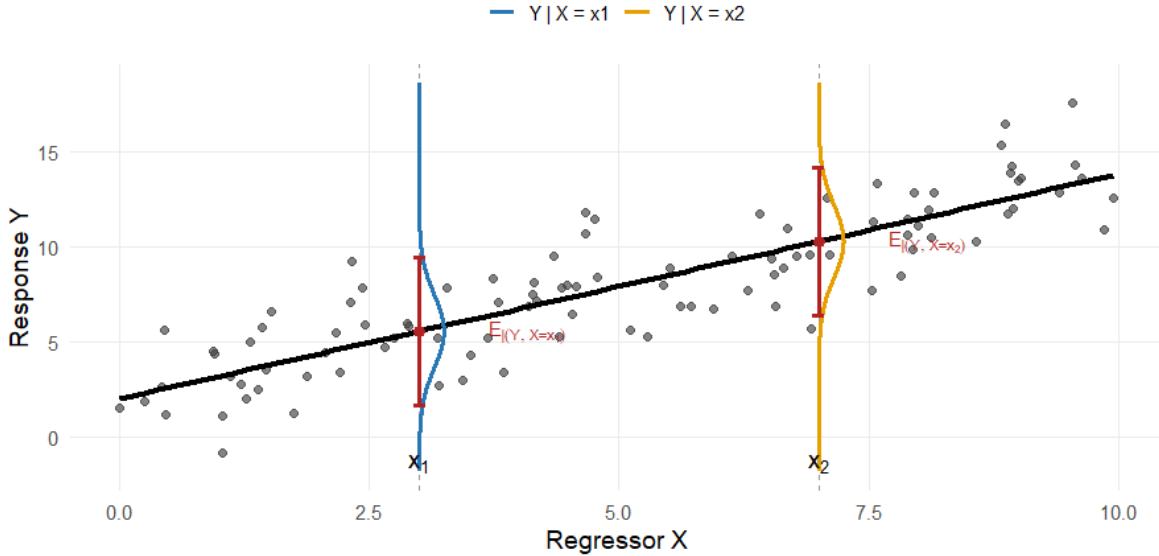
From (2.14), by taking $\underline{a} = \underline{x}^*$ we get

$$\mu^* \in \underline{x}^{*T} \hat{\beta}_{\text{OLS}} \pm t_{\alpha/2, n-p-1} \hat{\sigma}_{\text{UE}} \sqrt{\underline{x}^{*T} (X^T X)^{-1} \underline{x}^*} \quad \text{where} \quad \hat{\sigma}_{\text{UE}}^2 = \frac{\text{RSS}}{n-p-1}, \quad (2.16)$$

with probability $(1 - \alpha)$ for all β .

Normal Linear Regression

Conditional distributions and 95% prediction intervals



The above confidence interval is ideal if we know the future observation $\underline{x}^*$. However, usually we do

not know the future values of the regressors. Therefore, we need confidence interval for

$$\mu_0 = \mathbf{x}_0^T \boldsymbol{\beta},$$

simultaneously, for all possible values of $\mathbf{x}_0$.

Note that, the intersection of all the individual intervals would have much lower coverage (see the previous section). Therefore, we need to apply methods of obtaining simultaneous confidence intervals (SCIs). If we have a few known future observations, say, $\mathbf{x}_1, \dots, \mathbf{x}_k$, then one can attempt Bonferroni's corrected SCIs. However, if we need SCIs for all future values $\mathbf{x}_0$, then Scheffe's SCI provides the only feasible option.

Let $\mathbf{x}_0$ be any new observation (unknown). Then

$$\mathbf{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} \sim N(\mathbf{x}_0^T \boldsymbol{\beta} = \mu_0, \sigma^2 \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0) \Rightarrow \frac{\mathbf{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} - \mu_0}{\hat{\sigma}_{\text{UE}} \sqrt{\mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0}} \sim t_{n-p-1},$$

where $\hat{\sigma}_{\text{UE}}^2$ is as defined in (2.16).

Now, observe that, by Result 8

$$\sup_{\mathbf{x}_0 \neq \mathbf{0}} \frac{(\mathbf{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}))^2}{\sigma^2 \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0} = \frac{1}{\sigma^2} (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})^T (X^T X) (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \sim \chi_{p+1}^2$$

as

$$\hat{\boldsymbol{\beta}}_{\text{OLS}} \sim N(\boldsymbol{\beta}, \sigma^2 (X^T X)^{-1}) \quad \text{and} \quad \frac{1}{\sigma} (X^T X)^{1/2} (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta}) \sim N(\mathbf{0}, I_{p+1}).$$

Further, as $\hat{\boldsymbol{\beta}}_{\text{OLS}} \perp\!\!\!\perp RSS$,

$$\Rightarrow \sup_{\mathbf{x}_0 \neq \mathbf{0}} \frac{\{\mathbf{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\}^2}{(p+1) \hat{\sigma}_{\text{UE}}^2 \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0} \sim F_{p+1, n-p-1}.$$

Simultaneous Confidence Interval

Using the above result we obtain

$$P_{\hat{\boldsymbol{\beta}}} \left(\sup_{\mathbf{x}_0 \neq \mathbf{0}} \frac{\{\mathbf{x}_0^T (\hat{\boldsymbol{\beta}}_{\text{OLS}} - \boldsymbol{\beta})\}^2}{(p+1) \hat{\sigma}_{\text{UE}}^2 \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0} \leq F_{\alpha; p+1, n-p-1} \right) = (1 - \alpha)$$

$$\Leftrightarrow P_{\hat{\boldsymbol{\beta}}} \left(\mu_0 = \mathbf{x}_0^T \boldsymbol{\beta} \in \mathbf{x}_0^T \hat{\boldsymbol{\beta}}_{\text{OLS}} \pm \hat{\sigma}_{\text{UE}} \sqrt{(p+1) F_{\alpha; p+1, n-p-1} \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0} \quad \forall \mathbf{x}_0 \neq \mathbf{0} \right) = (1 - \alpha).$$

2.10 Prediction Interval

Consider a future observation for the response Y_0 corresponding to the predictor $\underline{x}_0$. We model

$$Y_0 = \underline{x}_0^T \underline{\beta} + \varepsilon_0.$$

with the **assumption** $\varepsilon_0 \sim N(0, \sigma^2)$ and $\varepsilon_0 \perp\!\!\!\perp \underline{Y}$. Here, the independence $\varepsilon_0 \perp\!\!\!\perp \underline{Y}$ indicates that the randomness of a future observation is independent of the randomness of the past observations. In other words, Y_0 is associated with the past observations $(X, \underline{Y})$, only through the systematic part.

In this section, we are interested in finding point and interval estimates of Y_0 . The interval estimates of Y_0 is called prediction interval.

Note that, unlike μ_0 , Y_0 is random. So, it is might be of interest to find simultaneous prediction intervals for k -different realizations of Y_0 , which we will also see later.

Point estimator of Y_0 :

$$\underline{x}_0^T \hat{\underline{\beta}}_{\text{OLS}} = \hat{Y}_0.$$

Observe that

$$Y_0 - \hat{Y}_0 = \underline{x}_0^T \underline{\beta} + \varepsilon_0 - \underline{x}_0^T \hat{\underline{\beta}}_{\text{OLS}}.$$

To obtain the prediction interval for Y_0 , it is required find the distribution of $Y_0 - \hat{Y}_0$. Towards that, recall

$$\underline{x}_0^T \hat{\underline{\beta}}_{\text{OLS}} \sim N(\underline{x}_0^T \underline{\beta}, \sigma^2 \underline{x}_0^T (X^T X)^{-1} \underline{x}_0 + \sigma^2 = \sigma^2 (1 + \underline{x}_0^T (X^T X)^{-1} \underline{x}_0))$$

As $\varepsilon_0 \perp\!\!\!\perp \underline{Y}$, $\varepsilon_0 \perp\!\!\!\perp g(\underline{Y})$ for any function g of $\underline{Y}$. In particular, $\varepsilon_0 \perp\!\!\!\perp (\text{RSS}, \hat{\underline{\beta}}_{\text{OLS}})$. Therefore,

$$\begin{aligned} Y_0 - \hat{Y}_0 &= \underline{x}_0^T \underline{\beta} + \varepsilon_0 - \underline{x}_0^T \hat{\underline{\beta}}_{\text{OLS}} \sim N(0, \sigma^2 (1 + \underline{x}_0^T (X^T X)^{-1} \underline{x}_0)) \\ \frac{\hat{Y}_0 - Y_0}{\sigma \sqrt{1 + \underline{x}_0^T (X^T X)^{-1} \underline{x}_0}} &\sim N(0, 1) \quad \text{and} \quad \frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2 \end{aligned}$$

(function of $(\hat{\underline{\beta}}_{\text{OLS}}, \varepsilon_0)$). As $\varepsilon_0 \perp\!\!\!\perp (\text{RSS}, \hat{\underline{\beta}}_{\text{OLS}})$ and $\text{RSS} \perp\!\!\!\perp \hat{\underline{\beta}}_{\text{OLS}}$, we have $\{\varepsilon_0, \hat{\underline{\beta}}_{\text{OLS}}, \text{RSS}\}$ are mutually independent.

Pairwise independence vs mutual independence

At this point, it is important to note the difference between pairwise independence and mutual independence. A set of k variables $\mathbf{Z} = \{Z_1, \dots, Z_k\}$ are said to be mutually independent if the joint distribution of $\mathbf{Z}$ is product of the marginal distribution of $Z_1, \dots, Z_k$ for all possible realizations. Further, pairwise independence, i.e., $Z_i \perp\!\!\!\perp Z_j$ for all $i \neq j$ does not guarantee mutual independence.

Here, we have a situation where we have 3 random variables, say, $\{Z_1, Z_2, Z_3\}$ where $Z_1 \perp\!\!\!\perp \{Z_2, Z_3\}$ and $Z_2 \perp\!\!\!\perp Z_3$. Then we claim that $\{Z_1, Z_2, Z_3\}$ are mutually independent.

Proof. For any realization $\{z_1, z_2, z_3\}$,

$$\begin{aligned} f_{Z_1, Z_2, Z_3}(z_1, z_2, z_3) &= f_{Z_1}(z_1) f_{Z_2, Z_3}(z_2, z_3) \quad [\text{as } Z_1 \perp\!\!\!\perp \{Z_2, Z_3\}], \\ &= f_{Z_1}(z_1) f_{Z_2}(z_2) f_{Z_3}(z_3) \quad [\text{as } Z_2 \perp\!\!\!\perp Z_3]. \end{aligned}$$

This proves the claim. □

Therefore,

$$\frac{\hat{Y}_0 - Y_0}{\hat{\sigma}_{UE} \sqrt{1 + \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0}} \sim t_{n-p-1}. \quad \text{where} \quad \hat{\sigma}_{UE}^2 = \frac{\text{RSS}}{n-p-1}.$$

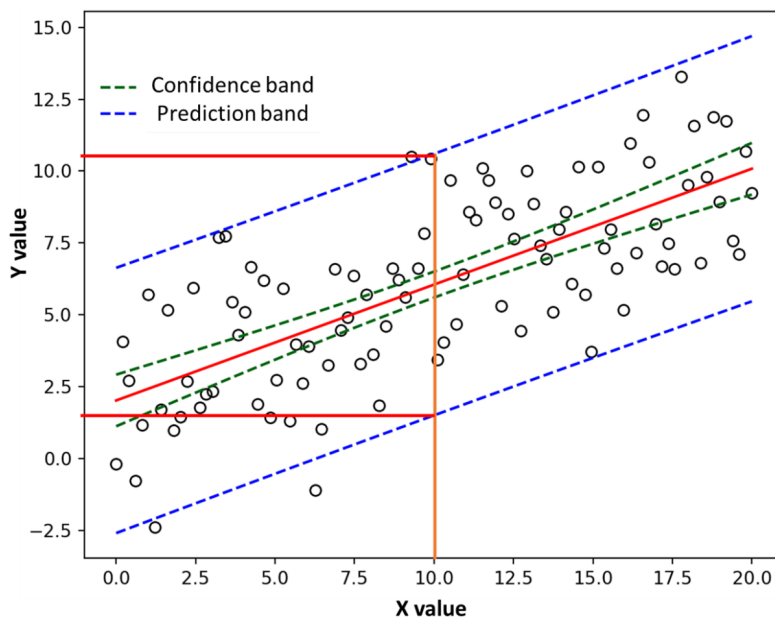
Prediction interval

So the prediction interval for Y_0 is

$$P_{\hat{\beta}} \left(Y_0 \in \hat{Y}_0 \pm t_{\alpha/2, n-p-1} \hat{\sigma}_{UE} \sqrt{1 + \mathbf{x}_0^T (X^T X)^{-1} \mathbf{x}_0} \right) = (1 - \alpha). \quad \forall \hat{\beta}$$

Note: From the above derivation it is evident that prediction intervals of Y_0 are wider than confidence intervals for μ_0 , although both are centered around $\hat{\mu}_0 = \hat{Y}_0 = \mathbf{x}_0^T \hat{\beta}$. This ordering is expected as a PI for Y_0 includes both the uncertainty in estimating the regression mean μ_0 and the random error of the new observation, whereas a CI for μ_0 includes only the estimation uncertainty.

Further, the difference of width between the CI and PI depends on $\hat{\sigma}_{UE}^2$. In fact, higher values of σ^2 would imply higher on an average deviation of Y_0 from its mean μ_0 (i.e., higher absolute value of ε_0 on an average), and $\hat{\sigma}_{UE}^2$ is a consistent estimator of σ^2 . Therefore, this association is expected.



This figure is obtained from [ISIXSIGMA](#)

2.10.1 Simultaneous Prediction Intervals

Next, we generalize the prediction intervals of Y_0 in two possible directions:

- to accommodate for any future observation x_0 ;
- to accommodate for k many future realizations of Y_0 for same x_0 ,

$$Y_{0j} = x_0^T \beta + \varepsilon_{0j}, \quad j = 1, \dots, k,$$

and also their combinations, for example $(Y_{01} + Y_{02})/2$.

There are two possible methods of these generalization (both proposed by Carlstein). We will discuss each of them below:

Method 1

Let $\mathbf{x}_0$ be an unknown future observations, with respect to which we want to predict for k realizations of Y_0 as

$$Y_{0j} = \mathbf{x}_0^T \beta + \varepsilon_{0j}, \quad j = 1, \dots, k.$$

Key assumption:

$$\varepsilon_0 = \begin{bmatrix} \varepsilon_{01} \\ \vdots \\ \varepsilon_{0k} \end{bmatrix} \perp\!\!\!\perp \varepsilon \quad (\text{and hence is independent of } \mathbf{Y}, \hat{\beta}_{\text{OLS}}, \text{RSS}).$$

Define

$$\mathbf{b} = \begin{bmatrix} \beta - \hat{\beta}_{\text{OLS}} \\ \varepsilon_0 \end{bmatrix} \sim N_{p+k+1} \left(\mathbf{0}, \sigma^2 \begin{bmatrix} (X^T X)^{-1} & \mathbf{0} \\ \mathbf{0} & I_k \end{bmatrix} \right).$$

Define

$$W := \begin{bmatrix} (X^T X)^{-1} & \mathbf{0} \\ \mathbf{0} & I_k \end{bmatrix}. \quad \text{Then} \quad \frac{1}{\sigma^2} \mathbf{b}^T W^{-1} \mathbf{b} \sim \chi_{p+k+1}^2, \quad \text{and} \quad \frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2 \quad (\text{independent of } \mathbf{b}).$$

This implies

$$\frac{\mathbf{b}^T W^{-1} \mathbf{b}}{(p+k+1)\hat{\sigma}_{\text{UE}}^2} \sim F_{p+k+1, n-p-1} \quad (\text{where } \hat{\sigma}_{\text{UE}}^2 \text{ is as in (2.16)}).$$

We know that

$$\mathbf{b}^T W^{-1} \mathbf{b} = \sup_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}^T \mathbf{b})^2}{\mathbf{a}^T W \mathbf{a}} \quad [\text{by Result 8}].$$

Therefore,

$$\begin{aligned} P_{\beta} \left(\frac{\mathbf{b}^T W^{-1} \mathbf{b}}{(p+k+1)\hat{\sigma}_{\text{UE}}^2} \leq F_{\alpha; p+k+1, n-p-1} \right) &= (1-\alpha) \quad \forall \beta \\ \Leftrightarrow P_{\beta} \left(|\mathbf{a}^T \mathbf{b}| \leq \hat{\sigma}_{\text{UE}} \sqrt{(p+k+1)F_{\alpha; p+k+1, n-p-1} \mathbf{a}^T W \mathbf{a}} \quad \forall \mathbf{a} \neq \mathbf{0} \right) &= (1-\alpha) \forall \beta. \end{aligned}$$

What choice of $\mathbf{a}$ will give simultaneous prediction intervals (SPIs) for Y_{0j} for all $j = 1, \dots, k$?

Ans. $\underline{a}_j = \begin{bmatrix} \underline{x}_0 \\ \underline{e}_j \end{bmatrix}$ where $\underline{e}_j = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \leftarrow j\text{-th}, \quad j = 1, \dots, k.$

For this particular choice $\underline{a}_j = \begin{bmatrix} \underline{x}_0 \\ \underline{e}_j \end{bmatrix}$ one obtains

$$\underline{a}_j^T W \underline{a}_j = \begin{pmatrix} \underline{x}_0^T & \mathbf{e}_j^T \end{pmatrix} \begin{bmatrix} (X^T X)^{-1} & \underline{0} \\ \underline{0} & I_k \end{bmatrix} \begin{pmatrix} \underline{x}_0 \\ \mathbf{e}_j \end{pmatrix} = \underline{x}_0^T (X^T X)^{-1} \underline{x}_0 + 1.$$

SPIs for new $\underline{x}_0$ and k realizations of Y_0

Therefore, the SPI obtained by method 1 is:

$$P_{\beta} \left(\underline{a}_j^{\top} b \in \pm \hat{\sigma}_{\text{UE}} \sqrt{\underbrace{(p+k+1)F_{\alpha; p+k+1, n-p-1}}_{F_{\alpha}^*(p, k)}} \underline{a}_j^{\top} \bar{W}^{-1} \underline{a}_j \quad \forall j = 1, \dots, k \right) \geq (1 - \alpha) \quad \forall \underline{\beta}$$

where $\underline{a}_j = (\underline{x}_0^{\top}, \underline{e}_j^{\top})^{\top}$. This is equivalent to

$$\begin{aligned} P_{\beta} \left(Y_{01} \in \hat{Y}_{01} \pm \hat{\sigma}_{\text{UE}} \sqrt{F_{\alpha}^*(p, k)} \left\{ 1 + \underline{x}_0^{\top} (X^{\top} X)^{-1} \underline{x}_0 \right\}, \right. \\ Y_{02} \in \hat{Y}_{02} \pm \hat{\sigma}_{\text{UE}} \sqrt{F_{\alpha}^*(p, k)} \left\{ 1 + \underline{x}_0^{\top} (X^{\top} X)^{-1} \underline{x}_0 \right\}, \dots, \\ \left. Y_{0k} \in \hat{Y}_{0k} \pm \hat{\sigma}_{\text{UE}} \sqrt{F_{\alpha}^*(p, k)} \left\{ 1 + \underline{x}_0^{\top} (X^{\top} X)^{-1} \underline{x}_0 \right\} \quad \forall \underline{x}_0 \neq \underline{0} \right) \geq (1 - \alpha) \quad \forall \underline{\beta}. \end{aligned}$$

What choice of $\underline{a}$ will give prediction interval for $(Y_{01} + Y_{02})/2$?

Ans.

$$\underline{a}^* = \begin{bmatrix} \underline{x}_0 \\ (\underline{e}_1 + \underline{e}_2)/2 \end{bmatrix}.$$

Verification:

$$\frac{Y_{01} + Y_{02}}{2} - \frac{\hat{Y}_{01} + \hat{Y}_{02}}{2} = \underline{x}_0^T (\underline{\beta} - \hat{\underline{\beta}}_{\text{OLS}}) + \frac{\varepsilon_{01} + \varepsilon_{02}}{2}.$$

Method 2

This method treats the uncertainty due to future $\underline{x}_0$ and due to randomness of ε_0 separately. Further, this is a more general method where one can have different confidence coefficients associated with the uncertainty of a future observation $\underline{x}_0$ and that associated with replications of Y_0 . The method is described below:

Let (α_1, α_2) be such that $\alpha_1 + \alpha_2 = \alpha$.

Step 1: Consider the simultaneous confidence interval (SCI) of regression mean $\mu_0 = \underline{x}_0^T \underline{\beta}$ with confidence coefficient $(1 - \alpha_1)$:

$$P_{\underline{\beta}} \left(\underbrace{\mu_0 \in \hat{\mu}_0 \pm \sqrt{(p+1)F_{\alpha_1; p+1, n-p-1}} \hat{\sigma}_{\text{UE}}^2 \underline{x}_0^T (X^T X)^{-1} \underline{x}_0}_{\mathcal{E}_1} \quad \forall \underline{x}_0 \neq \underline{0} \right) = (1 - \alpha_1) \quad \forall \underline{\beta}$$

where $\mu_0 = \underline{x}_0^T \underline{\beta}$ and $\hat{\mu}_0 = \underline{x}_0^T \hat{\underline{\beta}}_{\text{OLS}}$.

Now, $\underline{\varepsilon}_0 \sim N(\underline{0}, \sigma^2 I_k)$. Therefore,

$$\frac{1}{\sigma^2} \underline{\varepsilon}_0^T \underline{\varepsilon}_0 \sim \chi_k^2 \quad \text{and} \quad \frac{\text{RSS}}{\sigma^2} \sim \chi_{n-p-1}^2, \quad \text{independently,} \quad \implies \frac{\underline{\varepsilon}_0^T \underline{\varepsilon}_0}{k \hat{\sigma}_{\text{UE}}^2} \sim F_{k, n-p-1}.$$

By Result 8,

$$\underline{\varepsilon}_0^T \underline{\varepsilon}_0 = \sup_{\underline{\tilde{a}} \neq \underline{0}} \frac{(\underline{\tilde{a}}^T \underline{\varepsilon}_0)^2}{\underline{\tilde{a}}^T \underline{\tilde{a}}}.$$

Using this, we get

$$\begin{aligned} P \left(\frac{\underline{\varepsilon}_0^T \underline{\varepsilon}_0}{k \hat{\sigma}_{\text{UE}}^2} \leq F_{\alpha_2; k, n-p-1} \right) &= (1 - \alpha_2) \quad \forall \underline{\beta} \\ \Leftrightarrow P \left(|\underline{\tilde{a}}^T \underline{\varepsilon}_0| \leq \sqrt{k F_{\alpha_2; k, n-p-1}} \hat{\sigma}_{\text{UE}}^2 \underline{\tilde{a}}^T \underline{\tilde{a}} \quad \forall \underline{\tilde{a}} \neq \underline{0} \right) &= (1 - \alpha_2) \quad \forall \underline{\beta}. \end{aligned}$$

In particular, considering the choices $\underline{a} = \underline{e}_j$, $j = 1, \dots, k$, we get

$$P \left(\underbrace{\varepsilon_{0,j} \leq \pm \hat{\sigma}_{\text{UE}} \sqrt{k F_{\alpha_2; k, n-p-1}}}_{\mathcal{E}_2} \quad \forall j = 1, \dots, k \right) \geq (1 - \alpha_2) \quad \forall \underline{\beta}.$$

Now, observe that

$$P(\mathcal{E}_1 \cap \mathcal{E}_2) = P(\mathcal{E}_1) + P(\mathcal{E}_2) - P(\mathcal{E}_1 \cup \mathcal{E}_2) \geq P(\mathcal{E}_1) + P(\mathcal{E}_2) - 1 \geq (1 - \alpha_1) + (1 - \alpha_2) - 1 = 1 - (\alpha_1 + \alpha_2).$$

Using this fact we obtain the following SPIs.

SPIs for new $\underline{x}_0$ and k realizations of Y_0

$$P_{\underline{\beta}}(\mathcal{E}_1 \cap \mathcal{E}_2) = P_{\underline{\beta}} \left(\mu_0 + \varepsilon_{0,j} \in \hat{\mu}_0 \pm \hat{\sigma}_{\text{UE}} \left\{ \sqrt{(p+1)F_{\alpha_1; p+1, n-p-1}} \underline{x}_0^T (X^T X)^{-1} \underline{x}_0 + \sqrt{kF_{\alpha_2; k, n-p-1}} \right\} \quad \forall \underline{x}_0 \neq \underline{0}, \forall j = 1, \dots, p \right) \geq 1 - (\alpha_1 + \alpha_2) = 1 - \alpha \quad \forall \underline{\beta}.$$

By choosing different values of α_i , $i = 1, 2$, one can adjust the coverage of mean prediction and random realizations.

Chapter 3

Departure from the Assumptions of Normal Linear Regression Models

Recall the normal linear regression model with n samples on the response y and predictors $\{x_1, \dots, x_p\}$ as

$$\underline{Y} = X\underline{\beta} + \underline{\epsilon}; \quad \underline{\epsilon} \sim N(\underline{0}, \sigma^2 I).$$

Assumptions

The assumptions made in the above models are:

- (i) $E(\varepsilon_i) = 0 \quad \forall i$.
- (ii) $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0 \quad \forall i \neq j$.
- (iii) $\text{Var}(\varepsilon_i) = \sigma^2 \quad \forall i$.
- (iv) ε_i 's are normally distributed / independent.
- (v) X is non-random, and $(X'X)^{-1}$ exists.
- (vi) linearity of the parameters.

Questions to Explore

Now we will find answers to the following questions:

- What happens if any of these assumptions is violated?
- How to verify the assumptions?
- What to do if one of these assumptions does not hold?

3.1 Unbiasedness

What if the assumption $E(\varepsilon_i) = 0 \forall i$ is violated?

Suppose we consider the predictors $\{x_1, \dots, x_p\}$. But there are a few other predictors $\{z_1, \dots, z_k\}$ which we fail to consider.

The Actual vs. Working Model

The actual model should be:

$$\underline{Y} = \underset{n \times (p+1)}{X} \underset{\sim}{\beta} + \underset{n \times k}{Z} \underset{\sim}{\gamma} + \underline{\varepsilon}^*, \quad E(\underline{\varepsilon}^*) = \underline{0}, \quad \text{var}(\underline{\varepsilon}^*) = \sigma^2 I. \quad (\text{A})$$

Our working model is:

$$\underline{Y} = \underline{X} \underset{\sim}{\beta} + \underline{\varepsilon}, \quad E(\underline{\varepsilon}) = \underline{0}, \quad \text{Var}(\underline{\varepsilon}) = \sigma^2 I. \quad (\text{W})$$

Given the actual model is true, our proposed error term $\underline{\varepsilon} = \underline{Z}\underline{\gamma} + \underline{\varepsilon}^*$, which has expectation $\underline{Z}\underline{\gamma}$ and variance $\sigma^2 I$, i.e., $\underline{\varepsilon}$ does not have *zero* expectation.

Results obtained from the working model

$$\begin{aligned} \hat{\underset{\sim}{\beta}}_{OLS} &= (\underline{X}'\underline{X})^{-1} \underline{X}'\underline{Y}, \\ \hat{\underline{Y}}_{OLS} &= P(\underline{X})\underline{Y}, \\ RSS &= \underline{Y}'(I - P(\underline{X}))\underline{Y}. \end{aligned}$$

Prediction for a future point $\underline{x}_0$:

$$\hat{Y}_{0,OLS} = \underline{x}_0^T \hat{\underset{\sim}{\beta}}_{OLS}.$$

Bias in $\hat{\underset{\sim}{\beta}}_{OLS}$

Expectation:

$$E(\hat{\underset{\sim}{\beta}}_{OLS}) = \underset{\sim}{\beta} + \underbrace{(X'X)^{-1}X'Z\underset{\sim}{\gamma}}_{\text{bias term}} \neq \underset{\sim}{\beta} \quad (\text{biased})$$

Variance:

$$\text{Var}(\hat{\underset{\sim}{\beta}}_{OLS}) = \text{var}((X'X)^{-1}X'Y) = \sigma^2(X'X)^{-1} \quad (\text{Same as before})$$

Note

1. If X and Z are orthogonal, i.e., $\mathbf{x}_{(j)}^\top \mathbf{z}_{(\ell)} = 0 \quad \forall j = 1, \dots, p+1, \ell = 1, \dots, k$, then the bias is *zero*. This indicates that if X does not contain any information about Z , then $\hat{\beta}_{OLS}$ is unbiased. Similar phenomenon has been observed when we compare the OLS estimates of regression coefficients from a simple vs multiple linear regression model. We observe that, when the covariates are orthogonal, then both the estimates are equal. Here as well, if X and Z are orthogonal, then the OLS estimate of β remains same in from both the models in (A) and (W).
2. The bias is also *zero* if $\gamma = \mathbf{0}$. However, this implies that both the models (A) and (W) are the same.
3. Towards explaining the variance, observe that

$$\hat{\beta}_{OLS} = (X'X)^{-1}X'Y = \beta + (X'X)^{-1}X'Z\gamma + (X'X)^{-1}X'\epsilon^*.$$

Had the working model (W) been the true model, then $\hat{\beta}_{OLS} \stackrel{D}{=} \beta + (X'X)^{-1}X'\epsilon^*$. Therefore, the model misspecification has caused a location-shift in the distribution of $\hat{\beta}_{OLS}$. As the variance is not affected by a location shift, it remains unchanged even when the underlying model is (A).

Bias of the Fitted Values**Expectation:**

$$E(\hat{Y}_{OLS}) = P(X)E(Y) = \underbrace{P(X)X\beta}_{X\beta} + P(X)Z\gamma \neq E(Y) = X\beta.$$

If the underlying model were (W) then $E(\hat{Y}_{OLS})$ would be $X\beta$.

Variance:

$$\text{Var}(\hat{Y}_{OLS}) = P(X) \text{Var}(Y) P(X) = \sigma^2 P(X),$$

which is same irrespective of whether the underlying model is (A) or (W).

Note

1. If $P(X)Z = Z$, i.e., columns of Z lie in the column space of X , then

$$E(\hat{\mathbf{Y}}_{OLS}) = \mathbf{X}\beta + \mathbf{Z}\gamma = E(\mathbf{Y}),$$

even though the underlying model is not the working model. This is because, if $P(X)Z = Z$ implies that the information of Z is essentially captured by X . Thus, by dropping Z we do not lose any information.

2. On the other hand, if $P(X)Z = 0$, i.e., columns of Z are orthogonal to $\mathcal{C}(X)$, then $E(\hat{\mathbf{Y}}_{OLS}) = \mathbf{X}\beta \neq E(\mathbf{Y})$.
3. As for $\hat{\beta}_{OLS}$, here also one can show that, the distribution of $\hat{\mathbf{Y}}_{OLS}$, has a location shift of amount $P(X)Z\gamma$, when the underlying model is (A) compared to that when the underlying model is (W). Therefore, the variance remains unaltered.

RSS and Estimation of σ^2

The unbiased estimator of σ^2 is:

$$S_X := \frac{RSS}{n - p - 1}.$$

Is it unbiased when the working model is different from the true model?

$$\begin{aligned} E(RSS) &= E(\mathbf{Y}'(I - P(X))\mathbf{Y}) \\ &= E\left[(Z\gamma + \epsilon^*)'(I - P(X))(Z\gamma + \epsilon^*)\right] \\ &= \gamma'Z'(I - P(X))Z\gamma + E[\epsilon^{*'}(I - P(X))\epsilon^*] \\ &= \text{tr}((I - P(X))(\sigma^2 I)) + \gamma'Z'(I - P(X))Z\gamma \\ &= \sigma^2(n - p - 1) + \gamma'Z'(I - P(X))Z\gamma. \end{aligned}$$

Therefore,

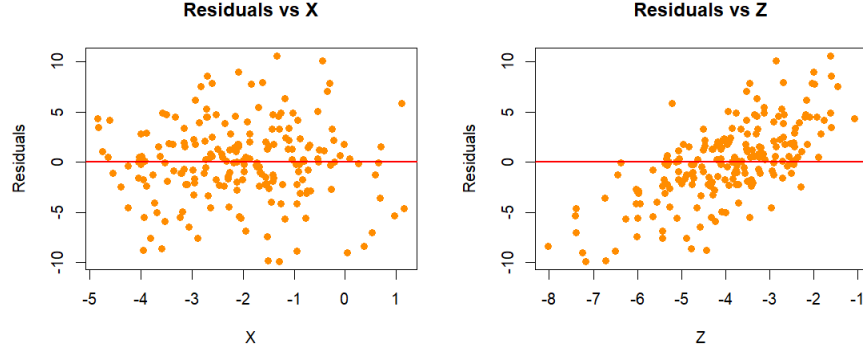
$$E(S_X) = \sigma^2 + \frac{\gamma'Z'(I - P(X))Z\gamma}{n - p - 1} \geq \sigma^2.$$

Residuals. The distribution of the residual is

$$\mathbf{e} = (I - P(X))\mathbf{y} \stackrel{(A)}{=} (I - P(X))Z\gamma + (I - P(X))\epsilon^*,$$

while under it would be $\stackrel{(W)}{=} (I - P(X))\epsilon$ if the working model was the true model.

Note: The variance of the residuals is $\sigma^2(I - P(X))$ in both the cases. However, when (A) is the true model, then there is a bias in the residual of order $(I - P(X))Z\gamma$. When σ^2 is estimated using $RSS = \mathbf{e}'\mathbf{e}$ ignoring the bias term (i.e., assuming $E(\mathbf{e}) = \mathbf{0}$) then it is over-estimated (as $E(\mathbf{e}'\mathbf{e}) = \text{trace}(\text{var}(\mathbf{e})) + E(\mathbf{e})'E(\mathbf{e})$).



Note

1. Had the working model (W) been the true model (i.e., unbiased) then RSS would be $\epsilon^{*'}(I - P(X))\epsilon^*$. However, since the underlying model is (A), now the RSS is

$$\gamma'Z'(I - P(X))Z\gamma + 2\epsilon^{*'}(I - P(X))Z\gamma + \epsilon^{*'}(I - P(X))\epsilon^*,$$

which is not merely a location. Thus, the expectation has increased by $\gamma'Z'(I - P(X))Z\gamma$ (since $E(\epsilon^*) = 0$). Further, unlike for $\hat{\beta}_{OLS}$ and $\hat{Y}_{OLS}$ it is expected that the variance of RSS will also change when the underlying true model is changed from (W) to (A).

2. The bias of S_X in estimating σ^2 will be zero if $(I - P(X))Z = 0$, i.e., columns of Z lie in the column space of X . If $\mathcal{C}(Z) \in \mathcal{C}(X)$, then there exists a matrix $p + 1 \times q$ matrix A such that $Z = XA$, and

$$\underline{X}\beta + \underline{Z}\gamma = \underline{X}\beta + \underline{X}A\gamma = \underline{X}(\beta + A\gamma) = \underline{X}\beta^*.$$

Thus, no information is lost by dropping Z from the model.

Prediction

Suppose $(\underline{x}_0, \underline{z}_0)$ is a new observation, where only $\underline{x}_0$ is observed. We need to estimate Y_0 where

$$Y_0 = \underline{x}_0'\beta + \underline{z}_0'\gamma + \varepsilon_0, \quad \varepsilon_0 \sim N(0, \sigma^2).$$

Our estimate of Y_0 based on the working model (W) is

$$\hat{Y}_{0,OLS} = \underline{x}_0'\hat{\beta}_{OLS} = \underline{x}_0'(X'X)^{-1}X'Y.$$

Expectation.

$$E(\hat{Y}_{0,OLS}) = \underline{x}_0'\beta + \underline{x}_0'(X'X)^{-1}X'Z\gamma \neq \underline{x}_0'\beta + \underline{z}_0'\gamma = E(Y_0).$$

Variance.

$$\text{Var}(\hat{Y}_{0,OLS}) = \sigma^2 \underline{x}'_0 (X'X)^{-1} X'X (X'X)^{-1} \underline{x}_0 = \sigma^2 \underline{x}'_0 (X'X)^{-1} \underline{x}_0. \quad (*)$$

Comparison with prediction Under the Actual Model (A): Define

$$W = (X : Z), \quad \underline{\hat{y}} = \begin{bmatrix} \underline{\beta} \\ \underline{\gamma} \end{bmatrix}_{(p+k+1)}, \quad \text{where} \quad W \in \mathbb{R}^{n \times (p+k+1)}, \quad X \in \mathbb{R}^{n \times (p+1)}, \quad Z \in \mathbb{R}^{n \times (k)}$$

If one considers the actual model (A) as the working model, then

$$\hat{\underline{y}}_{OLS} = (W'W)^{-1} W'Y, \quad \text{and} \quad \hat{Y}_{0,OLS}^A = (\underline{x}'_0 : \underline{z}'_0) \hat{\underline{y}}_{OLS}.$$

We will now compare the expectation and variance of $\hat{Y}_{0,OLS}$ with those of $\hat{Y}_{0,OLS}^A$.

$$E(\hat{Y}_{0,OLS}^A) = (\underline{x}'_0 : \underline{z}'_0) (W'W)^{-1} W'W \underline{\hat{y}} = (\underline{x}'_0 : \underline{z}'_0) \underline{\hat{y}} = \underline{x}'_0 \underline{\beta} + \underline{z}'_0 \underline{\gamma} = E(Y_0).$$

$$\text{Var}(\hat{Y}_{0,OLS}^A) = (\underline{x}'_0 : \underline{z}'_0) (W'W)^{-1} W' \text{Var}(\underline{Y}) W (W'W)^{-1} \begin{pmatrix} \underline{x}_0 \\ \underline{z}_0 \end{pmatrix} = \sigma^2 (\underline{x}'_0 : \underline{z}'_0) (W'W)^{-1} \begin{pmatrix} \underline{x}_0 \\ \underline{z}_0 \end{pmatrix}. \quad (**)$$

Next, we will show that, $\text{Var}(\hat{Y}_{0,OLS}^A)$ is larger than $\text{Var}(\hat{Y}_{0,OLS})$,

Block Matrix Inversion

Consider a symmetric block matrix

$$\begin{bmatrix} A & B \\ B' & D \end{bmatrix},$$

where A and D are symmetric matrices. Then

$$\begin{bmatrix} A & B \\ B' & D \end{bmatrix}^{-1} = \begin{bmatrix} A^{-1} + A^{-1} B H^{-1} B' A^{-1} & -A^{-1} B H^{-1} \\ -H^{-1} B' A^{-1} & H^{-1} \end{bmatrix}$$

where $H = (D - B' A^{-1} B)$, symmetric.

Now,

$$W = (X : Z), \quad W'W = \begin{bmatrix} X'X & X'Z \\ Z'X & Z'Z \end{bmatrix}.$$

Then, by the above formula

$$(W'W)^{-1} = \begin{bmatrix} \underbrace{(X'X)^{-1} + (X'X)^{-1} X'Z H^{-1} Z'X (X'X)^{-1}}_{(1,1)} & \underbrace{-(X'X)^{-1} X'Z H^{-1}}_{(1,2)} \\ \underbrace{-H^{-1} Z'X (X'X)^{-1}}_{(2,1)} & \underbrace{H^{-1}}_{(2,2)} \end{bmatrix} \quad (3.1)$$

and

$$H = Z'Z - Z'X (X'X)^{-1} X'Z = Z'(I - P(X))Z = \text{p.s.d. matrix.}$$

Therefore, (**) gives

$$\begin{aligned}\text{Var}(\hat{Y}_{0,OLS}^A) &= \sigma^2(\underline{z}_0' : \underline{z}_0')(W'W)^{-1} \begin{pmatrix} \underline{z}_0 \\ \underline{z}_0 \end{pmatrix} \\ &= \sigma^2 \left[\underline{z}_0'(X'X)^{-1}\underline{z}_0 + \underline{z}_0'(X'X)^{-1}X'ZH^{-1}Z'X(X'X)^{-1}\underline{z}_0 \right. \\ &\quad \left. - \underline{z}_0'H^{-1}Z'X(X'X)^{-1}\underline{z}_0 - \underline{z}_0'(X'X)^{-1}X'ZH^{-1}\underline{z}_0 + \underline{z}_0'H^{-1}\underline{z}_0 \right]\end{aligned}$$

Let $\underline{a}_0 = Z'X(X'X)^{-1}\underline{z}_0$, the above simplifies to:

$$\begin{aligned}\text{Var}(\hat{Y}_{0,OLS}^A) &= \text{Var}(\hat{Y}_{0,OLS}) + \sigma^2 \left[\underline{a}_0'H^{-1}\underline{a}_0 - 2\underline{z}_0'H^{-1}\underline{a}_0 + \underline{z}_0'H^{-1}\underline{z}_0 \right] \\ &= \text{Var}(\hat{Y}_{0,OLS}) + \sigma^2(\underline{a}_0 - \underline{z}_0)^\top H^{-1}(\underline{a}_0 - \underline{z}_0) \geq \text{Var}(\hat{Y}_{0,OLS}).\end{aligned}$$

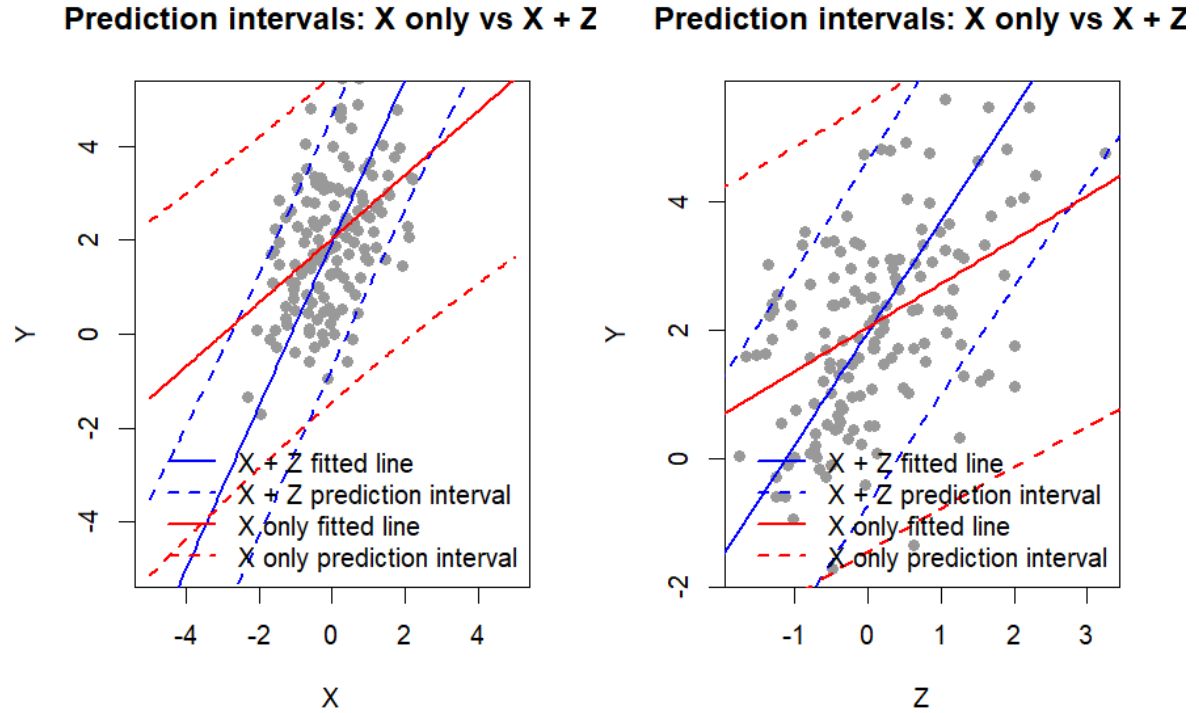


Figure 3.1: In the above figure, bias of the fit is clearly visible. For this data, var of the fitted values are ≈ 26 for the full model (A) and ≈ 4 for the wrong model (W). However, the prediction interval for (W) is wider as the estimate of σ^2 , $\hat{\sigma}_{UE}^2$ is larger under (W) compared to (A).

3.2 Including Additional Variables in the Model

What if $\{x_1, \dots, x_p\}$ are the *actual* predictors, but we also include $\{z_1, \dots, z_q\}$ in the model?

Model Specifications

The true underlying model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}^*; \quad \underline{\varepsilon}^* \sim N(\underline{0}, \sigma^2 I), \quad (\text{A}^*)$$

while the working model is

$$\underline{Y} = X\underline{\beta} + Z\underline{\gamma} + \underline{\varepsilon}; \quad \underline{\varepsilon} \sim N(\underline{0}, \sigma^2 I) \quad (\text{W}^*)$$

We write

$$W = [X : Z], \quad \underline{\delta} = \begin{bmatrix} \underline{\beta} \\ \underline{\gamma} \end{bmatrix}, \quad \text{so that} \quad (\text{W}^*) \Leftrightarrow \underline{Y} = W\underline{\delta} + \underline{\varepsilon}, \quad \text{and} \quad \hat{\underline{\delta}}_{\text{OLS}} = (W^T W)^{-1} W^T \underline{Y}.$$

Bias and Variance.

Since X is the actual predictor and Z mainly contains noise predictors, $\underline{\beta}$ is the main parameter of interest. We first check if $\underline{\beta}$ is predicted well. Observe that

$$\hat{\underline{\beta}}_{\text{OLS}} = [I_{p+1} : O] \hat{\underline{\delta}}_{\text{OLS}} = E_{p+1} \hat{\underline{\delta}}_{\text{OLS}}, \quad \text{and also} \quad X\underline{\beta} = [X : Z] \begin{pmatrix} \underline{\beta} \\ \underline{0} \end{pmatrix} = W \begin{pmatrix} \underline{\beta} \\ \underline{0} \end{pmatrix}.$$

Therefore, we have

$$E(\hat{\underline{\beta}}_{\text{OLS}}) = E_{p+1} (W^T W)^{-1} W^T E(\underline{Y}) = E_{p+1} \begin{pmatrix} \underline{\beta} \\ \underline{0} \end{pmatrix} = \underline{\beta}$$

and

$$\begin{aligned} \text{Var}(\hat{\underline{\beta}}_{\text{OLS}}) &= \text{Var}(E_{p+1} \hat{\underline{\delta}}_{\text{OLS}}) = E_{p+1} \text{Var}(\hat{\underline{\delta}}_{\text{OLS}}) E_{p+1}^T = \sigma^2 [I_{p+1} : O] (W^T W)^{-1} \begin{bmatrix} I_{p+1} \\ O \end{bmatrix} \\ &= \sigma^2 [\text{principal submatrix of } (W^T W)^{-1} \text{ of order } p+1] \\ &= \sigma^2 [(X'X)^{-1} + (X'X)^{-1} X^T Z \tilde{H}^{-1} Z' X (X'X)^{-1}] \geq \sigma^2 (X'X)^{-1}, \end{aligned} \quad (3.2)$$

from (3.1), where $H = Z'(I - P(X))Z$ is a p.s.d. matrix.

At this point, we define ordering of two symmetric matrices as follow:

Comparison of Square Matrices

Two symmetric matrices A, B can be compared as follows:

We say $A \geq B \Leftrightarrow A - B$ is p.s.d.

$\Leftrightarrow$ All eigenvalues of $A - B$ are non-negative.

$\Leftrightarrow$ All quadratic forms involving $(A - B)$ are non-negative.

$\Leftrightarrow$ For any normalized vector $\underline{a}$, $\underline{a}^\top A \underline{a} \geq \underline{a}^\top B \underline{a}$.

In (3.2) observe that,

$$\text{Var}(\hat{\beta}_{\text{OLS}}) - \sigma^2(X'X)^{-1} = \sigma^2 V^\top H^{-1} V, \quad \text{where } V = Z^\top X(X^\top X)^{-1}.$$

Thus, it is easy to see that $\text{Var}(\hat{\beta}_{\text{OLS}}) \geq \sigma^2(X'X)^{-1}$.

Comparison with model (A*)

Had one chosen the true model as the working model, then the additional noise variables would not appear. In that case $\text{Var}(\hat{\beta}_{\text{OLS}})$ would be $\sigma^2(X'X)^{-1}$.

This, further implies that estimator of any $\underline{a}^\top \beta$ has an higher variance when estimated by (W*), compared to (A*), for example, estimates of β_j , $j = 1, \dots, p$.

Therefore, including additional variables in the model increases variability of the OLS estimate $\hat{\beta}$.

Similarly, show that:

- 1) $\hat{Y}_{\text{OLS}}$ has **zero-bias** but higher variance.
- 2) $RSS/(n - p - q - 1)$ is unbiased for σ^2 .

In conclusion, although missing some important predictors introduces bias, adding unimportant predictors increases variance. Therefore, there is a bias variance trade off in selecting the right number of predictors. To maintain low MSE, one should include predictors parsimoniously, while not missing important predictors.

3.3 Heteroskedasticity

When the homoskedasticity assumption, i.e., $\text{var}(\varepsilon) = \sigma^2 I$, is violated, then regression model is called heteroskedastic. Under heteroskedasticity, we have $\text{Var}(\varepsilon) = \sigma^2 V$ where $V \neq cI$.

Model Specifications

The true underlying model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}^* ; \quad \underline{\varepsilon} \sim N(\underline{0}, \sigma^2 V), \quad (\text{A}^{**})$$

while the working model is

$$\underline{Y} = X\underline{\beta} + \underline{\varepsilon}^* ; \quad \underline{\varepsilon}^* \sim N(\underline{0}, \sigma^2 I) \quad (\text{W}^{**})$$

The expectation and variance of $\hat{\underline{\beta}}_{\text{OLS}} = (X'X)^{-1}X^T\underline{Y}$ would be:

- $E(\hat{\underline{\beta}}_{\text{OLS}}) = \underline{\beta}$ (unbiased)
- $\text{Var}(\hat{\underline{\beta}}_{\text{ols}}) = \sigma^2(X'X)^{-1}X^TVX(X'X)^{-1}$ (not same as in (W^{**}))

Similarly, for $\hat{\underline{Y}}_{\text{OLS}}$ and RSS , we have:

- $E(\hat{\underline{Y}}_{\text{OLS}}) = X\underline{\beta}$ (same as for (W^{**}))
- $\text{Var}(\hat{\underline{Y}}_{\text{ols}}) = \sigma^2 P(X)VP(X)$ (not same as in (W^{**}))
- $E(RSS) = E(\underline{Y}^T(I - P(X))\underline{Y}) = E(\underline{\varepsilon}^T(I - P(X))\underline{\varepsilon}) = \sigma^2 \text{tr}(V(I - P(X)))$.

Therefore, $\hat{\sigma}_{\text{UE}}^2$ is no-longer an unbiased estimator of σ^2 .

Therefore, heteroskedasticity does not affect the unbiasedness property of the OLS estimators. However, it affects the variances. Further, unless some idea on the structure of V is available, it is not possible to determine if (A^{**}) has an inflated variance, or a deflated variance compared to (W^{**}).

Remarks. There are two aspects of homoskedasticity, first, same variance of the ε_i s, $i = 1, \dots, n$, and zero-correlation among all pairs, i.e., $\text{cov}(\varepsilon_i, \varepsilon_j) = 0$. Either of these two can be violated. For example, the random part corresponding to larger data-points can have larger variance while ε_i s remain uncorrelated among themselves. In such a case, V will be a diagonal matrix with i -th diagonal as $g(\|\underline{x}_i\|)$, for some appropriate function $g : \mathbb{R} \rightarrow \mathbb{R}$. Alternatively, the ε_i s may have same variance, but they may be correlated. The correlation among the ε_i s is called *autocorrelation*. A common form of autocorrelation is given by the $AR(1)$ matrix described as

$$V = \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \vdots & & & & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{bmatrix}, \quad |\rho| < 1$$

This structure of V arises when the consecutive errors are highly correlated, and the correlation decays for far apart points, This naturally occurs in time-series data. Another, common form of autocorrelation is given by the *equi-correlation* matrix $V = (1 - \rho)I + \rho\underline{1}\underline{1}^T$, $-(n-1)^{-1} \leq \rho \leq 1$, where all the ε_i s are equally correlated.

3.4 Non-normality

In an OLS regression, normality of ε is not required for the OLS coefficient estimates to be unbiased or consistent. However, non-normality can create problems for small-sample statistical inference. In particular, the usual t-tests, F-tests, confidence intervals, and p -values rely on normality (or on large-sample approximations). If the errors are strongly skewed or heavy tailed, these procedures may give misleading significance levels and confidence intervals, especially when the sample size is small.

With a sufficiently large sample, the Central Limit Theorem often makes the sampling distribution of the OLS estimates approximately normal, so conventional inference becomes more reliable even when the errors themselves are non-normal. However, extreme non-normality, especially heavy-tailed errors or outliers, can make OLS estimates less precise and more sensitive to unusual observations, as we will see later.

3.5 X is Non-random

Non-randomness of the predictors is **NOT** an assumption of the OLS setup. Therefore, having random predictors does not violate any assumption of OLS. However, in that case, we have to redefine the regression model in such a way that accommodates random predictors:

When X is **random**, the $(Y, X_1, \dots, X_p)$ is a $p + 1$ -dimensional random variable.

Model Specifications

In the OLS setup we assume

$$\begin{aligned} E(\underline{Y} \mid \underline{X}_1 = \underline{x}_1, \dots, \underline{X}_p = \underline{x}_p) &= \beta_0 \underline{1} + \beta_1 \underline{x}_1 + \dots + \beta_p \underline{x}_p, \\ \text{and } \text{Var}(\underline{Y} \mid \underline{X}_1 = \underline{x}_1, \dots, \underline{X}_p = \underline{x}_p) &= \sigma^2 I. \end{aligned}$$

In normal linear regression model we assume

$$\underline{Y} \mid \underline{X}_1 = \underline{x}_1, \dots, \underline{X}_p = \underline{x}_p \sim N(\underline{0}, \sigma^2 I) \iff Y \mid X \stackrel{D}{=} X\beta + \varepsilon \quad \text{where } \varepsilon \sim N(0, \sigma^2 I).$$

Here as well the OLS estimate of β and σ^2 remain

$$\hat{\beta}_{\text{OLS}} = (X'X)^{-1}X^T\underline{Y}, \quad \text{and} \quad \hat{\sigma}_{\text{UE}}^2 = \frac{\text{RSS}}{n - p - 1}, \quad \text{where} \quad \text{RSS} = \underline{Y}^T(I - P(X))\underline{Y}.$$

Bias and Variance. OLS estimate of β is unbiased:

$$E(\hat{\beta}_{\text{OLS}} \mid X) = (X'X)^{-1}X^T \underbrace{E(\underline{Y} \mid X)}_{X\beta} = \beta \quad \text{and} \quad E(\hat{\beta}_{\text{OLS}}) = E_X(E(\hat{\beta}_{\text{OLS}} \mid X)) = \beta$$

The conditional variances remains unchanged:

$$\text{Var}(\hat{\beta}_{\text{OLS}} | X) = \sigma^2 (X'X)^{-1}$$

Further,

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = E_X \left(\text{Var}(\hat{\beta}_{\text{OLS}} | X) \right) + \underbrace{\text{Var}_X \left(E(\hat{\beta}_{\text{OLS}} | X) \right)}_{\beta} = \sigma^2 E((X'X)^{-1}) + 0 = \sigma^2 E((X'X)^{-1}).$$

Similarly, for $\hat{Y}_{\text{OLS}} = X\hat{\beta}_{\text{OLS}} = P(X)Y$, we have

$$E(\hat{Y}_{\text{OLS}}) = E_X(P(X)X\beta) = E_X(X\beta) = E(X)\beta$$

and

$$\begin{aligned} \text{Var}(\hat{Y}_{\text{OLS}}) &= E_X \left(\text{Var}(\hat{Y}_{\text{OLS}} | X) \right) + \text{Var} \left(E_{Y|X}(\hat{Y}_{\text{OLS}} | X) \right) \\ &= E_X(\sigma^2 P(X)) + \text{Var}(X\beta) = \sigma^2 E_X(P(X)) + \text{Var}(X\beta), \end{aligned}$$

as

$$\text{Var}_{Y|X}(\hat{Y}_{\text{OLS}}) = \text{Var}_{Y|X}(P(X)Y) = \sigma^2 P(X)$$

Observe that, the structure of

$$\text{Var}(X\beta) = E \left((X\beta - E(X)\beta)(X\beta - E(X)\beta)^T \right)$$

depends on the covariance matrix of X .

Here as well, $\hat{\sigma}_{\text{UE}}^2$ remains an unbiased estimate of σ^2 .

$$\begin{aligned} E(\text{RSS}) &= E_X(E_{Y|X}(Y^T(I - P(X))Y)) \\ &= E_X[E_{Y|X}(\varepsilon^T(I - P(X))\varepsilon)] \quad [\text{as } (I - P(X))X = 0 \text{ w.p.1}] \\ &= E_X[\text{tr}((I - P(X))\sigma^2 I)] = \sigma^2 E_X \left[\underbrace{\text{tr}(I - P(X))}_{n-p-1} \right] = \sigma^2(n - p - 1) \end{aligned}$$

Therefore, **no change in the results, as there is no departure from the assumptions.**

3.5.1 Fixed Predictors Observed with Errors

Let $\underline{x}_j$ be the j^{th} predictor. Instead of observing $x_{i,j} + \delta_{i,j} = z_{i,j}$, where $\delta_{i,j}$ is the error in observation.

Example: Let x be the age at death. While x is a continuous random variable, when we record it, we observe a spike around round figures, or multiples of five.

Model Specifications

$$\underset{n \times p}{Z} : \text{observed design matrix} = \underset{n \times p}{X} + \underset{n \times p}{\Delta}$$

Actual model: $y = X\beta + \varepsilon^*$, $\varepsilon^* \sim N(0, \sigma^2 I)$.

Working model: $y = Z\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$.

Continuing with the observed design matrix, one calculates the OLS estimate as $\hat{\beta}_{\text{OLS}} = (Z'Z)^{-1}Z'y$, which is biased as

$$E(\hat{\beta}_{\text{OLS}}) = E_{\Delta} \left((Z'Z)^{-1}Z' \left(\underbrace{X}_{(Z-\Delta)}\beta + E(\varepsilon^*) \right) \right) = \beta - E_{\Delta} \left[(Z'Z)^{-1}Z'\Delta\beta \right]$$

In such a situation, one requires an estimate of $\Delta\beta$ to measure this bias, as Z is observed. However, variance of $\hat{\beta}_{\text{OLS}}$ remains as expected

$$\text{Var}_{Y|Z}(\hat{\beta}_{\text{OLS}}) = (Z'Z)^{-1}Z'(\sigma^2 I)Z(Z'Z)^{-1} = \sigma^2(Z'Z)^{-1}.$$

Finally, in this case as well, $\hat{\sigma}_{\text{UE}}^2$ does not remain an unbiased estimate of σ^2 .

$$E_{Y|Z}(\text{RSS}) = E(Y'(I - P(Z))Y) = E\left[(X\beta + \varepsilon)^T(I - P(Z))(X\beta + \varepsilon)\right] \neq (n - p - 1)\sigma^2$$

as Now, $(I - P(Z))X \neq 0$.

3.6 Collinearity

Perfect Collinearity

$X'X$ is singular. We know that in such case, an unique OLS solution for the β is not available. The normal equation $X'X\beta = X'Y$ is consistent, and has multiple solutions. However, in this case, $P(X)$ is uniquely determined. Therefore, the measures based on $P(X)$, such as RSS, $\hat{Y}_{\text{OLS}}$, can be used conveniently.

Near Collinearity

$X'X$ is near singular.

- The data on the predictors show that the j -th predictor x_j is highly correlated with $\{x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_p\}$.
- $\det(X'X)$ is very close to 0.

- $(X'X)^{-1}$ exists, some of its eigenvalues are extremely large.

When $X'X$ is near-singular, none of the assumptions of OLS is violated. So, all the results associated with the OLS, or even the normal linear regression model are still valid. However, the OLS estimate of β becomes unstable and hence useless for practical purposes.

Suppose x_j is highly correlated to other predictors. Now

$$\text{Var}(\hat{\beta}_{\text{OLS}}) = \sigma^2(X'X)^{-1}, \quad \text{and} \quad \text{Var}(\hat{\beta}_{\text{OLS},j}) = \sigma^2 \underline{e}_j^T (X'X)^{-1} \underline{e}_j$$

Assume WLoG, $j = p$ and let $Z = [\underline{1}, \dots, x_{p-1}]$. Then

$$(X'X)^{-1} = \begin{bmatrix} Z'Z & Z'x_p \\ x_p^T Z & x_p^T x_p \end{bmatrix}^{-1} = \begin{bmatrix} ** & * \\ * & [x_p^T(I - P(Z))x_p]^{-1} \end{bmatrix}$$

Therefore,

$$\text{Var}(\hat{\beta}_{\text{OLS},p}) = \sigma^2 \underline{e}_p^T (X'X)^{-1} \underline{e}_p = \frac{\sigma^2}{x_p^T(I - P(Z))x_p}$$

If x_p is highly correlated with Z , then $(I - P(Z))x_p = \underline{s}_p$ will have $\|\underline{s}_p\|^2$ small (i.e., x_p is mostly explained by Z). Then $\|\underline{s}_p\|^2 = x_p^T(I - P(Z))x_p \approx 0$, and consequently, $\text{Var}(\hat{\beta}_{\text{OLS},p})$ becomes extremely large.

Variance Inflation Factor (VIF)

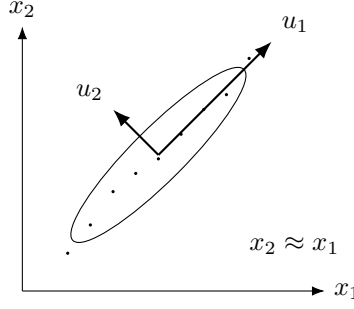
One possible way to detect near-collinearity or *multicollinearity* is through VIFs, where

$$\text{VIF}_j = \frac{\|x_j\|^2}{x_j^T(I - P(X_{(-j)}))x_j} \geq 1, \quad j = 1, \dots, p$$

and $X_{(-j)} = [\underline{1} : x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_p]$.

If VIF_j is very large for one or more j , then that indicates presence of **multicollinearity**.

An Illustrative Example Consider the situation when there are two predictors x_1 and x_2 , and $x_1 \approx x_2$. Further, assume that both the predictors and response are centered, so that there is no intercept in the model.


 Figure 3.2: Near-collinear covariates x_1 and x_2 .

Then, the spectral decomposition of $X^\top X$ will be

$$X^\top X = d_1 \underline{u}_1 \underline{u}_1^\top + d_2 \underline{u}_2 \underline{u}_2^\top, \quad \underline{u}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \underline{u}_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad d_1 \gg d_2, \quad d_2 \approx 0.$$

Consider the simple linear regression models:

$$Y = \alpha_0 + \beta_1 x_1 + \varepsilon_1 \quad \text{and} \quad Y = \alpha_1 + \beta_2 x_2 + \varepsilon_2, ,$$

For the above simple linear regression models, we expect $\hat{\beta}_{\text{OLS},1} \approx \hat{\beta}_{\text{OLS},2}$. However, if we consider the multiple linear regression model, then

$$Y = \alpha + \beta_1 x_1 + \beta_2 x_2 + \varepsilon \approx \alpha + \underbrace{(\beta_1 + \beta_2)}_{\beta} x_1 + \varepsilon,$$

i.e., it would behave like a simple linear regression model with $\beta := \beta_1 + \beta_2$ as the regression coefficient.

The **normal equations** would be:

$$(X^\top X) \hat{\beta} = X^\top \underline{Y}.$$

Writing the normal equations in terms of the $\{\underline{u}_1, \underline{u}_2\}$ basis, we get

$$\begin{aligned} & \frac{d_1}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} (\hat{\beta}_1 + \hat{\beta}_2) + \frac{d_2}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} (\hat{\beta}_1 - \hat{\beta}_2) = X^\top \underline{Y}. \\ \Rightarrow & \begin{bmatrix} d_1(\hat{\beta}_1 + \hat{\beta}_2) + d_2(\hat{\beta}_1 - \hat{\beta}_2) \\ d_1(\hat{\beta}_1 + \hat{\beta}_2) - \underbrace{d_2(\hat{\beta}_1 - \hat{\beta}_2)}_{\approx 0} \end{bmatrix} \approx \begin{bmatrix} \hat{\beta}_1 + \hat{\beta}_2 \\ \hat{\beta}_1 + \hat{\beta}_2 \end{bmatrix} = 2 \begin{bmatrix} \underline{x}_1^\top \underline{Y} \\ \underline{x}_2^\top \underline{Y} \end{bmatrix} \Rightarrow \hat{\beta}_1 + \hat{\beta}_2 \approx k_0 \quad (\text{approx}), \end{aligned}$$

where $k_0 = \underline{x}_1^\top \underline{y} \approx \underline{x}_2^\top \underline{y}$. Note that, as the normal equations are consistent, i.e., $X^\top \underline{Y} \in \mathcal{C}(X^\top X)$, there exists a solution of this equation. In this case, there exists infinitely many solutions to $(\hat{\beta}_1, \hat{\beta}_2)$. In fact, any choice of $(\hat{\beta}_1, \hat{\beta}_2)$ satisfying $\hat{\beta}_1 + \hat{\beta}_2 = k_0$ is a solution.

In particular, consider two particular choices

$$\hat{\beta}_1 = 1000 \quad \hat{\beta}_2 = -1000 + k_0 \quad \hat{\beta}_1 - \hat{\beta}_2 = 2000 - k_0$$

and

$$\hat{\beta}_1 = -1000 \quad \hat{\beta}_2 = 1000 + k_0 \quad \hat{\beta}_1 - \hat{\beta}_2 = -2000 - k_0$$

Thus, variance of $\hat{\beta}_1 - \hat{\beta}_2$ over these two choices is 8×10^6 (approx).

Therefore, while one linear combination $\hat{\beta}_1 + \hat{\beta}_2$ is uniquely obtained and has reasonable variance, another linear combination $\hat{\beta}_1 - \hat{\beta}_2$ is not obtained uniquely, and its variance is **huge**.

Problem of Variance Inflation under Near-collinearity

Now, consider the problem in a general context. Suppose

$$X^T X = d_1 \underline{u}_1 \underline{u}_1^T + \cdots + d_{p+1} \underline{u}_{p+1} \underline{u}_{p+1}^T.$$

Some of the d_j 's are very close to 0 [*near collinearity*]. Then

$$\text{Var}(\underline{a}^T \hat{\beta}_{OLS}) = \sigma^2 \underline{a}^T (X^T X)^{-1} \underline{a} = \sigma^2 \sum_{j=1}^{p+1} \frac{1}{d_j} (\underline{a}^T \underline{u}_j)^2 = \sigma^2 \sum_{j=1}^{p+1} \frac{1}{d_j} \alpha_j^2.$$

Note: $\alpha_j^2 \geq 0$ and $\sum \alpha_j^2 = \|\underline{a}\|^2$ ($\because \underline{a}^T U U^T \underline{a} = \underline{a}^T \underline{a}$).

If $d_j \approx 0$ for $j = j_0$, then $\frac{1}{d_j}$ is very large. Then the j_0 -th summand, $\frac{\alpha_{j_0}^2}{d_{j_0}}$, is also very large provided $\alpha_{j_0} = \underline{a}^T \underline{u}_{j_0}$ is not close to zero, i.e., $\underline{a}$ is **NOT** (nearly) orthogonal to $\underline{u}_{j_0}$.

In the above example, the choice $\underline{a} = (1, 1)$ was orthogonal to the singular direction, i.e., $(1, -1)$. Consequently, $\hat{\beta}_1 + \hat{\beta}_2$ did not have a large variance, while $\hat{\beta}_1 - \hat{\beta}_2$ had a large variance.

3.6.1 Detecting Collinearity

(1) Condition Number. In view of the above, it is easy to see that if at least one eigenvalue of $X^T X$ is very small, then there would exist at least one linear combination of β , whose OLS estimator will have large variance. Therefore, one way to detect multicollinearity is to consider the condition number

$$\kappa(X) = \sqrt{\frac{\lambda_{\max}(X^T X)}{\lambda_{\min}(X^T X)}} = \frac{\sigma_{\max}(X)}{\sigma_{\min}(X)}.$$

Note that, it is not possible that all the eigenvalues of $X^T X$ are small. Since the sum of all the eigenvalues of $X^T X$ is $\text{trace}(X^T X)$, which is fixed, and the eigenvalues are non-negative, the highest eigenvalue can't be small.

(2) **Variance Inflation Factor.** We have already discussed the VIF as

$$\text{VIF}_j = \frac{\|\tilde{x}_j\|^2}{\tilde{x}_j^T (I - P(X_{-j})) \tilde{x}_j} > 1, \quad j = 1, \dots, p + 1.$$

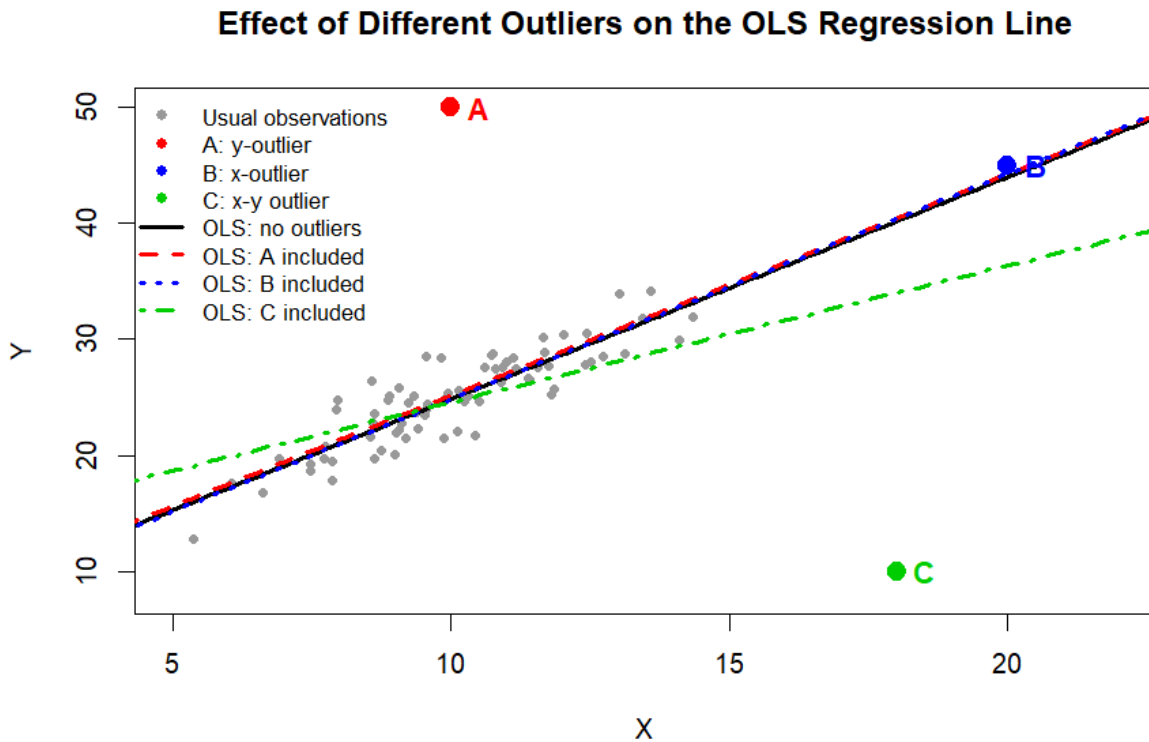
Since

$$\text{Var}(\hat{\beta}_{OLS,j}) = \sigma^2 x_j^T (I - P(x_{-j})) x_j,$$

VIF verifies if the variances of the individual regression coefficients are small enough. The numerator $\|\tilde{x}_j\|^2$ makes VIF scale-free, and sets the lowerbound of VIF as *one*.

It can be shown that, when near-collinearity exists, then slight perturbation $\tilde{Y}$ may result in a huge change in the OLS estimated $\tilde{\beta}$, i.e., $\hat{\tilde{\beta}}_{OLS}$. (PS)

3.7 Outliers



There are 3 types of outliers:

1 Residual-outlier: The points having unusually high error (unusual in y -axis). For example, point

A in the figure.

2. high-leverage points. The points which are unusual with respect to the X data cloud. For example, point B in the figure.

3. influential points. The points which are unusual, both with respect to the X data cloud and residual. For example, point C in the figure.

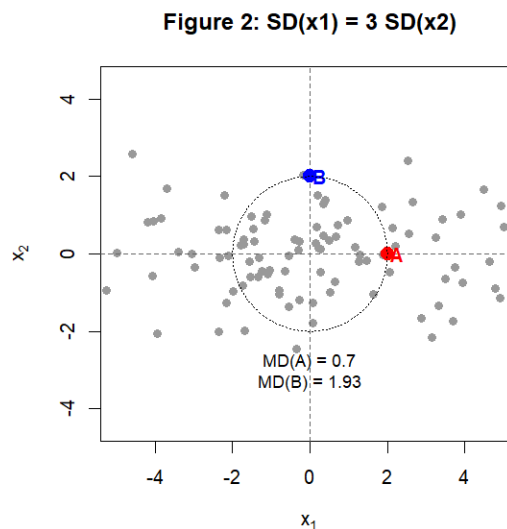
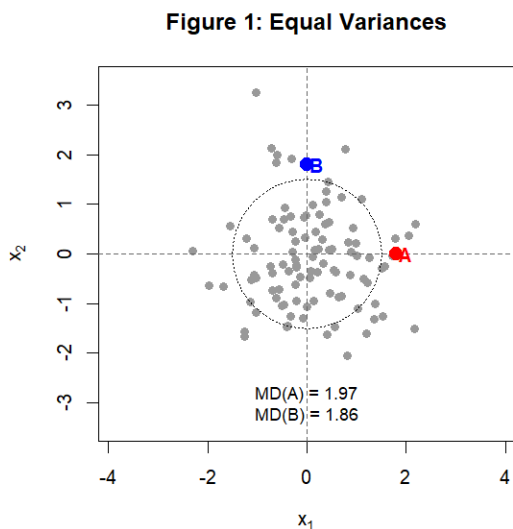
3.7.1 Detecting a point which is unusual with respect to the X -data cloud.

Mahalanobis distance is a tool to detect how far a data point is located with respect to the center of the data cloud.

Mahalanobis Distance (MD)

Suppose $\mathbf{x}_{(1)}, \dots, \mathbf{x}_{(n)}$ are the n data points, $\mathbf{x}_{(i)} \in \mathbb{R}^{p+1}$. Then the MD of the i -th data point $\mathbf{x}_{(i)}$ from the center of the data cloud is given by

$$MD_i = MD(\mathbf{x}_{(i)}) = \sqrt{(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^T S_x^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}})}. \quad \text{where} \quad S_x = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_{(i)} - \bar{\mathbf{x}})(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^T = ((\text{cov}(\mathbf{x}_j, \mathbf{x}_k))).$$



Relation between the projection matrix $P(X)$ and MD

As $\hat{\mathbf{y}}_{OLS} = P(X)\mathbf{y}$, $P(X) = ((p_{ij}))$, $P(X)$ is also called the *Hat Matrix*. Next, we will explore a connection between the MD_i and the i -diagonal of $P(X)$.

Recall, from Section 2.4, we have $P(X) = P(X_c)$ where X_c is the matrix of centered predictors (while the first column is $\underline{1}$). We write $X_c = \begin{bmatrix} \underline{1} & Z_c \end{bmatrix}$, where $Z_c = \begin{bmatrix} (x_1 - \bar{x}_1)\underline{1} & \dots & (x_p - \bar{x}_p)\underline{1} \end{bmatrix}$. Further, it has been shown that

$$P(X) = P(X_c) = \frac{1}{n} \underline{1}\underline{1}^\top + P(Z_c).$$

Using this, we express the i -th diagonal element $P(X)$ as:

$$\begin{aligned} p_{ii} &= \underline{e}_i^\top P(X) \underline{e}_i = \underline{e}_i^\top \left[\frac{1}{n} \underline{1}\underline{1}^\top + P(Z_c) \right] \underline{e}_i, \quad \text{where } P(Z_c) = Z_c(Z_c^\top Z_c)^{-1} Z_c^\top \\ &= \frac{1}{n} + \underline{e}_i^\top Z_c S_x^{-1} Z_c^\top \underline{e}_i / n, \\ &= \frac{1}{n} + \frac{1}{n} [(x_{i,1} - \bar{x}_1) \cdots (x_{i,p} - \bar{x}_p)] S_x^{-1} \begin{bmatrix} (x_{i,1} - \bar{x}_1) \\ \vdots \\ (x_{i,p} - \bar{x}_p) \end{bmatrix}, \end{aligned}$$

since

$$\underbrace{Z_c^\top Z_c}_{p \times p} = \left(\left(\sum_{i=1}^n (x_{i,j} - \bar{x}_j)(x_{i,k} - \bar{x}_k) \right) \right)_{\substack{j=1, \dots, p \\ k=1, \dots, p}} = ((\text{cov}(\underline{x}_j, \underline{x}_k))) = n S_x,$$

and

$$P(Z_c) = Z_c(n S_x)^{-1} Z_c^\top = n^{-1} Z_c S_x^{-1} Z_c^\top.$$

Thus, for the i -th observation

$$p_{ii} = \frac{1}{n} \{1 + (\underline{x}_i - \bar{\underline{x}})^\top S_x^{-1} (\underline{x}_i - \bar{\underline{x}})\} = \frac{1}{n} (1 + \text{MD}_i^2). \quad (3.3)$$

Range of $p_{i,i}$: From (3.3), as $\text{MD}_i \geq 0$, we have $p_{i,i} \geq n^{-1}$. Further, observe that, $P(X)$ is symmetric and

$$\begin{aligned} P(X)^2 = P(X) &\Rightarrow \underline{p}_{i,\cdot}^\top \underline{p}_{i,\cdot} = p_{i,i}, \quad \text{where } \underline{p}_{i,\cdot} \text{ is the } i\text{-th row of } P(X), \\ &\Rightarrow p_{i,i}^2 + \sum_{j \neq i} p_{i,j}^2 = p_{i,i} \Rightarrow p_{i,i}^2 \leq p_{i,i} \Rightarrow p_{i,i} \leq 1. \end{aligned} \quad (3.4)$$

Therefore, we have

$$\frac{1}{n} \leq p_{i,i} \leq 1. \quad (3.5)$$

Next, observe that

$$\hat{Y}_{OLS,i} = \underline{e}_i^\top \hat{\underline{Y}}_{OLS} = \underline{e}_i^\top P(X) \underline{Y} = \underline{p}_{i,\cdot} \underline{Y} = p_{i,1} Y_1 + \cdots + p_{i,n} Y_n = p_{i,i} Y_i + \sum_{j \neq i} p_{i,j} Y_j, \quad (3.6)$$

where $\underline{p}_{i,\cdot}$: i -th row of $P(X)$ and $\underline{e}_i = \begin{bmatrix} 0 & \cdots & 0 & \underbrace{1}_{i\text{-th}} & 0 & \cdots & 0 \end{bmatrix}^\top$.

1. If the i -th point is a high-leverage point, then it will have higher MD_i value. Then, by (3.3), the corresponding $p_{i,i}$ would be large. Therefore, a high-leverage point can be identified by a high $p_{i,i}$ value.
2. From (3.4), we have $p_{i,i}(1 - p_{i,i}) = \sum_{j \neq i} p_{i,j}^2$. This implies if $p_{i,i}$ is large, i.e., close to 1, then the remaining $(n - 1)$ components of $\underline{p}_{i,\cdot}$, $\{p_{i1}, \dots, p_{i,i-1}, p_{i,i+1}, \dots, p_{in}\}$, are close to zero.
3. Consequently, if $p_{i,i}$ is close to one, then from (3.6), the value of $\hat{Y}_{OLS,i}$ is largely determined by Y_i only (as the weight of Y_i is close to one and that of all Y_j s, for $j \neq i$, is close to zero).
4. Further, $\sum_i p_{i,i} = \text{trace}(P(X)) = p + 1$. Therefore, if for some i , $p_{i,i}$ is close to one, then the remaining $(n - 1)$ many $p_{j,j}$ s, $j \neq i$, share the total weight p . Therefore, all the points can not have $p_{i,i}$ -values close to one. This implies that the $\hat{Y}_{OLS,i}$ -values need not be close to their corresponding Y_i -values, for all i .
5. Finally, if an observation with a high $p_{i,i}$ has a large residual, the regression line may be substantially influenced by that observation, shifting in such a way that $\hat{Y}_{OLS,i}$ becomes closer to Y_i .

Thus, a **high-leverage point** has **higher influence** on the regression line.

3.7.2 High-Residual Point but Not High-Leverage Point

If a point is a high-residual point then it can be well-identified by the residuals, as long as it is not a high-leverage point.

To see this, suppose WLoG that the last observation Y_n is a high-residual point:

$$\left. \begin{array}{l} Y_i = \underline{x}_i^T \underline{b} + \varepsilon_i, \quad i = 1, \dots, n-1 \\ Y_n = \underline{x}_n^T \underline{b} + \delta + \varepsilon_n, \quad |\delta| \text{ is large.} \end{array} \right\} \Rightarrow \underline{y} = X\underline{b} + \underline{\Delta} + \underline{\varepsilon}, \quad \text{where } \underline{\Delta} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \delta \end{bmatrix}, \quad \underline{\varepsilon} \sim N(0, \sigma^2 I).$$

Then the **residual vector** would be:

$$\underline{e} = (I - P(X))\underline{y} = (I - P(X))\underline{\Delta} + (I - P(X))\underline{\varepsilon},$$

with expectation

$$E(\underline{e}) = (I - P(X))\underline{\Delta} = Q(X)\underline{\Delta} = \delta Q(X)\underline{e}_n = \delta \cdot \underline{q}_n, \quad \text{where } \underline{q}_n : \text{last column of } Q(X).$$

Observe that

$$Q(X) = I - P(X) = \begin{bmatrix} 1 - p_{11} & -p_{12} & \cdots & -p_{1n} \\ \vdots & & & \vdots \\ -p_{n1} & -p_{n2} & \cdots & 1 - p_{nn} \end{bmatrix},$$

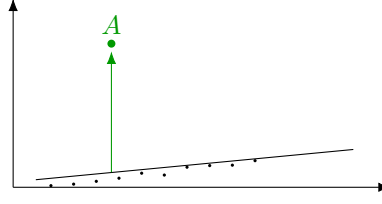
Therefore,

$$E(\underline{e}) = \begin{bmatrix} E(e_1) \\ \vdots \\ E(e_n) \end{bmatrix} = \begin{bmatrix} -\delta p_{1n} \\ \vdots \\ -\delta(1 - p_{nn}) \end{bmatrix}.$$

- If $(\underline{x}_n, Y_n)$ is **not** a high-leverage point, then $p_{n,n}$ is not unusually large. If all the n observations are equally spread over the data cloud, then we expect that the MD_i are nearly same for all $i = 1, \dots, n$, and consequently, $p_{i,i}$ are nearly same.
- Since $\sum_i p_{i,i} = p + 1$, we expect each $p_{i,i} \sim (p + 1)/n$. Further, from properties of Mahalanobis distance (see PS3, Q4), we expect $p_{i,j} \sim 1/n(1 - p/(n - 1))$.
- Therefore, $1 - p_{n,n} \gg |p_{i,n}|$ for all $i = 1, \dots, n$ (particularly, when n is large). Therefore, for a large enough $|\delta|$, $E(e_n) \gg E(e_i)$ for $i \neq n$.
- Further, note that the variance of $\underline{e}$, $\text{Var}(\underline{e}) = \sigma^2(I - P(X))$ remains unaltered.
- Consequently, unusually large residual indicates high-residual outlier, unless the outlier is also a high-leverage point. If $(\underline{x}_n, Y_n)$ is also a high-leverage point, then $p_{n,n}$ would be close to *one*, and in that case $E(e_n)$ might be comparable with $E(e_i)$ for $i \neq n$.

Infludential point

1. An observation that has an unusually large residual relative to the regression line fitted without that observation, and that is also unusual with respect to the X -data cloud, is called an **infludential point**.
2. Unlike a high-residual outlier, which can be identified solely by its large residual, an infludential point cannot be identified from the residual plot based on the regression line fitted to the full data set (including that point).
3. To identify an infludential point, we compare the regression line fitted using all observations with the regression line obtained after excluding that observation. A substantial change in the regression line indicates that the observation has a strong influence on the fitted model.

Figure 3.3: High-residual outlier A , identified via the residual plot.

3.7.3 Influential Points

Based on the above discussion, one may define a point as **influential** if the regression line including that point is very different from the one excluding that point.

To determine whether a point is influential, we need to compare the regression line fitted excluding that observation with the regression line fitted using the full data. This raises the question of whether the regression line needs to be fitted separately after removing each observation.

The following result shows that this is not necessary. In particular, the regression coefficients obtained after excluding the i -th observation can be expressed in terms of the regression coefficients fitted on the full-data, the residual of the i -th observation from the full-data regression line, and p_{ii} . Thus, the regression line needs to be fitted only once.

Result

Result: Let $\hat{\beta}_{OLS}$ and $\hat{\beta}_{OLS,(-i)}$ be the OLS regression coefficients of $\underline{Y}$ on X using the full data set, and of $\underline{Y}_{(-i)}$ on $X_{(-i)}$, respectively, where $(\underline{Y}_{(-i)}, X_{(-i)})$ is the data set excluding the i -th observation $(\underline{x}_{(i)}, Y_i)$. Further, let e_i is the residual of the i -th fitted Y_i using $\hat{\beta}_{OLS}$. Then we have

$$\hat{\beta}_{OLS} - \hat{\beta}_{OLS,(-i)} = (X^T X)^{-1} \underline{x}_i \left(\frac{e_i}{1 - p_{ii}} \right).$$

Woodbury Matrix Identity

$$(A + \underline{u}\underline{v}^T)^{-1} = A^{-1} - A^{-1}\underline{u} \left(1 + \underline{v}^T A^{-1} \underline{u} \right)^{-1} \underline{v}^T A^{-1}.$$

Proof of the Result

$\hat{\beta}_{\sim OLS, (-i)}$ = OLS estimate of β considering $(n - 1)$ observations excluding $(\mathbf{x}_{(i)}, Y_i)$. Let

$$\underbrace{X_{(-i)}}_{(n-1) \times (p+1)} = \begin{bmatrix} \mathbf{x}_{(1)}^T \\ \vdots \\ \mathbf{x}_{(i-1)}^T \\ \mathbf{x}_{(i+1)}^T \\ \vdots \\ \mathbf{x}_{(n)}^T \end{bmatrix}, \quad \text{then} \quad X^T X = \begin{bmatrix} \mathbf{x}_{(1)} & \cdots & \mathbf{x}_{(i)} & \cdots & \mathbf{x}_{(n)} \end{bmatrix} \begin{bmatrix} \mathbf{x}_{(1)}^T \\ \vdots \\ \mathbf{x}_{(i)}^T \\ \vdots \\ \mathbf{x}_{(n)}^T \end{bmatrix}$$

$$= \mathbf{x}_{(1)} \mathbf{x}_{(1)}^T + \cdots + \mathbf{x}_{(n)} \mathbf{x}_{(n)}^T = X_{(-i)}^T X_{(-i)} + \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T.$$

Hence,

$$X_{(-i)}^T X_{(-i)} = X^T X - \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T,$$

Now, by Woodbury matrix identity, we have

$$\begin{aligned} \left(X_{(-i)}^T X_{(-i)} \right)^{-1} &= \left(X^T X - \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T \right)^{-1} \\ &= (X^T X)^{-1} - (X^T X)^{-1} \mathbf{x}_{(i)} \left(-1 + \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)} \right)^{-1} \mathbf{x}_{(i)}^T (X^T X)^{-1} \\ &= (X^T X)^{-1} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T (X^T X)^{-1}}{1 - \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)}}. \end{aligned}$$

Also,

$$X^T \mathbf{y} = \begin{bmatrix} \mathbf{x}_{(1)} & \cdots & \mathbf{x}_{(n)} \end{bmatrix} \mathbf{y} = Y_1 \mathbf{x}_{(1)} + \cdots + Y_n \mathbf{x}_{(n)},$$

Therefore,

$$X_{(-i)}^T \mathbf{y}_{(-i)} = X^T \mathbf{y} - Y_i \mathbf{x}_{(i)},$$

and note that $\mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)} = p_{i,i}$. Then

$$\begin{aligned} \hat{\beta}_{\sim OLS, (-i)} &= \left(X_{(-i)}^T X_{(-i)} \right)^{-1} X_{(-i)}^T \mathbf{y}_{(-i)} \\ &= \left[(X^T X)^{-1} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^T (X^T X)^{-1}}{1 - \mathbf{x}_{(i)}^T (X^T X)^{-1} \mathbf{x}_{(i)}} \right] (X^T \mathbf{y} - Y_i \mathbf{x}_{(i)}) \\ &= \hat{\beta}_{OLS} - Y_i (X^T X)^{-1} \mathbf{x}_{(i)} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \left\{ \mathbf{x}_{(i)}^T \hat{\beta}_{OLS} - Y_i p_{i,i} \right\}}{1 - p_{i,i}} \\ &= \hat{\beta}_{OLS} + \frac{(X^T X)^{-1} \mathbf{x}_{(i)} \left\{ -(1 - p_{i,i}) Y_i + \left(\mathbf{x}_{(i)}^T \hat{\beta}_{OLS} - Y_i \right) + Y_i (1 - p_{i,i}) \right\}}{1 - p_{i,i}} \\ &= \hat{\beta}_{OLS} - \frac{e_i}{1 - p_{i,i}} (X^T X)^{-1} \mathbf{x}_{(i)}. \end{aligned}$$

Hence the result follows.

3.7.4 A Remark on Robustness

While identifying a high-leverage point by MD_i or p_{ii} , note that MD_i depends on the measures $(\bar{x}, S_x)$, and both $\bar{x}$ and S_x are affected by an extreme observation. So, we have to replace $\bar{x}, S_x$ by some **robust estimators** of location and scale.

Thus, we may consider a robust version of MD by considering a robust versions of location and scale parameters:

$$\widetilde{MD}_i^2 = (\underline{x}_i - \underline{m}(X))^T C(X)^{-1} (\underline{x}_i - \underline{m}(X)),$$

where $\underline{m}(X)$ is a **robust location estimate** (replacing $\bar{x}$) and $C(X)$ is a **robust scale estimate** (replacing S_x).

Choices of $\underline{m}(X)$ and $C(X)$

- $\underline{m}(X)$ and $C(X)$ can be the **trimmed mean and variance**, i.e., $\bar{x}_{(n-h)}$ and $S_{x,n-h}$ (i.e., $\bar{x}$ and S_x calculated using a subset of $\{\underline{x}_{(1)}, \dots, \underline{x}_{(n)}\}$ of $(n-h)$ observations).
- Which $(n-h)$ observations to retain?

Minimum Covariance Determinant (MCD) method: Consider that subset for which $S_{x,n-h}$ has the **lowest determinant**.

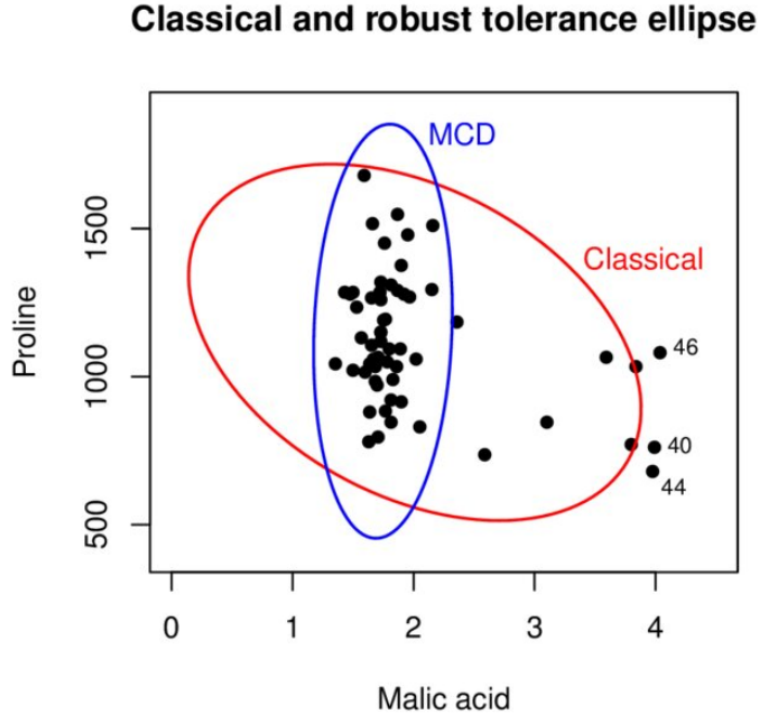


Figure 3.4: Illustration of MCD method in comparison with usual Mahalanobis distance based ellipsoid

The lowest determinant is intuitive as the volume of the ellipse

$$\mathcal{E}_k = \{\underline{x} : (\underline{x} - \underline{m})^T C^{-1} (\underline{x} - \underline{m}) \leq k\},$$

is proportional to $\det(C)$, i.e., $\text{vol}(\mathcal{E}_k) \propto \det(C)$, and it becomes the lowest, for a fixed k , if $\det(C)$ is the lowest.